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Orbital Motion in Strongly Perturbed Environments

Applications to Asteroid, Comet and Planetary Satellite Orbiters

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At the heart of our dynamical analyses are known solutions to fundamental dynamical problems of orbital and rotational motion. In this chapter we present the two core solutions necessary for more in-depth study of the problem: the two-body problem under mutual gravitation and the rotational dynamics of a torque-free rigid body. These are both classical solutions, but we detail their derivation here for completeness and to introduce specific concepts and notations used in this book.

3.1 The Two-Body Problem

An understanding of the two-body problem is required to motivate our dynamical understanding of motion about small bodies, as this solution serves as the starting point for many theories of motion. Thus, instead of providing a description of the solution we provide a brief but thorough derivation of the two-body problem.

3.1.1 The Two-Body Problem Statement

Assume that two bodies P_1 and P_2, with mass M_1 and M_2 and position vectors $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$, respectively, are attracting each other according to Newton's law of gravitation. If the mutual gravitational potential between these bodies is derived assuming both bodies have spherical mass distributions we find the following dynamical system:

$$M_i \ddot{\boldsymbol{r}}_i = \frac{\partial \mathcal{U}}{\partial \boldsymbol{r}_i} \tag{3.1}$$

$$i = 1, 2$$

$$\mathcal{U} = \frac{\mathcal{G} M_1 M_2}{|\boldsymbol{r}_2 - \boldsymbol{r}_1|} \tag{3.2}$$

where $\mathcal{U}$ is the gravitational force potential (opposite in sign to the potential energy).

3.1.2 Classical Integrals of Motion

The system conserves translational and angular momentum, as well as energy. Conservation of translational momentum allows us to reduce the problem by three degrees of freedom by recasting it in terms of its center of mass and the relative motion between the two bodies. Specifically, by choosing new coordinates:

$$\boldsymbol{r}_C = \frac{1}{M_1 + M_2}\left(M_1 \boldsymbol{r}_1 + M_2 \boldsymbol{r}_2\right) \tag{3.3}$$

$$\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1 \tag{3.4}$$

it is easy to show that the new equations of motion for these variables becomes

$$\ddot{\boldsymbol{r}}_C = 0 \tag{3.5}$$

$$\ddot{\boldsymbol{r}} = \frac{M_1 + M_2}{M_1 M_2}\frac{\partial \mathcal{U}}{\partial \boldsymbol{r}} \tag{3.6}$$

Define the relative potential for the system as:

$$U = \frac{M_1 + M_2}{M_1 M_2}\mathcal{U} \tag{3.7}$$

$$= \frac{\mathcal{G}(M_1 + M_2)}{|\boldsymbol{r}|} \tag{3.8}$$

which leads to the usual statement of the two-body problem in relative form:

$$\ddot{\boldsymbol{r}} = \frac{\partial U}{\partial \boldsymbol{r}} \tag{3.9}$$

The resulting system still conserves energy and angular momentum

$$\mathcal{E} = \frac{1}{2}\frac{M_1 M_2}{M_1 + M_2}\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} - \mathcal{U}(\boldsymbol{r}) \quad \text{energy} \tag{3.10}$$

$$\boldsymbol{\mathcal{H}} = \frac{M_1 M_2}{M_1 + M_2}\boldsymbol{r} \times \dot{\boldsymbol{r}} \quad \text{ang. momentum} \tag{3.11}$$

More commonly used are the specific energy and angular momentum of the system, found by dividing through by the effective mass $M_1 M_2/(M_1 + M_2)$,

$$E = \frac{1}{2}\boldsymbol{v} \cdot \boldsymbol{v} - U(\boldsymbol{r}) \tag{3.12}$$

$$\boldsymbol{H} = \boldsymbol{r} \times \boldsymbol{v} \tag{3.13}$$

where $\boldsymbol{v} = \dot{\boldsymbol{r}}$. Conservation of energy E relies on the potential U not being a function of time, while conservation of the angular momentum $\boldsymbol{H}$ relies on U corresponding to a central field, i.e., only a function of the distance between the two bodies. The simple two-body potential satisfies both of these conditions, as can be verified.

The geometrical interpretation of the angular momentum integral can be split into two pieces, and is most easily noted by rewriting it as $\boldsymbol{H} = H\hat{\boldsymbol{H}}$, the product of the magnitude and the unit vector in which it points, both of which must be constant. Constancy of $\hat{\boldsymbol{H}}$ implies that the solution to the two-body problem lies on a fixed plane perpendicular to $\hat{\boldsymbol{H}}$. This plane can be oriented relative to an inertial coordinate system by two angles, traditionally chosen as the inclination, i, which is the angle between $\hat{\boldsymbol{H}}$ and the $\hat{\boldsymbol{z}}$-axis of the coordinate frame, and the longitude of the ascending node Ω, which is the angle from the $\hat{\boldsymbol{x}}$-axis to the vector $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{H}}$, which lies in the $\hat{\boldsymbol{x}} - \hat{\boldsymbol{y}}$ plane. The angular momentum unit vector can be specified using these two angles as:

$$\hat{\boldsymbol{H}} = \sin i \sin \Omega \hat{\boldsymbol{x}} - \sin i \cos \Omega \hat{\boldsymbol{y}} + \cos i \hat{\boldsymbol{z}} \tag{3.14}$$

The angular momentum being a constant of the motion, these two "orbital elements" are also constants.

The constancy of the orbit energy and angular momentum magnitude provide explicit relationships between the velocity and the position of the relative solution in the plane defined by $\hat{\boldsymbol{H}}$. If we define the angle between the position and velocity vector as $\pi/2 - \gamma$, where γ is commonly termed the flight path angle (see Fig. 3.1), then $H = rv\cos\gamma$. It is interesting to consider the projection of the velocity vector along the radius vector and perpendicular to it:

$$v_r = \boldsymbol{v} \cdot \hat{\boldsymbol{r}} \qquad \text{radial} \tag{3.15}$$

$$= v \sin \gamma \tag{3.16}$$

$$v_h = v \cos \gamma \qquad \text{horizontal} \tag{3.17}$$

$$= H/r \tag{3.18}$$

where v_r stands for radial and v_h stands for horizontal. If we introduce the angular rotation rate $\dot{\theta}$ of the position vector relative to inertial space, using an implied polar coordinate system in the orbit plane with center at body P_1, and note the familiar kinematic relationship $v_h = r\dot{\theta}$, we find

$$H = r^2 \dot{\theta} \tag{3.19}$$

Next consider the orbit energy, E, and what constraints it places on the motion. For natural systems there is a strong distinction between positive and negative energy solutions, with fundamentally different behavior resulting between the two cases. Consider a system with total negative energy, $E < 0$. Then from the energy equation it is simple to derive an upper bound on the distance between the two bodies. For the two-body problem our specific argument rests on the fact that $\lim_{r\to\infty} U(\boldsymbol{r}) = 0$, $\partial U/\partial r < 0$ for all r, and $\lim_{r\to 0} U(\boldsymbol{r}) \to \infty$. Then, there exists a radius R_{max} such that $U(R) + E \leq 0$ for $R \geq R_{max}$. But $U(r) + E = \frac{1}{2}v^2 \geq 0$. Thus, $U(r) \geq U(R)$ which leads to $r \leq R_{max}$, an upper bound on the distance between the two bodies. When the system reaches this upper bound, the kinetic energy of the system is zero and the energy is completely contained in the potential energy of the system. For the case of positive energy there are no upper bounds on the motion, and for the two-body problem all solutions will eventually travel to arbitrarily large separations between the bodies.

Combining the angular momentum magnitude and the energy integrals we can establish a lower bound on the energy for a given angular momentum. Specifically we find that for a given angular momentum H, the energy must be greater than or equal to $E_{min} = -\frac{\mu^2}{2H^2}$. To show this start with the supposition

$$E = \frac{1}{2}v^2 - \frac{\mu}{r} \geq -\frac{\mu^2}{2H^2} \tag{3.20}$$

Removing the speed v from the above inequality with the substitution $v = H/(r\cos\gamma)$, we can find the quadratic equation

$$r^2 - \frac{2H^2}{\mu}r + \frac{H^4}{\mu^2\cos^2\gamma} \geq 0 \tag{3.21}$$

and its factorization

$$r = \frac{H^2}{\mu}\left[1 \pm \sqrt{1 - \frac{1}{\cos^2\gamma}}\right] \tag{3.22}$$

Note that the radicand is less than or equal to 0, and thus that a physical value of radius r can never equal its root except when $\gamma = 0$ and $r = H^2/\mu$. At this value of γ and r, however, the energy is identically equal to E_{min}. For any other value of γ the energy is greater than E_{min}, establishing the result.

Continuing with the angular momentum and energy we can also find a lower bound on the radius. First, we note that $H > 0$ and that a minimum separation distance between the bodies exists. Then, at a minimum distance we must have $d(\boldsymbol{r}\cdot\boldsymbol{r})/dt = 0$, or $\boldsymbol{r}\cdot\boldsymbol{v} = 0$, and thus the flight path angle $\gamma = 0$ at a minimum of the separation radius leading to $H = r_p v_p$. Eliminating v_p from the energy and angular momentum then provides the result:

$$E = \frac{H^2}{2r_p^2} - \frac{\mu}{r_p} \tag{3.23}$$

which can be changed into the quadratic equation and its solution

$$r_p^2 + \frac{\mu}{E}r_p - \frac{H^2}{2E} = 0 \tag{3.24}$$

$$r_p = \frac{-\mu}{2E}\left[1 \pm \sqrt{1 + \frac{2EH^2}{\mu^2}}\right] \tag{3.25}$$

For a physical orbit the radicand must always be greater or equal to zero, given the lower limit on energy. Thus the solution always has at least one physical (i.e., real) root. If $E > 0$, then the radicand is always real and greater than 1, and the minus sign has a positive separation between the two bodies. If $E < 0$ we note that $E \geq E_{min}$ and the radicand is positive and less than 1. Thus in this case there are two real solutions; inspection indicates that one is a minimum and one a maximum. Thus the minimum radius of the system can be denoted as:

$$r \geq r_p \tag{3.26}$$

$$r_p = \frac{\mu}{2E}\left[\sqrt{1 + \frac{2EH^2}{\mu^2}} - 1\right] \tag{3.27}$$

Conversely, whether there is a maximum radius depends on the energy of the system, leading to the result $r \leq r_a$ with

$$r_a = \begin{cases} \frac{-\mu}{2E}\left[1 + \sqrt{1 + \frac{2EH^2}{\mu^2}}\right] & E < 0 \\ \infty & E \geq 0 \end{cases} \tag{3.28}$$

3.1.3 Additional Integrals and the Orbit Trajectory

While strong constraints can be placed on the planar motion for the two-body motion, the solution is not completely described as of yet. The general n-body problem has the same integrals of motion discussed to this point, translational momentum, angular momentum, and energy, resulting in 10 separate integrals of motion that can be used to reduce the dimension of the system and place constraints on its motion. In the two-body problem we note that the system can be reduced to a one-degree-of-freedom system by use of the Clairaut solution substitution [6]. We will proceed in a different direction. First note the existence of an additional integral of motion which is only defined for the two-body problem, the Laplace vector or equivalently the eccentricity vector. Second note Jacobi's final integral theorem, which states that any system that has been reduced to a one-dimensional dynamical system can always be solved by quadratures.

The Laplace vector can be formed by the combination:

$$\boldsymbol{b} = \boldsymbol{v} \times \boldsymbol{H} - \mu\hat{\boldsymbol{r}} \tag{3.29}$$

while the eccentricity vector is just:

$$\boldsymbol{e} = \frac{\boldsymbol{b}}{\mu} \tag{3.30}$$

$$= \frac{1}{\mu}\boldsymbol{v} \times \boldsymbol{H} - \hat{\boldsymbol{r}} \tag{3.31}$$

It can be shown that $\dot{\boldsymbol{e}} = 0$ by direct substitution of the previous integrals of motion and the equations of motion. Specifically,

$$\dot{\boldsymbol{e}} = -\frac{1}{r^3}\boldsymbol{r} \times \boldsymbol{H} - \frac{1}{r}\left[\boldsymbol{U} - \hat{\boldsymbol{r}}\hat{\boldsymbol{r}}\right] \cdot \boldsymbol{v} \tag{3.32}$$

Note that $\boldsymbol{r} \times \boldsymbol{H} = \tilde{\boldsymbol{r}} \cdot \tilde{\boldsymbol{r}} \cdot \boldsymbol{v} = r^2 \left[\hat{\boldsymbol{r}}\hat{\boldsymbol{r}} - \boldsymbol{U}\right] \cdot \boldsymbol{v}$, showing that these terms are equal and opposite, leading to $\dot{\boldsymbol{e}} = 0$.

The eccentricity vector only has one additional item of information apart from the classical integrals. As a vector, it lies in the orbit plane of motion and its magnitude equals $e^2 = 1 + 2H^2E/\mu^2$, neither of which provides new information. The additional information that $\boldsymbol{e}$ provides is related to its orientation in the plane of motion, which provides an inertially fixed reference point for the relative motion between the bodies. Denote the angle between $\boldsymbol{e}$ and the node vector defined by the unitized vector $\hat{\boldsymbol{z}} \times \boldsymbol{H}$ as ω, this is called the argument of periapsis and is another orbital element.

To solve for the trajectory of the bodies, take the dot product of the constant vector $\boldsymbol{e}$ with the radius vector $\boldsymbol{r}$ to find:

$$\boldsymbol{r} \cdot \boldsymbol{e} = \frac{H^2}{\mu} - r \tag{3.33}$$

$$re\cos f = \frac{H^2}{\mu} - r \tag{3.34}$$

f - true anomaly

and where the angle between $\boldsymbol{r}$ and $\boldsymbol{e}$ is denoted as f and called the true anomaly. This equation can be explicitly solved for the distance between the two bodies to find:

$$r = \frac{H^2/\mu}{1 + e\cos f} \tag{3.35}$$

which gives us the relative distance between the two bodies as a function of the angle f measured from the $\boldsymbol{e}$ vector, which in turn is specified by the angle ω measured in the orbital plane from the node crossing (see Fig. 3.1 for a description of the geometry). This simple relationship captures Kepler's first law of planetary motion.

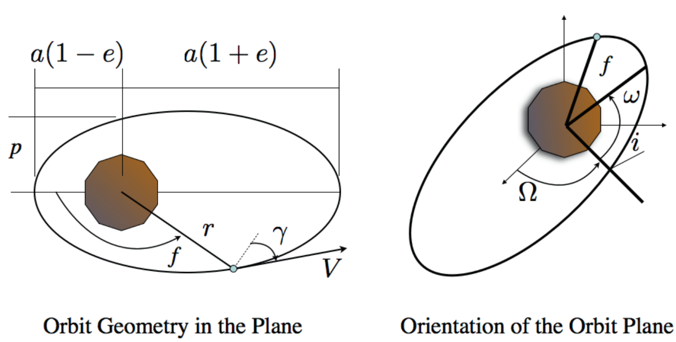

Fig. 3.1 Geometric definition of the orbit elements.

The eccentricity, e, equals $\sqrt{1 + 2H^2E/\mu^2}$. Due to the minimum energy value constraint $e \geq 0$, with $e = 0$ occurring at the minimum energy for a given angular momentum. Thus, the eccentricity is a measure of the excess energy that an orbit has. If $E \geq 0$ then $e \geq 1$, which yields a fundamental change in the radius formula as now the denominator can disappear and, apparently $r \to \infty$. Denote the value H^2/μ as the orbit parameter, p, a sole function of the angular momentum of the orbit with units of length. With this new notation the trajectory solution for the two-body problem takes on its standard form:

$$r = \frac{p}{1 + e\cos f} \tag{3.36}$$

Using this relation it is now possible to relate the speed and flight path angle as functions of the true anomaly f. From the energy equation the speed can be evaluated, and from the angular momentum the flight-path angle can be evaluated

γ- fpa

$$v = \sqrt{\frac{\mu}{p}\left(1 + 2e\cos f + e^2\right)} \tag{3.37}$$

$$\tan\gamma = \frac{e\sin f}{1 + e\cos f} \tag{3.38}$$

Again, these formula are valid for all energy and angular momentum. The only constraint which occurs is when $e \geq 1$, as then $1 + e\cos f = 0$ can occur at two particular values of true anomaly, $f = \pm f_\infty$ where $f_\infty = \arccos(-1/e)$, at which point the radius formula is undefined. Thus $f \in (-f_\infty, f_\infty)$. The additional quantities p, e and ω are all considered to be "classical" orbital elements in addition to the previously defined inclination i and longitude of the ascending node Ω. Note that another, alternate, orbit element is usually used known as the semi-major axis, a. This is a direct function of the energy only and can be computed as $a = -\frac{\mu}{2E}$, or alternately $a = p/(1 - e^2)$. Thus when $E < 0$, $a > 0$ and vice versa. When $E = 0$ the quantity a is undefined.

For an orbit with zero energy, a parabolic orbit, $f_\infty = \pi$. The radius, velocity and flight path angle all take on special forms, arising from trigonometric reductions of the formula:

$$r = \frac{p}{2\cos^2(f/2)} \tag{3.39}$$

$$v = 2\sqrt{\frac{\mu}{p}}\cos(f/2) \tag{3.40}$$

$$\tan\gamma = \tan(f/2) \tag{3.41}$$

In this case as the true anomaly approaches $\pm f_\infty$ the speed approaches zero and the flight path angle approaches $\pi/2$.

For an orbit with positive energy, a hyperbolic orbit, f_∞ lies in the interval $(\pi, \pi/2)$, with its value approaching π as $e \to 1$ and $\pi/2$ as $e \to \infty$. The latter situation occurs when $\mu \to 0$, which in fact mimics the situation when a spacecraft has a flyby of a small asteroid. While the form of the radius, speed and flight path angle formulae do not change for this case, their limiting values do. Now as true anomaly approaches $\pm f_\infty$ the following limits occur, $v \to v_\infty$ and $\gamma \to \gamma_\infty$, where

$$v_\infty = \sqrt{\frac{\mu(e^2-1)}{p}} \tag{3.42}$$

$$= \sqrt{\frac{\mu}{-a}} \tag{3.43}$$

$$\gamma_\infty = \frac{\pi}{2} \tag{3.44}$$

For $E > 0$ recall that $e > 1$ and $a < 0$, thus yielding positive quantities within all of the relevant square roots.

With the results found to this point it is possible to define the three-dimensional motion of the two-body solution. Using the derived directions of $\hat{\boldsymbol{H}}$ and $\hat{\boldsymbol{e}}$ the vector $\hat{\boldsymbol{e}}_\perp = \hat{\boldsymbol{H}} \times \hat{\boldsymbol{e}}$ can be defined, and using $\hat{\boldsymbol{r}}$ can also define $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{H}} \times \hat{\boldsymbol{r}}$. Then, by applying these definitions the full position and velocity vector describing the orbit is:

$$\boldsymbol{r} = r\left[\cos f \hat{\boldsymbol{e}} + \sin f \hat{\boldsymbol{e}}_\perp\right] \tag{3.45}$$

$$\boldsymbol{v} = \frac{H}{r}\left[\hat{\boldsymbol{\theta}} + \tan\gamma \hat{\boldsymbol{r}}\right] \tag{3.46}$$

In these expressions r, v, γ, $\hat{\boldsymbol{r}}$, and $\hat{\boldsymbol{\theta}}$ all vary with true anomaly, with the appropriate limits for the cases of $E \geq 0$.

3.1.4 Motion in Time

What has been defined is the path, or trajectory, that a solution to the two-body problem takes in position and velocity space, also called phase space. The link between the true anomaly and time has not been developed yet. Also, the reader can note that we have only defined 5 independent integrals to this point. These are the angular momentum $\boldsymbol{H}$, energy E, and eccentricity vector $\boldsymbol{e}$, which have 5 independent quantities that can be expressed in terms of the classical orbital elements. These are the orientation of the orbit plane, inclination i and longitude of ascending node Ω, orientation of the trajectory in the orbit plane, the argument of periapsis ω, and the shape and size of the orbit, the eccentricity e and either the semi-major axis a or the orbit parameter p.

What is lacking is the integral that locates where the solution is located within the trajectory at a given time. This final integral always exists due to a theorem by Jacobi. This theorem states that a one-dimensional ordinary differential equation $\dot{x} = f(x)$ can always be solved by quadratures as

$$t - \tau_o = \int_{x_o}^{x} \frac{dx'}{f(x')} \tag{3.47}$$

with the epoch time τ_o serving as the final integral of motion. This result can either hold globally or locally between zeros of $f(x)$. In the two-body problem it is easiest to find it by noting a simple relationship between the time rate of change of the true anomaly and the angular momentum magnitude.

Recall the angular momentum magnitude formula, $H = rv\cos\gamma$, and its alternate statement $H = r^2\dot{\theta}$, where $\dot{\theta}$ is the time rate of change of the angular location of the solution. For the two-body problem this will equal the time rate of change of the true anomaly, $\dot{f}$. The quantity $r^2\dot{\theta}$ equals twice the rate at which the radius vector sweeps out area, which is the classical statement of Kepler's Second Law. Solving for $\dot{f}$ and evaluating r yields

$$\dot{f} = \sqrt{\frac{\mu}{p^3}}(1 + e\cos f)^2 \tag{3.48}$$

This differential equation can be solved by separation of variables, putting it into the form

$$\sqrt{\frac{\mu}{p^3}}(t - \tau_o) = \int_{f_o}^{f} \frac{df'}{(1 + e\cos f')^2} \tag{3.49}$$

We have the freedom to choose one of the initial conditions τ_o or f_o, and following convention choose $f_o = 0$, at periapsis, with τ_o then signifying the time at which the orbit was at periapsis, which then becomes the final integral of motion.

First consider the case $E < 0$ and $0 < e < 1$. The quadrature can be carried out for this case, yielding

$$\begin{aligned}\sqrt{\frac{\mu}{p^3}}\,(t - \tau_o) &= \left.\frac{-e}{1 - e^2}\frac{\sin f'}{1 + e\cos f'}\right|_{f_o}^{f} \\ &\quad + \left.\frac{2}{(1 - e^2)^{3/2}}\arctan\left\{\sqrt{\frac{1 - e}{1 + e}}\tan\frac{1}{2}f'\right\}\right|_{f_o}^{f}\end{aligned} \tag{3.50}$$

Evaluating this integral from $f = 0$ to 2π defines the period of the orbit, T, which takes on a particularly simple form when stated in terms of the semi-major axis:

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}} \tag{3.51}$$

and is Kepler's Third Law. Thus the period is only a function of the orbit energy. From this the mean motion can be defined as $n = \sqrt{\mu/a^3} = 2\pi/T$, as this is by definition the average rate at which the orbit is traversed.

The expression in Eq. 3.50 is not convenient to work with, leading to the definition of the eccentric anomaly, E. The arc-tangent function will define an angle, leading to the natural definition

$$\tan E/2 = \sqrt{\frac{1 - e}{1 + e}}\tan f/2 \tag{3.52}$$

Since this is a half-angle formula it yields angles resolved over the full interval $[0, 2\pi)$. Corresponding formulae, and their inverses, for $\cos E$ and $\sin E$ in terms of $\cos f$ and $\sin f$ are stated in Appendix A. Using these definitions gives us the standard form of Eq. 3.50, which is usually called Kepler's equation:

$$n(t - \tau_o) = E - e \sin E \tag{3.53}$$

The mean anomaly is defined as $M = nt$, an additional angle. Then the final integral of motion can also be restated as $M_o = n\tau_o$. This gives the usual definition of Kepler's equation

$$M - M_o = E - e \sin E \tag{3.54}$$

Given a true anomaly the mean anomaly can be directly computed, first transforming into the eccentric anomaly and then using Kepler's equation. Evaluation of the true anomaly given a mean anomaly is not as straightforward, as Kepler's equation must be solved to find the eccentric anomaly that corresponds to a given mean anomaly. Despite the simplicity of this equation, it cannot be solved in closed form given its transcendental nature. Significant research and analysis of this equation has occurred historically, and indeed still occurs today. See [26] for a detailed history of this equation and methods for its inversion.

Equivalent quadratures and definitions can be carried out for the case $e = 1$ and $e > 1$, for parabolic and hyperbolic orbits. These relations are needed when modeling the flyby of a small body relative to a planet or of a spacecraft relative to a small body. As mentioned above, the other trajectory relations still hold, the only modification needed is to Kepler's equation. Going back to the fundamental quadrature,

$$\sqrt{\frac{\mu}{p^3}}(t - \tau_o) = \int_0^f \frac{df}{(1 + e\cos f)^2}$$

when $e > 1$ the result is

$$\sqrt{\frac{\mu}{p^3}}(t - \tau_o) = \frac{e}{e^2 - 1}\frac{\sin f'}{1 + e\cos f'}\bigg|_{f_o}^{f} - \frac{2}{(e^2 - 1)^{3/2}} \operatorname{arctanh}\left\{\sqrt{\frac{e-1}{e+1}}\tan\frac{1}{2}f'\right\}\bigg|_{f_o}^{f} \tag{3.55}$$

Following the same approach for the elliptic case, define the Hyperbolic Anomaly, F, as

$$\tanh F/2 = \sqrt{\frac{e-1}{e+1}}\tan f/2 \tag{3.56}$$

with similar relationships between $\cos f$ and $\sin f$ and $\cosh F$ and $\sinh F$. Applying these substitutions yields the hyperbolic version of Kepler's equation:

$$\sqrt{\frac{\mu}{|a|^3}}(t - \tau_o) = e \sinh F - F \tag{3.57}$$

For parabolic orbits, the quadrature takes on a simpler form and yields a direct relationship between time and true anomaly:

$$\sqrt{\frac{\mu}{p^3}}\,(t-\tau_o) = \frac{1}{2}\tan\left(\frac{f}{2}\right)\left[1+\frac{1}{3}\tan^2\left(\frac{f}{2}\right)\right] \tag{3.58}$$

where p is generally well defined (as it is computed from the angular momentum) even if $e = 1$.

Finally, note that for orbits with zero angular momentum, $p = 0$, the entire analysis must be redone given the singular definition of the true anomaly in this case. Such a derivation is not performed here, however, as it is not needed in the following. We only note that the eccentric anomaly remains well-defined and can be used to describe the geometry of motion in this case.

3.2 Rotational Dynamics of Small Bodies

For many applications it is of interest to have a compact, analytical description of rotational motion which closely matches the expected motion of a body rotating in space. For precision modeling and estimation the rotational dynamics will usually be modeled using numerical integration incorporating all relevant torques acting on the body [107]. However, the torques acting on a rotating body will often be small in magnitude, meaning that over reasonable lengths of time it may be feasible to use the Euler solution for the torque-free rotation of a rigid body to describe rigid body rotation. Such an analytical solution is of interest as it allows the body's orientation to be specified at arbitrary epochs without requiring numerical integration of Euler's equations from an initial epoch. In this section the classical solution for all possible cases of rotational motion is restated – generalizing the analysis given by MacMillan [94]. The solution is not derived in as much detail as for the two-body problem, as the remainder of the text only deals with orbital motion, and introducing a detailed discussion of the solution of the rotational equations of motion would take us a bit far afield. Closed-form solutions for the orientation of the rigid body are also provided in the form of Euler angles as a function of time. The solutions are stated in terms of elliptic integrals and functions, which are usually available from standard mathematical libraries. The following derivation avoids the use of theta functions to express the Euler angles of the body (which the classical solution relies on, [195]). The qualitative specification of a small body's rigid body rotation state is shown to be given by only two numbers once the moments of inertia are defined for the body.

3.2.1 The Inertia Dyadic

First define the mass moments and products of inertia of a general body. These arise from the fundamental definition of the rotational angular momentum of a body. Consider a collection of mass elements $dm(\boldsymbol{\rho})$ located by position vectors $\boldsymbol{\rho}$

and rigidly connected to each other. Assume these elements are all rotating about the center of mass of this rigid body collection with an instantaneous rotation rate $\boldsymbol{\omega}$. Then the velocity of each mass element equals $\boldsymbol{\omega} \times \boldsymbol{\rho} = \tilde{\boldsymbol{\omega}} \cdot \boldsymbol{\rho}$ and enables us to define the rotational angular momentum of the body as an integral over the mass distribution denoted by β

$$\boldsymbol{H} = \int_\beta \boldsymbol{\rho} \times \boldsymbol{v}\, dm \tag{3.59}$$

$$= -\int_\beta \tilde{\boldsymbol{\rho}} \cdot \tilde{\boldsymbol{\rho}}\, dm \cdot \boldsymbol{\omega} \tag{3.60}$$

where the angular velocity vector can be pulled out of the integral as it is constant throughout the body. The integral in front of the angular velocity is then defined as the inertia dyadic, or if resolved into a specific basis the inertia matrix where the diagonal elements are the moments of inertia and the off-diagonal elements are the products of inertia. The inertia dyadic $\boldsymbol{\mathcal{I}}$ can then be defined as

$$\boldsymbol{\mathcal{I}} = \int_\beta \left[\rho^2 \boldsymbol{U} - \boldsymbol{\rho}\boldsymbol{\rho}\right] dm(\boldsymbol{\rho}) \tag{3.61}$$

where $\boldsymbol{U}$ is the unity dyadic and the second term is a dyad. Using this notation the angular momentum vector of the body is then $\boldsymbol{H} = \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega}$. A frame for the inertia dyadic has not been defined as of yet, but it is convenient to fix it in the body-frame, as then the integral quantities are all constant.

3.2.2 Orientation of the Rigid Body

A second crucial definition is the orientation of the rigid body relative to inertial space. The rotation dyadic (or equivalently matrix), denoted as $\boldsymbol{C}$, represents a linear operator that takes a vector from a body-fixed frame and expresses it in inertial space. The specification of the rotation dyadic as a function of time is the ultimate goal for describing the attitude motion of a body. To carry this out the relevant differential equations must be posed and integrated.

Consider the operation of $\boldsymbol{C}$ taking a vector specified in a body-fixed frame, $\boldsymbol{\rho}_B$, into a vector specified in an inertial frame, $\boldsymbol{\rho}_I = \boldsymbol{C} \cdot \boldsymbol{\rho}_B$. Taking the time derivative of this expression and recalling the transport theorem on time derivatives in a rotating frame, the equivalence $\boldsymbol{C} \cdot [\boldsymbol{\omega} \times \boldsymbol{\rho}_B + \dot{\boldsymbol{\rho}}_B] = \dot{\boldsymbol{C}} \cdot \boldsymbol{\rho}_B + \boldsymbol{C} \cdot \dot{\boldsymbol{\rho}}_B$ can be established. As this relation must hold for arbitrary vector $\boldsymbol{\rho}_B$, the relevant differential equation can be extracted

$$\dot{\boldsymbol{C}} = \boldsymbol{C} \cdot \tilde{\boldsymbol{\omega}} \tag{3.62}$$

3.2.3 Euler's Equations

Now consider the derivation of Euler's equations for the rigid body. It can be assumed that there are no external torques operating on the body, and thus the total rotational angular momentum must be conserved in inertial space. This conserved vector is specified, using the above notations and definitions, as

$$\boldsymbol{H} = \boldsymbol{C} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega} \tag{3.63}$$

Noting that $\dot{\boldsymbol{H}} = 0$ yields

$$0 = \dot{\boldsymbol{C}} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega} + \boldsymbol{C} \cdot \dot{\boldsymbol{\mathcal{I}}} \cdot \boldsymbol{\omega} + \boldsymbol{C} \cdot \boldsymbol{\mathcal{I}} \cdot \dot{\boldsymbol{\omega}} \tag{3.64}$$

but $\dot{\boldsymbol{\mathcal{I}}} = 0$ in a body-fixed frame, and substituting the differential equation for $\boldsymbol{C}$ yields $\boldsymbol{C} \cdot [\boldsymbol{\mathcal{I}} \cdot \dot{\boldsymbol{\omega}} + \tilde{\boldsymbol{\omega}} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega}] = 0$. This must hold for any rotation dyadic and thus defines the general statement of Euler's equations in the body-fixed frame

$$\boldsymbol{\mathcal{I}} \cdot \dot{\boldsymbol{\omega}} + \tilde{\boldsymbol{\omega}} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega} = 0 \tag{3.65}$$

3.2.4 Conserved Quantities

The fundamental conserved quantities that are defined for the torque-free rotation of a body are its rotational angular momentum and kinetic energy. The rotational kinetic energy is defined as

$$T = \frac{1}{2} \boldsymbol{\omega} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega} \tag{3.66}$$

Taking its time derivative yields $\dot{T} = \boldsymbol{\omega} \cdot \boldsymbol{\mathcal{I}} \cdot \dot{\boldsymbol{\omega}}$, and substituting Euler's equation from above immediately verifies that $\dot{T} \equiv 0$.

It is more instructive, for the moment, to consider the magnitude of the angular momentum vector, or $H^2 = \boldsymbol{H} \cdot \boldsymbol{H}$, which is naturally conserved, even in the body-fixed frame. Assume that the inertia dyadic is diagonal and is aligned along the principal moments of inertia with axes $\boldsymbol{x}$, $\boldsymbol{y}$ and $\boldsymbol{z}$ following the convention $I_x \leq I_y \leq I_z$. Then when written in expanded form both the kinetic energy and the angular momentum each constrain the angular velocity vector (in the body-fixed frame) to lie on an ellipsoid

$$H^2 = I_x^2 \omega_x^2 + I_y^2 \omega_y^2 + I_z^2 \omega_z^2 \tag{3.67}$$

$$2T = I_x \omega_x^2 + I_y \omega_y^2 + I_z \omega_z^2 \tag{3.68}$$

For these two ellipsoids to be consistent with each other, the following condition must hold

$$\boldsymbol{\omega} \cdot \boldsymbol{\mathcal{I}} \cdot \left[\boldsymbol{U} - \frac{H^2}{2T} \boldsymbol{\mathcal{I}}^{-1} \right] \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega} = 0 \tag{3.69}$$

where the inertia dyadic always has an inverse for finite density distributions. To simplify the discussion, define the ratio $H^2/(2T)$ to be the "dynamic inertia" I_D, which turns out to be a fundamental quantity. Then for the equality to hold the dyadic $\boldsymbol{U} - I_D \boldsymbol{\mathcal{I}}^{-1}$ must be negative semi-definite. Since the inertia dyadic is diagonal, this reduces to the condition that the determinant be ≤ 0, which can be modified to

$$(I_x - I_D)(I_y - I_D)(I_z - I_D) \leq 0 \tag{3.70}$$

The condition is identically equal to zero if I_D equals any of the principal moments of inertia. Also if $I_D < I_x$ or $I_D > I_z$ then the inequality is violated. Thus the dynamic inertia must satisfy (with the usual order of moments of inertia)

$$I_x \leq I_D \leq I_z \tag{3.71}$$

From this fundamental inequality a restriction on the possible values of kinetic energy for a given angular momentum can be inferred.

$$\frac{1}{2}\frac{H^2}{I_x} \geq T \geq \frac{1}{2}\frac{H^2}{I_z} \tag{3.72}$$

The lower bound on T corresponds to the body rotating about its maximum moment of inertia, which is the minimum energy state and is the stable final rotational state in the presence of energy dissipation. Conversely, the upper bound corresponds to rotation about the minimum moment of inertia, which is the maximum energy state for a given angular momentum and which is unstable in the presence of energy dissipation.

To finish, define the effective spin rate of the body

$$\omega_l = \frac{2T}{\sqrt{H^2}} \tag{3.73}$$

The effective spin rate can, in combination with the dynamic inertia, be used as an alternate statement of energy and angular momentum

$$H = \sqrt{\boldsymbol{\omega} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega}} = I_D \omega_l \tag{3.74}$$

$$T = \frac{1}{2}\boldsymbol{\omega} \cdot \boldsymbol{\mathcal{I}} \cdot \boldsymbol{\omega} = \frac{1}{2} I_D \omega_l^2 \tag{3.75}$$

Both of these parameters figure prominently in our analytic solution and represent the rotation rate and moment of inertia of a "sphere" with the given angular momentum and kinetic energy.

3.2.5 Problem Statement and Parameter Definitions

There are five special solution cases to be considered depending on the value of I_D relative to the principal moments of inertia.

1. If $I_D = I_z$ then the small body is in principal axis rotation about the z-axis, the largest moment of inertia, and is in its minimum energy state for a given angular momentum.
2. If $I_y < I_D < I_z$ then the small body has a non-zero nutation and precesses about the z-axis, commonly termed a short-axis rotation mode (SAM), as the maximum moment of inertia axis $\boldsymbol{z}$ is generally the shortest axis of a constant density body.
3. If $I_y = I_D$ then the small body is in principal axis rotation about the y-axis, or its angular velocity moves along a heteroclinic connection between rotations about the y-axis in opposite directions.
4. If $I_x < I_D < I_y$ then the small body has a non-zero nutation and precesses about the x-axis, commonly termed a long-axis rotation mode (LAM), as the minimum moment of inertia axis $\boldsymbol{x}$ is generally the longest axis of a constant density body.
5. If $I_x = I_D$ then the small body in in principal axis rotation about the x-axis, which is the smallest moment of inertia, and is in its maximum energy state for a given angular momentum.

The solution for cases 1 and 5 are trivial and just give uniform rotation about a principal axis of the body, that axis being fixed in inertial space. For cases 2 and 4 the motion becomes more complex and explicit solutions are needed and given here. For case 3 the uniform rotation case is also trivial, although the case for motion on the heteroclinic orbit has added complication. This solution is a special case of 2 or 4 and is discussed for completeness.

3.2.6 Angular Velocity in the Body-Fixed Frame

Solutions for Short-Axis Mode (SAM) and Long-Axis Mode (LAM)

If the dynamic inertia lies in the interval $I_y < I_D < I_z$ the body is in a short-axis rotation mode and the angular velocity vector will circulate about the z-axis of the body. This is thought to be the usual case for Solar System objects, as any body dissipating energy eventually enters this rotation mode on its way to uniform rotation about its maximum moment of inertia. If instead the dynamic inertia lies in the interval $I_x < I_D < I_y$ the angular velocity vector will circulate about the x-axis. Under energy dissipation this mode will asymptotically transition to a SAM mode of rotation and finally to uniform rotation about the z-axis [20]. Despite this, it is important to note that at least some asteroids reside in a LAM rotation state ([76], [63]).

The angular velocity vector of the body can be expressed in the principal axis, body-fixed frame in terms of elliptic functions:

$$\omega_x = \omega_l \sqrt{\frac{I_D\,(I_z - I_D)}{I_x\,(I_z - I_x)}}\sqrt{1 - \sigma^2\,\mathrm{sn}(\tau)^2} \tag{3.76}$$

$$\omega_y = \omega_l \sigma \sqrt{\frac{I_D\,(I_z - I_D)}{I_y\,(I_z - I_y)}}\,\mathrm{sn}(\tau) \tag{3.77}$$

$$\omega_z = \omega_l \sqrt{\frac{I_D\,(I_D - I_x)}{I_z\,(I_z - I_x)}}\sqrt{1 - \gamma^2\,\mathrm{sn}(\tau)^2} \tag{3.78}$$

where

$$\sigma = \begin{cases} 1 & I_y < I_D < I_z \\ k & I_x < I_D < I_y \end{cases} \tag{3.79}$$

$$\gamma = \begin{cases} k & I_y < I_D < I_z \\ 1 & I_x < I_D < I_y \end{cases} \tag{3.80}$$

$$k^2 = \begin{cases} \frac{(I_y - I_x)(I_z - I_D)}{(I_D - I_x)(I_z - I_y)} & I_y < I_D < I_z \\ \\ \frac{(I_D - I_x)(I_z - I_y)}{(I_y - I_x)(I_z - I_D)} & I_x < I_D < I_y \end{cases} \tag{3.81}$$

The parameter k is the usual "parameter" used in the computation and definition of the elliptic sine function sn. See Appendix F for a brief discussion and definition of elliptic functions and integrals. Note that $\mathrm{sn}(\tau)$ is periodic in τ with a period of $4K(k)$, where $K(k)$ is the complete elliptic integral of the first kind and k is its parameter. It should be noted that $k_{LAM} = 1/k_{SAM}$ and that for the general case $k < 1$.

The relation of the time parameter τ to the "real time" t is:

$$\tau - \tau_o = \sqrt{\frac{I_D\,(I_z - I_y)\,(I_D - I_x)}{I_x I_y I_z}}\,\frac{\omega_l\,(t - t_o)}{\sigma} \tag{3.82}$$

where t_o is the initial epoch. The period of the vector $\boldsymbol{\omega}$ in the body-fixed frame is thus

$$P = \sqrt{\frac{I_x I_y I_z}{I_D\,(I_z - I_y)\,(I_D - I_x)}}\,\frac{4\sigma}{\omega_l} K(k) \tag{3.83}$$

Motion along the Heteroclinic Connection ($I_D = I_y$)

For the special case of $I_D = I_y$ either formulation can be used to arrive at a solution. First note that $k = 1$ for this special case and that $\mathrm{sn}(\tau, k = 1) = \tanh(\tau)$, where tanh is the hyperbolic tangent function. The explicit solution for the angular rates become:

$$\omega_x = \omega_l \sqrt{\frac{I_y\,(I_z - I_y)}{I_x\,(I_z - I_x)}} \frac{1}{\cosh(\tau)} \tag{3.84}$$

$$\omega_y = \omega_l \tanh(\tau) \tag{3.85}$$

$$\omega_z = \omega_l \sqrt{\frac{I_y\,(I_y - I_x)}{I_z\,(I_z - I_x)}} \frac{1}{\cosh(\tau)} \tag{3.86}$$

$$\tau - \tau_o = \sqrt{\frac{(I_z - I_y)\,(I_y - I_x)}{I_x I_z}} \omega_l (t - t_o) \tag{3.87}$$

where cosh is the hyperbolic cosine function. The period of motion becomes infinite, of course. As $\tau \to \pm\infty$ the functions $\tanh(\tau) \to \pm 1$ and $1/\cosh(\tau) \to 0$ so that the motion reduces to a uniform rotation about the y-axis with rotation rate ω_l in either direction.

Principal Axis Rotations

Finally, note that substitution of $I_D = I_x$ or $I_D = I_z$ makes the corresponding k parameters equal to zero. Further, $\mathrm{sn}(\tau, k = 0) = \sin(\tau)$ and that $K(0) = \pi/2$. Thus when $I_D = I_z$ then $\omega_x = \omega_y = 0$ and $\omega_z = \omega_l$. It is interesting to note that the rotation period in the body frame approaches $\sqrt{\left\{[I_x I_y]/[(I_z - I_y)(I_z - I_x)]\right\}}\, 2\pi/\omega_l$, which is different than the inertial period of $2\pi/\omega_l$.

Analogously, when $I_D = I_x$, $\omega_y = \omega_z = 0$ and $\omega_x = \omega_l$. The rotation period in the body frame is again different than the inertial period.

Symmetric Moment of Inertia Rotations

If any two of the moments of inertia of the rotating body are equal a significant simplification occurs in the rotational dynamics. These cases are widely discussed in textbooks, and hence we will only state the simplified form of their rotational dynamics for this case. Two different cases can be distinguished, where the minimum moments of inertia are equal to each other, which corresponds to an oblate body, and where the two maximum moments of inertia are equal to each other, which is a prolate body.

Oblate Bodies Setting $I_x = I_y = I_m$ in the solution has a number of consequences. First, since $I_m \leq I_D \leq I_z$ the body is always in the SAM rotation mode. Note that $k = 0$, $\gamma = 0$ and $\sigma = 1$. Thus, the elliptic sine function degenerates to a circular sine function. Substituting these changes gives us the new simplified equations for rotation in the body-fixed frame

$$\omega_x = \omega_l \sqrt{\frac{I_D\left(I_z - I_D\right)}{I_m\left(I_z - I_m\right)}} \cos(\tau) \tag{3.88}$$

$$\omega_y = \omega_l \sigma \sqrt{\frac{I_D\left(I_z - I_D\right)}{I_m\left(I_z - I_m\right)}} \sin(\tau) \tag{3.89}$$

$$\omega_z = \omega_l \sqrt{\frac{I_D\left(I_D - I_m\right)}{I_z\left(I_z - I_m\right)}} \tag{3.90}$$

The spin rate is constant about the maximum moment of inertia and varies sinusoidally between the two minimum axes. It is important to note that the angular velocity is still moving in the body-fixed frame and remains periodic. The relationship between τ and time becomes

$$\tau - \tau_o = \sqrt{\frac{I_D\left(I_z - I_m\right)\left(I_D - I_m\right)}{I_z}} \frac{\omega_l\left(t - t_o\right)}{I_m} \tag{3.91}$$

The period of the angular velocity vector $\boldsymbol{\omega}$ in the body-fixed frame is now

$$P = I_m \sqrt{\frac{I_z}{I_D\left(I_z - I_m\right)\left(I_D - I_m\right)}} \frac{2\pi}{\omega_l} \tag{3.92}$$

Prolate Bodies Setting $I_y = I_z = I_M$ in the solution yields different changes. Now $I_x \leq I_D \leq I_M$ and the body is always in a LAM rotation mode. The parameter $k = 0$ again, but now $\gamma = 1$ and $\sigma = 0$. For the prolate case, care must be made in setting these moments of inertia equal to each other, as there are cancellations of zeros that occur. Substituting these into the angular velocity equations now yields

$$\omega_x = \omega_l \sqrt{\frac{I_D\left(I_M - I_D\right)}{I_x\left(I_M - I_x\right)}} \tag{3.93}$$

$$\omega_y = \omega_l \sqrt{\frac{I_D\left(I_D - I_x\right)}{I_M\left(I_M - I_x\right)}} \sin(\tau) \tag{3.94}$$

$$\omega_z = \omega_l \sqrt{\frac{I_D\left(I_D - I_x\right)}{I_M\left(I_M - I_x\right)}} \cos(\tau) \tag{3.95}$$

Similar dynamics occur, except now the minimum moment of inertia has a constant spin rate and the circulation occurs across the two maximum moments. The relation of the time parameter τ to t is

$$\tau - \tau_o = \sqrt{\frac{I_D\left(I_M - I_x\right)\left(I_M - I_D\right)}{I_x}} \frac{\omega_l\left(t - t_o\right)}{I_M} \tag{3.96}$$

The period of the vector $\boldsymbol{\omega}$ in the body-fixed frame is thus

$$P = I_M \sqrt{\frac{I_x}{I_D\left(I_M - I_x\right)\left(I_M - I_D\right)}} \frac{2\pi}{\omega_l} \tag{3.97}$$

3.2.7 Analytic Solution for Type-II Euler Angles

To complete the analytical specification of torque-free rigid body solution an expression must also be found for the rotation matrix. We choose to express this using Euler angles of the body as a function of time, which can be solved for completely in terms of elliptic functions and integrals. The specific realization of these angles as a function of time will depend on the set of Euler angles specified, in the following the standard Type-II Euler angles are used [55]. Starting from an inertial frame X–Y–Z the first angle is a ccw rotation about the Z-axis by an angle ϕ (the right ascension), followed by a ccw rotation about the X'-axis by an angle θ (the declination), followed by a ccw rotation about the Z''-axis by an angle ψ (the hour angle). The range of the angles are $\phi \in [0, 2\pi]$, $\theta \in [0, \pi]$, and $\psi \in [0, 2\pi)$. The rotation matrix from the inertial frame to the body-fixed frame, $\boldsymbol{C}^T$ in our notation, is specified as:

$$\boldsymbol{C}^T = \begin{bmatrix} \begin{array}{c}\cos\phi\cos\psi \\ -\sin\phi\cos\theta\sin\psi\end{array} & \begin{array}{c}\sin\phi\cos\psi \\ +\cos\phi\cos\theta\sin\psi\end{array} & \sin\theta\sin\psi \\ \\ \begin{array}{c}-\cos\phi\sin\psi \\ -\sin\phi\cos\theta\cos\psi\end{array} & \begin{array}{c}-\sin\phi\sin\psi \\ +\cos\phi\cos\theta\cos\psi\end{array} & \sin\theta\cos\psi \\ \\ \sin\phi\sin\theta & -\cos\phi\sin\theta & \cos\theta \end{bmatrix} \tag{3.98}$$

The rotation matrix from the body-fixed frame to the inertial frame is the transpose.

Solution for Euler Angles $\boldsymbol{\theta}$ and $\boldsymbol{\psi}$

First note that the hour angle and declination (Euler angles ψ and θ) can be solved using only algebraic manipulations. To see this choose the inertial coordinate frame orientation such that the inertial z-axis is aligned with the angular momentum vector $\boldsymbol{H}$ and note that $|\boldsymbol{H}| = I_D\omega_l$. Then, the angular momentum vector in the body-fixed frame is $\boldsymbol{H} = I_D\omega_l \left[\sin\theta\sin\psi; \sin\theta\cos\psi; \cos\theta\right]^T$, which must be equal to the vector $\boldsymbol{I}\cdot\boldsymbol{\omega} = \left[I_x\omega_x; I_y\omega_y; I_z\omega_z\right]^T$. Equate the two vectors to find the solutions:

$$\tan\psi = \frac{I_x\omega_x}{I_y\omega_y} \tag{3.99}$$

$$\cos\theta = \frac{I_z\omega_z}{I_D\omega_l} \tag{3.100}$$

The specific form of these relations will depend on the rotational mode the asteroid is in.

Evaluating the ratios for the case of SAM and LAM rotation yields:

$$\tan\psi = \frac{1}{\sigma}\sqrt{\frac{I_x\left(I_z-I_y\right)}{I_y\left(I_z-I_x\right)}}\frac{\sqrt{1-\sigma^2\mathrm{sn}^2(\tau)}}{\mathrm{sn}(\tau)} \tag{3.101}$$

$$\cos\theta = \sqrt{\frac{I_z\left(I_D-I_x\right)}{I_D\left(I_z-I_x\right)}}\sqrt{1-\gamma^2\mathrm{sn}^2(\tau)} \tag{3.102}$$

For a SAM rotation the angle ψ will circulate through all possible values as τ increases monotonically. For a LAM rotation this angle will instead librate about $\pm\pi/2$, never crossing the angles 0 or π. For the angle θ note that the quantity $\sqrt{\left\{[I_z(I_D-I_x)]/[I_D(I_z-I_x)]\right\}} \leq 1$ for $I_D \leq I_z$. Thus the relationship is always defined in terms of real arguments. For the case of SAM rotation the angle θ is constrained to lie in the interval $(0,\pi/2)$ while for the case of a LAM rotation the angle will instead lie in the interval $(0,\pi)$ and librate about $\pi/2$.

Solution for Euler Angle ϕ

The expression of the right ascension (Euler angle ϕ) is not as simple. Classical discussions usually give the closed form for this final angle in terms of ratios of theta functions [195], which are generally not available for numerical evaluation in standard packages. This final angle can be found in terms of an elliptic integral of the third kind, which has a much simpler form than the classical solutions and can be easily implemented on a computer.

First note the equation for $\dot{\phi}$ [55]:

$$\dot{\phi} = \frac{1}{\sin\theta}\left[\sin\psi\omega_x + \cos\psi\omega_y\right] \tag{3.103}$$

From Eqs. 3.99 and 3.100

$$I_D\omega_l\sin\theta\sin\psi = I_x\omega_x \tag{3.104}$$

$$I_D\omega_l\sin\theta\cos\psi = I_y\omega_y \tag{3.105}$$

Multiply Eqs. 3.104 and 3.105 by ω_x and ω_y respectively and add, and square Eqs. 3.104 and 3.105 and add. Then substitute the resultant quantities into Eq. 3.103 to find:

$$\dot{\phi} = I_D\omega_l\frac{I_x\omega_x^2 + I_y\omega_y^2}{I_x^2\omega_x^2 + I_y^2\omega_y^2} \tag{3.106}$$

Next, use relations $I_D^2\omega_l^2 = \omega_x^2 I_x^2 + \omega_y^2 I_y^2 + \omega_z^2 I_z^2$ and $I_D\omega_l^2 = \omega_x^2 I_x + \omega_y^2 I_y + \omega_z^2 I_z$ to simplify Eq. 3.106 further to find:

$$\dot{\phi} = \frac{I_D\omega_l}{I_z}\left[1 + \frac{I_D\left(I_z - I_D\right)\omega_l^2}{I_D^2\omega_l^2 - I_z^2\omega_z^2}\right] \tag{3.107}$$

where ω_z is the only function of time in the expression. Making the substitution for ω_z from Eq. 3.78 and carrying out a series of algebraic manipulations puts Eq. 3.107 into the form:

$$\dot{\phi} = \frac{I_D\omega_l}{I_z}\left[1 + \frac{I_z - I_x}{I_x}\frac{1}{1 + n\,\mathrm{sn}^2}\right] \tag{3.108}$$

$$n = \sigma^2\left(\frac{I_z}{I_x}\right)\left(\frac{I_y - I_x}{I_z - I_y}\right) \tag{3.109}$$

Equation 3.108 consists of a linear increase in time plus a quadrature involving an elliptic function. The quadrature can be stated as:

$$\begin{aligned}\phi - \phi_o = &\frac{I_D\omega_l}{I_z}(t - t_o)\\ &+\sigma\sqrt{\frac{I_D I_y}{I_z I_x}}\frac{\left(I_z - I_x\right)}{\sqrt{\left(I_z - I_y\right)\left(I_D - I_x\right)}}\int_{\tau_o}^{\tau}\frac{d\tau}{1 + n\,\mathrm{sn}^2(\tau)}\end{aligned} \tag{3.110}$$

The integral $\mathit{\Pi}(\tau, n) = \int_0^\tau \frac{d\tau}{1+n\,\mathrm{sn}^2(\tau)}$ is the elliptic integral of the third kind, for which robust evaluation routines exist [136]. A few brief properties should be noted here:

$$\mathit{\Pi}(\tau + 2K(k), n) = \mathit{\Pi}(\tau, n) + 2\mathit{\Pi}(K(k), n) \tag{3.111}$$

$$\mathit{\Pi}(\tau + K(k), n) = 2\mathit{\Pi}(K(k), n) - \mathit{\Pi}(K(k) - \tau, n) \tag{3.112}$$

Thus, to have the function $\mathit{\Pi}(\tau, n)$ defined unambiguously and to accumulate the appropriate increase in time, we use the following definition:

$$\begin{aligned}\bar{\mathit{\Pi}}(\tau, n, k) = &\,2\mathit{\Pi}\left(K(k), n\right)\mathrm{int}\left[\tau/2K(k)\right]\\ &+\mathit{\Pi}\left(\tau - 2K(k)\,\mathrm{int}\left[\tau/2K(k)\right], n\right)\end{aligned} \tag{3.113}$$

The function int$[x]$ denotes the truncation of decimal points of the number x, defined for positive or negative values.

Thus, the right ascension is described as:

$$\begin{aligned}\phi = &\,\phi_o + \frac{I_D\omega_l}{I_z}(t - t_o) + \\ &\sigma\sqrt{\frac{I_D I_y}{I_z I_x}}\frac{\left(I_z - I_x\right)}{\sqrt{\left(I_z - I_y\right)\left(I_D - I_x\right)}}\left[\bar{\mathit{\Pi}}(\tau, n, k) - \bar{\mathit{\Pi}}(\tau_o, n, k)\right]\end{aligned} \tag{3.114}$$

Note that ϕ consists of two main terms, a linear drift in time described by the angular momentum and the I_z moment of inertia, and a more complex increasing term involving an elliptic integral. The linear drift in ϕ is significant as it is the rotation rate of an arbitrary body whose kinetic energy has dissipated to a minimum value while keeping its angular momentum constant [20].

It is of interest to compute the average rate of change of the right ascension. This can be computed by differencing the value of ϕ at τ_o and $\tau = \tau_o + 4K(k)$ and dividing by the corresponding period P. This yields the result:

$$\overline{\left(\frac{d\phi}{dt}\right)} = \frac{\omega_l I_D}{I_z}\left[1 + \frac{I_z - I_x}{I_x}\frac{\Pi(K(k), n)}{K(k)}\right] \tag{3.115}$$

The Euler angles ψ and θ are periodic with period P. The angle ϕ, on the other hand, will not (in general) increase by a multiple of 2π over this time period. Thus the inertial attitude of the body will be quasi-periodic in time.

Finally, in the limit as $I_D \rightarrow I_z$ the Euler angles approach a singular value for $\theta = 0$ with $\phi + \psi$ equalling the total rotation angle. Taking the appropriate limits and adding, both ϕ and ψ will contribute non-trivial computations to this sum that converge to the total rotation angle $\phi + \psi = \phi_o + \pi/2 + \omega_l(t - t_o)$. Alternate angle definitions can, of course, be used to avoid some of the singular aspects of the Euler angles as specific configurations are approached. It is important to note that non-singular algorithms can be robustly coded to handle all possible issues with the Euler angles.

3.2.8 Specification of the Rotational Dynamics

Finally, it is important to specify the rotational dynamics of a given body relative to an inertial frame. There are two ways to do this, either by specifying the qualitative nature of the dynamics or by specifying the initial conditions of the rotational dynamics. Both procedures are reviewed below.

Starting from the Qualitative Nature of the Rotational Dynamics

The qualitative nature of its motion is controlled by the effective rotation rate ω_l and its dynamic inertia I_D. By specifying these two quantities the qualitative properties of the body's rotation motion are at once known, and by using the relationships given herein the quantitative values for its motion are also specified. By specifying the initial epoch τ_o the location of the angular velocity vector in its periodic orbit is defined. Next the initial epoch of the motion t_o must be specified as well as the initial right ascension (ϕ_o). Once these are chosen the attitude of the body at the initial epoch is fixed relative to the initial angular momentum vector.

The above four constants specify the rotational motion of the body relative to its initial angular momentum vector. To completely specify its motion one needs only specify the orientation of the angular momentum vector in inertial space, which can be done with the choice of two angles. This defines an additional, constant, transformation matrix from the angular momentum relative system to inertial space that must be applied whenever the attitude of the body is required in inertial space. These six initial conditions then completely specify the ideal rotational dynamics of a body, such as an asteroid or comet.

Specifying the rotational dynamics in this way has benefits for simulation, as the rotation mode for the asteroid is directly chosen as well as the direction about which the body will precess and nutate (i.e., the angular momentum vector of the body).

Specifying the Initial Conditions

Now consider the reverse problem, the specification of the rotational motion of the body given an initial attitude and body-fixed angular velocity. Assume the initial attitude of the body is specified as a transformation matrix from the body-fixed frame to the inertial frame, and is specified as $\boldsymbol{\mathcal{T}}_o$. The initial body-fixed angular velocity of the body is specified as $\boldsymbol{\omega}_o$.

Given the initial body-fixed angular velocity $\boldsymbol{\omega}_o$ and inertia tensor $\boldsymbol{\mathcal{I}}$ the constants ω_l and I_D are computed using Eqs. 3.74 and 3.75. Once this computation is performed the qualitative aspects of the rotational motion are known and all the relevant parameters of the motion can be computed, such as k, σ, and γ. The specification of the initial parameter τ_o must be made more carefully, however. First find the values of $\mathrm{sn}(\tau_o)$ and $\mathrm{cn}(\tau_o)$ from Eqs. 3.76–3.78.

$$\mathrm{sn}(\tau_o) = \frac{\omega_y(t_o)}{\sigma\omega_l}\sqrt{\frac{I_y(I_z - I_y)}{I_D(I_z - I_D)}} \tag{3.116}$$

where the value of σ depends on which mode the rotation state is in. The evaluation of the $\mathrm{cn}(\tau_o)$ term also depends on which rotation mode the asteroid is in:

$$\mathrm{cn}(\tau_o) = \begin{cases} \dfrac{\omega_x(t_o)}{\omega_l}\sqrt{\dfrac{I_x(I_z - I_D)}{I_D(I_z - I_D)}} & I_y < I_D \leq I_z \\[2ex] \dfrac{\omega_z(t_o)}{\omega_l}\sqrt{\dfrac{I_z(I_z - I_x)}{I_D(I_D - I_x)}} & I_x \leq I_D < I_y \end{cases} \tag{3.117}$$

These elliptic functions must be inverted to compute τ_o.

To perform the inversion note the following definitions $\mathrm{sn}(\tau_o) = \sin\varphi_o$ and $\mathrm{cn}(\tau_o) = \cos\varphi_o$, and derive the amplitude angle $\varphi_o = \arctan\left(\mathrm{sn}(\tau_o)/\mathrm{cn}(\tau_o)\right)$, where this angle will lie in the interval $[0, 2\pi)$ in general. The quantities τ_o and φ_o are related via the elliptic integral of the first kind, as defined in Eq. F.2:

$$\tau_o = F(\varphi_o, k) \tag{3.118}$$

To properly carry out this computation requires the proper accounting for which quadrant the angle φ_o lies in

$$\tau_o = \begin{cases} F(\varphi_o, k) & \text{if } \varphi \leq \pi/2 \\ 2F(\pi/2, k) - F(\pi - \varphi_o, k) & \text{if } \pi/2 < \varphi \leq \pi \\ 2F(\pi/2, k) + F(\varphi_o - \pi, k) & \text{if } \pi < \varphi \leq 3\pi/2 \\ 4F(\pi/2, k) - F(2\pi - \varphi_o, k) & \text{if } 3\pi/2 < \varphi < 2\pi \end{cases} \tag{3.119}$$

where $F(\pi/2, k) = K(k)$ is the complete elliptic integral of the first kind. If the case $I_D = I_y$ should occur, then τ_o can be solved uniquely from the initial condition $\omega_y(t_o) = \omega_l \tanh(\tau_o)$ without using elliptic functions or integrals.

Having computed ω_l, I_D, and τ_o the Euler angles are defined and hence that the transformation matrix from the inertial frame with angular momentum along the z-axis to the body-fixed frame at the initial epoch, $\boldsymbol{C}_o^T$, is defined. The matrix $\boldsymbol{C}_o$ takes us from the body-frame to the inertial frame at the initial epoch. Note that, in general, there is a disconnect between the inertial frame defined by the $\boldsymbol{C}_o$ matrix and the initial inertial frame in which the body orientation is defined. To close the loop an additional transformation matrix must be applied to $\boldsymbol{C}_o$ to rotate from the inertial frame with angular momentum aligned with the z-axis to the inertial frame the body's attitude is specified in, specifically the transformation $\boldsymbol{T}$ such that:

$$\boldsymbol{T}\boldsymbol{C}_o = \boldsymbol{\mathcal{T}}_o \tag{3.120}$$

$$\boldsymbol{T} = \boldsymbol{\mathcal{T}}_o \boldsymbol{C}_o^T \tag{3.121}$$

Then the inertial attitude of the body is specified by $\boldsymbol{T} \cdot \boldsymbol{C}(t)$.

4. Equations of Motion for a Small Body Orbiter

This chapter considers the basic equations of motion for a spacecraft or particle relative to a small body. Several forms of the equations are derived, each with their own special applications.

4.1 Newtonian Equations of Motion

The most fundamental form for the equations of motion are the standard Newton's equations. In the following we assume that the particle has no effect on the motion of the central body. Assume a frame fixed to the central mass distribution, and relative to a fixed orientation with respect to inertial space. We also assume gravitational attraction by the Sun and other bodies, and solar radiation pressure from the Sun acting on the body. Combining these different accelerations together the standard equations of motion can be stated as

$$\ddot{\boldsymbol{r}} = \frac{\partial U(\boldsymbol{C}^T \cdot \boldsymbol{r})}{\partial \boldsymbol{r}} + \sum_{i=0}^{N} \frac{\partial \mathcal{R}_{p_i}(\boldsymbol{r}, \boldsymbol{d}_i)}{\partial \boldsymbol{r}} + \frac{\partial \mathcal{R}_{SRP}(\boldsymbol{r}, \boldsymbol{d}_0)}{\partial \boldsymbol{r}} \tag{4.1}$$

where $\boldsymbol{r}$ is the spacecraft position vector relative to the small body and referred to an inertial frame (i.e., the time derivatives are relative to an inertially oriented frame), U is the gravitational potential of the body discussed in Section 2.5, $\mathcal{R}_{p_i}$ represents the perturbation from the ith attracting body, and $\mathcal{R}_{SRP}$ represents the solar radiation pressure perturbation, both of these are defined in Section 2.6. The vector $\boldsymbol{d}_0$ represents the position of the small body relative to the Sun and the vectors $\boldsymbol{d}_i$ for $i \geq 1$ represent the position of additional perturbing bodies relative to the small body. The transformation dyadic $\boldsymbol{C}^T$ takes vectors from an inertial frame to the small body-fixed frame indicating that the gravitational potential of the central body is always stated in a body-fixed frame.

It is theoretically important, at this point, to establish that this full set of equations can be restated in a Lagrangian or Hamiltonian formulation. This allows us to invoke properties of these general classes of systems later in our discussions.

4.2 Lagrangian Form of the Equations

Starting from the Newtonian frame, it is simple to write out the general Lagrangian. Once this is given, changes in coordinates and coordinate frames are simply made with the appropriate substitution into the Lagrangian. Denote the specific kinetic energy of the spacecraft relative to the asteroid-centered frame and the full potential:

$$T = \frac{1}{2}\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} \tag{4.2}$$

$$\mathcal{U} = U + \sum_{i=0}^{N} \mathcal{R}_{p_i} + \mathcal{R}_{SRP} \tag{4.3}$$

The potential is ultimately a function of position and time, and thus does not lead to a time-invariant system in general.

The Lagrangian is then formed as $L = T + \mathcal{U}$, with a positive sign as the potential is used instead of the potential energy. The Lagrange equations of motion are then [55]:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i} \; ; \; i = 1, 2, \ldots, n \tag{4.4}$$

where for the original system the q_i and $\dot{q}_i$ are the components of the $\boldsymbol{r}$ and $\dot{\boldsymbol{r}}$ vectors. Once the Lagrangian is defined, it is relatively simple to rewrite the equations of motion in an alternate frame (such as a rotating frame) or in an alternate set of coordinates, so long as the specific transformation equations can be solved for the original position and velocity vectors in terms of the new coordinates and their time derivatives.

The equations of motion as stated above are convenient for direct numerical simulation provided that the relative ephemeris of all the bodies, $\boldsymbol{d}_i$, are known and the rotational dynamics of the central body is known and specified via the transformation dyadic $\boldsymbol{C}$. This form of the equations are not, however, particularly useful in terms of understanding the system analytically, nor in interpreting the output of any numerical simulation. To address these concerns a number of different reference frames are considered from which to observe the dynamics of the system or coordinates for reporting the motion. Which frame or coordinates are chosen depends on what aspect of the problem is of interest. The main frames of interest are a frame fixed to the central body and a frame fixed to the orbit of the small body about the Sun. The former frame is more useful for surface and close proximity motions. The latter is more useful for discussing orbital mechanics perturbed by solar radiation pressure and solar gravity. The specific methods used to study these frames are significantly different, and thus the following derivations are carried out independently of each other. For each frame it is still possible, and often relevant, to carry the full effect of the small body gravity or solar perturbations.

4.2.1 Body-Fixed Frame

The equations of motion can be restated under the assumption that the particle position vector is specified relative to the central body-fixed frame, and that this body is most generally in a state of complex rotational motion. In terms of the gravitational perturbations, this formulation actually simplifies the situation as it removes the transformation from body-fixed to inertial frame that otherwise must appear in the gravitational potential. It does not simplify the external perturbations, as there must still be a transformation that maps the external perturbations from their frame of definition into the body-fixed frame. If the body's rotation state is specified as a function of time, however, these just represent the time-varying transformation $\boldsymbol{C}(t)$ discussed in the previous chapter. To make this example more specific, imagine that the central body is in a state of torque-free tumbling. This is the situation for the asteroid Toutatis, studied in [76] and Chapter 8.

If the location of the particle is $\boldsymbol{r}$ relative to an inertial frame, the particle location with respect to the body-fixed frame is then $\boldsymbol{q} = \boldsymbol{C}(t)^T \cdot \boldsymbol{r}$ and the substitution $\boldsymbol{r} = \boldsymbol{C}(t) \cdot \boldsymbol{q}$ is made everywhere. The time rate of change of the position vector can also be computed as $\dot{\boldsymbol{r}} = \boldsymbol{C}(t) \cdot (\dot{\boldsymbol{q}} + \boldsymbol{\omega} \times \boldsymbol{q})$, where $\dot{\boldsymbol{q}}$ is taken with respect to the rotating frame and is expressed in that frame. These terms can be substituted into the Lagrangian to find:

$$L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t) = \frac{1}{2} \left[\dot{\boldsymbol{q}} + \boldsymbol{\omega} \times \boldsymbol{q}\right] \cdot \left[\dot{\boldsymbol{q}} + \boldsymbol{\omega} \times \boldsymbol{q}\right] + \mathcal{U}\left(\boldsymbol{C}(t) \cdot \boldsymbol{q}\right) \tag{4.5}$$

$$\mathcal{U} = U(\boldsymbol{q}) + \sum_{i=0}^{N} \mathcal{R}_{p_i}(\boldsymbol{C}(t) \cdot \boldsymbol{q}, \boldsymbol{d}_i) + \mathcal{R}_{SRP}(\boldsymbol{C}(t) \cdot \boldsymbol{q}, \boldsymbol{d}_0) \tag{4.6}$$

Application of Lagrange's equations using $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ as the independent coordinates and velocities then yields the correct form of the equations as stated in a rotating reference frame:

$$\ddot{\boldsymbol{q}} + \dot{\boldsymbol{\omega}} \times \boldsymbol{q} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{q}} + \boldsymbol{\omega} \times \boldsymbol{\omega} \times \boldsymbol{q} = \frac{\partial \mathcal{U}}{\partial \boldsymbol{q}} \tag{4.7}$$

It is interesting to note that if the central body is in a torque-free rotation state, but not rotating about its principal axes of inertia, then the angular velocity is periodic in the body-fixed state. Thus, the equations of motion have time-varying, periodic coefficients represented by the angular velocity $\boldsymbol{\omega}$.

If the exogenous perturbations are neglected the potential $\mathcal{U} = U(\boldsymbol{q})$ and is independent of the transformation dyadic $\boldsymbol{C}$. If the body rotates about a principal moment of inertia, the angular velocities $\boldsymbol{\omega}$ are constant and the $\dot{\boldsymbol{\omega}}$ term disappears. For this case the system has an integral of motion, a discussion that is left for later.

4.2.2 Orbit-Fixed Frame

The orbit-fixed frame is defined as the frame that follows the heliocentric orbit position of the small body. In this frame the Sun remains fixed in an angular sense, although its distance from the small body will in general vary periodically as will the frame rotation rate. It is physically distinct from the body-fixed frame, however, in that the angular velocity always has a fixed direction, normal to the orbital plane.

Now denote the orbit-fixed frame with a coordinate $\boldsymbol{Q} = \boldsymbol{C}_O(t) \cdot \boldsymbol{r}$. If the inertial z axis is aligned with the heliocentric orbit angular momentum and the x-axis lies along the Sun–body position vector, the transformation matrix takes a simple form:

$$\boldsymbol{C}_O = \begin{bmatrix} \cos(\omega_0 + f) & \sin(\omega_0 + f) & 0 \\ -\sin(\omega_0 + f) & \cos(\omega_0 + f) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{4.8}$$

where the small body heliocentric true anomaly f is an implicit function of time. The angular velocity is aligned along the z-axis and is:

$$\boldsymbol{\Omega} = \dot{f} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{4.9}$$

where $\dot{f} = \sqrt{\frac{\mu}{p_0^3}} \left(1 + e_0 \cos(f)\right)^2$.

The transformed Lagrangian in this frame is specified as:

$$L(\boldsymbol{Q}, \dot{\boldsymbol{Q}}, t) = \frac{1}{2} \left[\dot{\boldsymbol{Q}} + \dot{f}\hat{\boldsymbol{z}} \times \boldsymbol{Q}\right] \cdot \left[\dot{\boldsymbol{Q}} + \dot{f}\hat{\boldsymbol{z}} \times \boldsymbol{Q}\right] + \mathcal{U}\left(\boldsymbol{C}_O^T \cdot \boldsymbol{Q}\right) \tag{4.10}$$

Given this form of the angular velocity, the general equations in this frame are:

$$\ddot{\boldsymbol{Q}} + \ddot{f}\hat{\boldsymbol{z}} \times \boldsymbol{Q} + 2\dot{f}\hat{\boldsymbol{z}} \times \dot{\boldsymbol{Q}} + \dot{f}^2\hat{\boldsymbol{z}} \times \hat{\boldsymbol{z}} \times \boldsymbol{Q} = \frac{\partial \mathcal{U}}{\partial \boldsymbol{Q}} \tag{4.11}$$

Neglecting the mass distribution of the central body, i.e., by choosing $U = \mu/|\boldsymbol{Q}|$ and neglecting non-solar perturbations the equations of motion are periodic in general with period $T = 2\pi a_0^{3/2}/\sqrt{\mu_0}$. This period is very long with respect to the usual orbit period about the central body, and only becomes significant for motion far from the central body. This is consistent with the simplifying assumption where the central body mass distribution in neglected and the central body is treated as a point mass. This assumption is made in the following as it provides a good motivation to derive a significantly different set of equations for motion in this regime.

As $\dot{f}$ is a function of f, one is tempted to make the true anomaly the independent parameter for these equations. Performing this transformation alone does not yield any significant simplifications. Rather, following tradition in the elliptic restricted three-body problem [97] it is easier to simultaneously scale the vector $\boldsymbol{Q}$ by the Sun–body distance $d = p_0/(1 + e_0 \cos f)$ and change the independent parameter from time to true anomaly. Then, the transformation sequence can be stated as $\boldsymbol{Q} \rightarrow d\boldsymbol{q}$, and the time transformation can be stated as $\dot{\boldsymbol{Q}} = d\boldsymbol{Q}/dt = d\boldsymbol{Q}/df\dot{f} = \boldsymbol{Q}'\dot{f}$. This leads to the following sequence (note, do not confuse the $d\boldsymbol{q}$ notation with a differential, it is just a factor d multiplied by $\boldsymbol{q}$):

$$\boldsymbol{Q} = d\boldsymbol{q} \tag{4.12}$$

$$\dot{\boldsymbol{Q}} = d'\boldsymbol{q}\dot{f} + d\boldsymbol{q}'\dot{f} \tag{4.13}$$

$$= \sqrt{\frac{\mu_0}{p_0}}\left[e_0 \sin f\boldsymbol{q} + (1 + e_0 \cos f)\boldsymbol{q}'\right] \tag{4.14}$$

$$\ddot{\boldsymbol{Q}} = \sqrt{\frac{\mu_0}{p_0}}\left[e_0 \cos f\boldsymbol{q} + e_0 \sin f\boldsymbol{q}' - e_0 \sin f\boldsymbol{q}' + (1 + e_0 \cos f)\boldsymbol{q}''\right]\dot{f} \tag{4.15}$$

$$= \frac{\mu_0}{p_0^2}(1 + e_0 \cos f)^3\left[\boldsymbol{q}'' + \frac{e_0 \cos f}{1 + e_0 \cos f}\boldsymbol{q}\right] \tag{4.16}$$

Substituting these into Eq. 4.11, a common factor can be extracted

$$\boldsymbol{q}'' + 2\hat{\boldsymbol{z}} \times \boldsymbol{q}' + \frac{e_0 \cos f}{1 + e_0 \cos f}\boldsymbol{q} + \hat{\boldsymbol{z}} \times \hat{\boldsymbol{z}} \times \boldsymbol{q} = \frac{p_0}{\mu_0}\frac{1}{(1 + e_0 \cos f)^2}\frac{\partial \mathcal{U}}{\partial \boldsymbol{q}} \tag{4.17}$$

Now consider the potential and its specification in this particular frame (i.e., just a point mass plus the solar perturbation). In terms of the original variables

$$\mathcal{U} = \frac{\mu}{|\boldsymbol{Q}|} + \mu_0\left[\frac{1}{|\boldsymbol{d} - \boldsymbol{Q}|} - \frac{\boldsymbol{d}\cdot\boldsymbol{Q}}{|\boldsymbol{d}|^3}\right] - \boldsymbol{Q}\cdot\frac{P_0\boldsymbol{d}}{Bd^3} \tag{4.18}$$

where the simple lumped solar radiation pressure model is assumed and the vector $\boldsymbol{d}$ points towards the small body from the Sun. Implementing the new transformation changes the potential to

$$\mathcal{U} = \frac{\mu}{d|\boldsymbol{q}|} + \frac{\mu_0}{d}\left[\frac{1}{|\hat{\boldsymbol{d}} - \boldsymbol{q}|} - \hat{\boldsymbol{d}}\cdot\boldsymbol{q}\right] - \frac{1}{d}\boldsymbol{q}\cdot\frac{P_0}{B}\hat{\boldsymbol{d}} \tag{4.19}$$

Defining $\mathcal{V} = \frac{p_0}{\mu}\frac{1}{(1+e_0 \cos f)^2}\mathcal{U}$,

$$\mathcal{V} = \frac{1}{1 + e_0 \cos f}\left[\frac{\mu/\mu_0}{|\boldsymbol{q}|} + \frac{1}{|\hat{\boldsymbol{d}} - \boldsymbol{q}|} - \hat{\boldsymbol{d}}\cdot\boldsymbol{q} - \frac{P_0/\mu_S}{B}\boldsymbol{q}\cdot\hat{\boldsymbol{d}}\right] \tag{4.20}$$

and the associated equations of motion:

$$\boldsymbol{q}'' + 2\hat{\boldsymbol{z}} \times \boldsymbol{q}' + \frac{e_0 \cos f}{1 + e_0 \cos f}\boldsymbol{q} + \hat{\boldsymbol{z}} \times \hat{\boldsymbol{z}} \times \boldsymbol{q} = \frac{\partial \mathcal{V}}{\partial \boldsymbol{q}} \tag{4.21}$$

The form of these equations is somewhat simpler than the original statement, and the periodic terms are explicitly identified. Careful attention must be given to the interpretation to the solution of these equations, as they are relative to a length scale that changes in time. Specifically, when these solutions are to be transformed into an inertial frame or into a set of osculating orbit elements, the transformations in Eqs. 4.12 and 4.14 must first be applied. Additionally, as the independent parameter is true anomaly, to switch to the time domain one must make the transformations from true anomaly through Kepler's equation, to time. In this direction there is no need to solve Kepler's equation iteratively.

This statement of the problem has no approximations other than the neglect of the central body mass distribution and the simplified solar radiation pressure term. Thus it is similar to the elliptic restricted three-body problem, except the coordinate center is not at the center of mass. It is useful to introduce an additional approximation to this set of equations, assuming that the particle is close to the central body, or that $|\boldsymbol{q}| \ll 1$ and expanding the term $1/|\hat{\boldsymbol{d}} - \boldsymbol{q}|$ up to the second order. This is essentially the "Hill approximation" and was derived in Chapter 2. The potential $\mathcal{V}$ can then be expressed as

$$\mathcal{V} = \frac{1}{1 + e_0 \cos f} \left[\frac{\mu/\mu_0}{|\boldsymbol{q}|} + \frac{3}{2} \left(\hat{\boldsymbol{d}} \cdot \boldsymbol{q} \right)^2 - \frac{1}{2} \boldsymbol{q} \cdot \boldsymbol{q} - \frac{P_0/\mu_0}{B} \boldsymbol{q} \cdot \hat{\boldsymbol{d}} + \dots \right] \tag{4.22}$$

To isolate the time periodic terms in the potential, extract the $1/2q^2$ for reasons that will be evident in a moment and truncate the higher-order terms. The resulting potential is $\mathcal{V}_H$, where the H stands for the Hill approximation. Evaluating the equations of motion then yield:

$$\boldsymbol{q}'' + 2\hat{\boldsymbol{z}} \times \boldsymbol{q}' + \frac{e_0 \cos f}{1 + e_0 \cos f} \boldsymbol{q} + \hat{\boldsymbol{z}} \times \hat{\boldsymbol{z}} \times \boldsymbol{q} = -\frac{1}{1 + e_0 \cos f} \boldsymbol{q} + \frac{\partial \mathcal{V}_H}{\partial \boldsymbol{q}} \tag{4.23}$$

and simplified by moving the exposed term from the right to the left:

$$\boldsymbol{q}'' + 2\hat{\boldsymbol{z}} \times \boldsymbol{q}' + \boldsymbol{q} + \hat{\boldsymbol{z}} \times \hat{\boldsymbol{z}} \times \boldsymbol{q} = \frac{\partial \mathcal{V}_H}{\partial \boldsymbol{q}} \tag{4.24}$$

Finally, noting the identity $\tilde{\hat{\boldsymbol{z}}} \cdot \tilde{\hat{\boldsymbol{z}}} = \hat{\boldsymbol{z}}\hat{\boldsymbol{z}} - \boldsymbol{U}$ leads to the final form of the equations

$$\boldsymbol{q}'' + 2\hat{\boldsymbol{z}} \times \boldsymbol{q}' + (\hat{\boldsymbol{z}} \cdot \boldsymbol{q})\hat{\boldsymbol{z}} = \frac{\partial \mathcal{V}_H}{\partial \boldsymbol{q}} \tag{4.25}$$

$$\mathcal{V}_H = \frac{1}{1 + e_0 \cos f} \left[\frac{\mu/\mu_0}{|\boldsymbol{q}|} + \frac{3}{2} \left(\hat{\boldsymbol{d}} \cdot \boldsymbol{q} \right)^2 - \frac{P_0/\mu_0}{B} \boldsymbol{q} \cdot \hat{\boldsymbol{d}} \right] \tag{4.26}$$

Thus, the periodic portion is isolated to one term on the right-hand side.

In the Hill three-body problem, it is customary to make one additional normalization to remove all parameters from the problem. This is to scale the length by the quantity $(\mu/\mu_0)^{1/3}$. Doing so does not change the form of the left-hand side of the equations, and thus can be modeled completely by defining a new potential that has been so normalized. Let us call this potential $\tilde{\mathcal{V}}_H$

$$\tilde{\mathcal{V}}_H = \frac{1}{1 + e_0 \cos f} \left[\frac{1}{|\boldsymbol{q}|} + \frac{3}{2} \left(\hat{\boldsymbol{d}} \cdot \boldsymbol{q} \right)^2 - \tilde{\beta} \boldsymbol{q} \cdot \hat{\boldsymbol{d}} \right] \tag{4.27}$$

$$\tilde{\beta} = \frac{P_0}{\mu_0 B} \left(\frac{\mu}{\mu_0} \right)^{-1/3} \tag{4.28}$$

This basic form of the equations of motion was originally derived in [171].

4.3 Hamiltonian Form of the Equations

The Lagrangian frame allows for simple restatement of the equations of motion in different coordinate frames and coordinate systems, so long as the dynamical system is specified by a set of coordinates and their time derivatives. For physical intuition and spatial geometry, this works quite well and almost all of the fundamental properties of astrodynamical systems can be expressed using a Lagrangian set of the equations of motion. There are a number of deeper properties that exist for these systems, however, that are not as easily realized in the Lagrangian format. The first step away from these is via Hamilton's equations, which replaces the idea of a coordinate and a velocity with a set of conjugate quantities that are related to each other as coordinates and their corresponding momenta.

Given a set of equations in a Lagrangian frame, it is rather simple to transform them into a Hamiltonian form. One can leave the coordinates as-is and only operate on the momenta and velocities if one wishes to make the transformation in a minimum number of steps. An efficient way to perform this transformation is via the Legendre Transformation.

Assume that a well defined Lagrangian $L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$ exists from which the equations of motion can be derived using Lagrange's equations. Define the momentum, p_i, conjugate to the coordinate q_i as $p_i = \partial L / \partial \dot{q}_i$, or in vector form:

$$\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}} \tag{4.29}$$

Recalling the standard form of Lagrange's equations this becomes

$$\dot{\boldsymbol{p}} = \frac{\partial L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)}{\partial \boldsymbol{q}} \tag{4.30}$$

This does not simplify matters, however, as we wish to completely eliminate $\dot{\boldsymbol{q}}$ from the Lagrangian to find a statement of the equations of motion independent of the velocities. In principle this can be done if Eq. 4.29 can be solved for the velocity, or $\dot{\boldsymbol{q}} = f(\boldsymbol{q}, \boldsymbol{p})$. A technical condition for this is that $|\partial^2 L / \partial \dot{\boldsymbol{q}}^2| \neq 0$, which is generally satisfied for the systems considered here.

The Legendre Transformation naturally handles such substitutions and indicates the functional form of the desired expression $\dot{\boldsymbol{q}} = f(\boldsymbol{q}, \boldsymbol{p})$. To carry this out form the new functional expression:

$$H(\boldsymbol{q}, \boldsymbol{p}, t) = \boldsymbol{p} \cdot \dot{\boldsymbol{q}} - L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t) \tag{4.31}$$

where the $\dot{\boldsymbol{q}}$ expressions in the above are functions of $\boldsymbol{q}$ and $\boldsymbol{p}$. Now consider variations in the new variable $\boldsymbol{p}$ while keeping $\boldsymbol{q}$ fixed:

$$H_{\boldsymbol{p}} \cdot \delta \boldsymbol{p} = \dot{\boldsymbol{q}} \cdot \delta \boldsymbol{p} + \boldsymbol{p} \cdot \delta \dot{\boldsymbol{q}} - L_{\dot{\boldsymbol{q}}} \cdot \delta \dot{\boldsymbol{q}} \tag{4.32}$$

where $\delta \dot{\boldsymbol{q}}$ can be further expressed as a variation with respect to $\boldsymbol{p}$. This is not needed, however, as the two terms dotted with $\delta \dot{\boldsymbol{q}}$ are equal and opposite (Eq. 4.29) and cancel. This leaves the defining relationship for the transformation:

$$\dot{\boldsymbol{q}} = H_{\boldsymbol{p}} \tag{4.33}$$

Using a similar variation technique it can be shown that $H_{\boldsymbol{q}} = -L_{\boldsymbol{q}}$, leading to the complete set of equations of motion in terms of the coordinates and momenta:

$$\dot{\boldsymbol{p}} = -H_{\boldsymbol{q}} \tag{4.34}$$

Collecting the coordinates and momenta into a single state $\boldsymbol{x} = [\boldsymbol{q}; \boldsymbol{p}] \in \mathbf{R}^{2n}$ with the attendant equations of motion defines the state space form of the equations of motion

$$\dot{\boldsymbol{x}} = J H_{\boldsymbol{x}} \tag{4.35}$$

where J is defined as:

$$J = \begin{bmatrix} 0_n & I_n \\ -I_n & 0_n \end{bmatrix} \tag{4.36}$$

and 0_n is an $n \times n$ matrix of zeros and I_n is the $n \times n$ identity matrix. The matrix J plays a significant role in Hamiltonian dynamics.

A reapplication of the Legendre Transformation with the defining quantity $\dot{\boldsymbol{q}} = H_{\boldsymbol{p}}$ and assuming that $|H_{\boldsymbol{pp}}| \neq 0$ will yield the Lagrangian equations of motion. Thus the Hamiltonian and Lagrangian expressions of the equations of motion are fundamentally linked to each other. While the Legendre Transformation tells us what the new functional for deriving the equations of motion is and indicates the relationship between the parameters, one must still solve for the $\dot{\boldsymbol{q}}$ in terms of the $\boldsymbol{p}$ and make the substitutions.

The importance of the Hamiltonian form of the equations of motion is that they allow the structure of the equations of motion to be preserved through a systematic set of transformations between different sets of variables called "canonical variables." A simple definition of a canonical transformation of a Hamiltonian dynamical system is one that preserves the same structure of the equations of motion. Specifically, a transformation between old variables $\boldsymbol{q}$ and $\boldsymbol{p}$ and new variables $\boldsymbol{Q}(\boldsymbol{q}, \boldsymbol{p})$ and $\boldsymbol{P}(\boldsymbol{q}, \boldsymbol{p})$ is a canonical transformation if the transformation equations are unique and invertible and the resulting equations of motion are:

$$\dot{\boldsymbol{Q}} = \frac{\partial K(\boldsymbol{Q}, \boldsymbol{P})}{\partial \boldsymbol{P}} \tag{4.37}$$

$$\dot{\boldsymbol{P}} = -\frac{\partial K(\boldsymbol{Q}, \boldsymbol{P})}{\partial \boldsymbol{Q}} \tag{4.38}$$

where $K(\boldsymbol{Q}, \boldsymbol{P}) = H(\boldsymbol{q}(\boldsymbol{Q}, \boldsymbol{P}), \boldsymbol{p}(\boldsymbol{Q}, \boldsymbol{P}))$.

To show that a proposed transformation is canonical it is not necessary to completely reconstruct the equations of motion. Rather, there are a number of tests that can be carried out on the transformation to ascertain whether it is canonical. A particularly useful test is that if the Jacobian of the transformation is a symplectic matrix, then the transformation is canonical. This test is used exclusively in this text, as it has a deeper meaning when discussing the dynamical evolution of a Hamiltonian system. The Jacobian of an arbitrary transformation is defined as:

$$M = \frac{\partial(\boldsymbol{Q}, \boldsymbol{P})}{\partial(\boldsymbol{q}, \boldsymbol{p})} \tag{4.39}$$

and is a matrix of dimension $2n \times 2n$.

For a matrix M to be symplectic it must satisfy two simple rules. It must be a square matrix of even dimension and it must satisfy the matrix identity:

$$J = M^T J M \tag{4.40}$$

The matrix J satisfies this equation and is thus symplectic itself.

Symplectic matrices form a group and have distinctive properties. These are discussed in many texts, but a particularly clear statement of their properties is found in [132], which are summarized in Appendix D. One important property is that the determinant of any symplectic matrix always equals 1, and thus a symplectic matrix is always non-singular and invertible. Next is that the inverse of a symplectic matrix can always be stated in closed form as

$$M^{-1} = -J M^T J \tag{4.41}$$

Both of these results can be directly proven from the definition of a symplectic matrix, as discussed in the appendix.

Beyond the derivation of the equations of motion in some standard forms, the main application of canonical transformations used here is to note the properties of solution flows. Specifically, the solution flow of a Hamiltonian dynamical system is, itself, a canonical transformation from the state at a specified epoch (i.e., the initial conditions) to the state at a different time. To establish this consider a Hamiltonian dynamical system evolved forward in time by a small time interval, Δt, using a Taylor series expansion in time to evolve the flow forward [56]

$$\boldsymbol{x}(t + \Delta t) = \boldsymbol{x}(t) + \dot{\boldsymbol{x}}(t)\Delta t + \frac{1}{2!}\ddot{\boldsymbol{x}}(t)\Delta t^2 + \ldots \tag{4.42}$$

For small Δt we only consider the linear terms in Δt, as the limit $\Delta t \to 0$ will be taken in a moment. Substituting the equations of motion then yields the mapping between the two states:

$$\boldsymbol{x}(t + \Delta t) = \boldsymbol{x}(t) + J H_{\boldsymbol{x}} \Delta t \tag{4.43}$$

Taking the partial of $\boldsymbol{x}(t + \Delta t)$ with respect to $\boldsymbol{x}$, denoted as Φ, yields

$$\Phi = \frac{\partial \boldsymbol{x}(t + \Delta t)}{\partial \boldsymbol{x}(t)} \tag{4.44}$$

$$= I + J H_{\boldsymbol{x}\boldsymbol{x}} \Delta t \tag{4.45}$$

Now the symplectic test is applied to Φ,

$$\begin{aligned}\Phi^T J \Phi &= [I - H_{\boldsymbol{xx}} J \Delta t]\, J\, [I + J H_{\boldsymbol{xx}} \Delta t] && (4.46)\\ &= [I - H_{\boldsymbol{xx}} J \Delta t]\, [J - H_{\boldsymbol{xx}} \Delta t] && (4.47)\\ &= J + H_{\boldsymbol{xx}} \Delta t - H_{\boldsymbol{xx}} \Delta t + \mathcal{O}(\Delta t^2) && (4.48)\\ &= J && (4.49)\end{aligned}$$

The derivation is only for an infinitesimal Δt; however, canonical transformations form a group, meaning that any combination of canonical transformations in sequence results in a canonical transformation. Thus, this result can be extended to an arbitrary time interval by successive application of the transformation, meaning that the solution flow, denoted as $\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}_o, t_o)$, defines a canonical transformation from $\boldsymbol{x}_o$ to $\boldsymbol{x}$.

Define the Jacobian of the solution flow as the matrix Φ, commonly termed the state transition matrix

$$\Phi(t, t_o, \boldsymbol{x}_o) = \frac{\partial \phi(t; \boldsymbol{x}_o, t_o)}{\partial \boldsymbol{x}_o} \tag{4.50}$$

Then this matrix is a symplectic matrix, a fact that becomes important later.

4.4 Lagrange Planetary and Gauss Equations

Derivation of equations of motion can be generalized beyond the structured Lagrangian and Hamiltonian approaches. In the following two very important alternate statements of the equations of motion for a particle are derived, the Lagrange Planetary Equations and the Gauss Equations. The heart of these equations of motion is the identification of a problem that can be fully integrated in terms of constants of motion, in this case the two-body problem with no additional perturbations. The inclusion of additional perturbation forces generates changes to the constants of motion, causing them to become time-varying coordinates. However, the original transformation between the constants and the solution still stands and allows the time-varying coordinates to describe the solution. The method by which such equations can be formally developed is called Variation of Parameters. Two simple derivations are provided, relying on Brouwer and Clemence for a detailed discussion of the derivations [19].

We take two different approaches to the Lagrange Planetary and Gauss equations, as each derivation and set of equations has important distinctions between them. The Lagrange Planetary equations are derived assuming that the perturbation functions arise from a potential. The Gauss equations, on the other hand, are derived assuming an arbitrary acceleration is applied to the particle. It is possible to relate these two assumptions to each other; however, for completeness independent derivations will be given for both.

4.4.1 Lagrange Planetary Equations

Consider the dynamical system defined by a position vector $\boldsymbol{r}$ and a velocity vector $\boldsymbol{v} = \dot{\boldsymbol{r}}$. Assume the system is governed by a set of equations of motion that can be decomposed as:

$$\ddot{\boldsymbol{r}} = \frac{\partial U}{\partial \boldsymbol{r}} + \frac{\partial R}{\partial \boldsymbol{r}} \tag{4.51}$$

where $U(\boldsymbol{r})$ and $R(\boldsymbol{r}, t)$. The potential function U is the main potential and R is the perturbing potential. The basic assumption is that the problem can be integrated when $R \equiv 0$ and expressed in terms of constants of motion, denoted here as $\boldsymbol{\alpha} \in \mathbf{R}^6$. Specifically, the solution of the dynamical system can be written as $\boldsymbol{r}(t) = \boldsymbol{r}(t, \boldsymbol{\alpha})$ and $\boldsymbol{v}(t) = \boldsymbol{v}(t, \boldsymbol{\alpha})$. Note that $\boldsymbol{v} = \partial \boldsymbol{r}/\partial t$ and that $\partial \boldsymbol{v}/\partial t = \partial U/\partial \boldsymbol{r}$. For example, if the main potential is $U = \mu/r$, then the $\boldsymbol{\alpha}$ could be the classical orbital elements.

The main idea behind the Variation of Constants approach is to let the constants of integration become the new variables of the dynamical system. This is a well-posed transformation so long as the $\boldsymbol{\alpha}$ comprise a full, linearly independent solution of the system, i.e., that the Jacobian $\partial \boldsymbol{\alpha}/\partial(\boldsymbol{r}, \boldsymbol{v})$ is non-singular. Then, under the perturbation term R, which need not be small, the "constants" of motion become functions of time and are no longer constant. A future state of the system $\boldsymbol{\alpha}^*$ can always be transformed to the position and velocity of the point at a new time through the transformations $\boldsymbol{r}(t) = \boldsymbol{r}(t; \boldsymbol{\alpha}^*)$ and $\boldsymbol{v}(t) = \boldsymbol{v}(t; \boldsymbol{\alpha}^*)$. The $\boldsymbol{\alpha}$ are sometimes called "osculating orbit elements" as they are the set of elements that to describe the true trajectory state in phase space at each point in time.

To derive the equations, first take the full time derivative of the position vector:

$$\dot{\boldsymbol{r}} = \frac{\partial \boldsymbol{r}}{\partial \boldsymbol{\alpha}} \dot{\boldsymbol{\alpha}} + \frac{\partial \boldsymbol{r}}{\partial t} \tag{4.52}$$

Now note that $\dot{\boldsymbol{r}} = \boldsymbol{v}$ by definition and that $\partial \boldsymbol{r}/\partial t = \boldsymbol{v}$ by choice. Efroimsky [33] has noted that this choice is not necessary, and that by generalizing this relation one can introduce a so-called "gauge function" into the theory. In the current description the more conventional derivation is presented that assumes $\partial \boldsymbol{r}/\partial t = \boldsymbol{v}$. Due to this, the first relation for $\dot{\boldsymbol{\alpha}}$ is

$$\frac{\partial \boldsymbol{r}}{\partial \boldsymbol{\alpha}} \dot{\boldsymbol{\alpha}} = 0 \tag{4.53}$$

where $\boldsymbol{r}_{\boldsymbol{\alpha}} \in \mathbf{R}^{3\times 6}$.

Next take the full time derivative of the velocity vector

$$\dot{\boldsymbol{v}} = \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{\alpha}} \dot{\boldsymbol{\alpha}} + \frac{\partial \boldsymbol{v}}{\partial t} \tag{4.54}$$

$$= \frac{\partial U}{\partial \boldsymbol{r}} + \frac{\partial R}{\partial \boldsymbol{r}} \tag{4.55}$$

By definition $\frac{\partial \boldsymbol{v}}{\partial t} = \frac{\partial U}{\partial \boldsymbol{r}}$ and can be cancelled, leaving the second relation

$$\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{\alpha}} \dot{\boldsymbol{\alpha}} = \frac{\partial R}{\partial \boldsymbol{r}} \tag{4.56}$$

where $\boldsymbol{v}_{\boldsymbol{\alpha}} \in \mathbf{R}^{3\times 6}$.

These two relations can be combined into a single matrix equation:

$$\begin{bmatrix} \boldsymbol{r}_{\boldsymbol{\alpha}} \\ \boldsymbol{v}_{\boldsymbol{\alpha}} \end{bmatrix} \dot{\boldsymbol{\alpha}} = \begin{bmatrix} 0 \\ R_{\boldsymbol{r}}^T \end{bmatrix} \tag{4.57}$$

As the constants $\boldsymbol{\alpha}$ are assumed to provide a complete solution of the system, the matrix is invertible and can be formally solved for the equations $\dot{\boldsymbol{\alpha}}$. This is not a particularly useful approach to the problem, however, as the functions $\boldsymbol{r}$ and $\boldsymbol{v}$ must be used at every step of the way.

Lagrange introduced an elegant simplification of this problem, as explained in [19]. Pre-multiply each side of the equation by the matrix $[\boldsymbol{r}_{\boldsymbol{\alpha}}^T; \boldsymbol{v}_{\boldsymbol{\alpha}}^T]J$, where J is the symplectic matrix introduced earlier. Carrying out the multiplications yields:

$$\left[\boldsymbol{r}_{\boldsymbol{\alpha}}^T \boldsymbol{v}_{\boldsymbol{\alpha}} - \boldsymbol{v}_{\boldsymbol{\alpha}}^T \boldsymbol{r}_{\boldsymbol{\alpha}}\right] \dot{\boldsymbol{\alpha}} = \left[\boldsymbol{r}_{\boldsymbol{\alpha}}^T R_{\boldsymbol{r}}^T\right] \tag{4.58}$$

Now note a few items that will simplify these results. First, $R_{\boldsymbol{\alpha}} = R_{\boldsymbol{r}} \cdot \boldsymbol{r}_{\boldsymbol{\alpha}}$ under the assumption that R is a function of position only, thus the term on the right-hand side can be replaced by $R_{\boldsymbol{\alpha}}^T$. The implication of this is that the relation for $\boldsymbol{r}$ can be substituted as a function of $\boldsymbol{\alpha}$ into the perturbing potential and one only needs deal with a perturbing potential $R(\boldsymbol{\alpha})$.

Next note that the general term i, j in the matrix on the left-hand side can be expressed as

$$[\alpha_i, \alpha_j] = \boldsymbol{r}_{\alpha_i} \cdot \boldsymbol{v}_{\alpha_j} - \boldsymbol{v}_{\alpha_i} \cdot \boldsymbol{r}_{\alpha_j} \tag{4.59}$$

The operator $[-, -]$ is called a Lagrange bracket and serves as the inverse to the Poisson bracket. Its properties are discussed in detail in [19], and will be discussed to a limited extent here. An important feature of the Lagrange bracket is that, for a given dynamical system, it is only a function of the constants of motion $\boldsymbol{\alpha}$ that have been chosen. Indeed, it is important to note that we have not chosen the specific constants yet in this problem, and have only relied on their existence. Using the Lagrange brackets the equation can be rewritten for the variation of parameters as

$$[[\alpha_i, \alpha_j]]\, \dot{\boldsymbol{\alpha}} = R_{\boldsymbol{\alpha}}^T \tag{4.60}$$

where $[[\alpha_i, \alpha_j]]$ represents a matrix with elements $[\alpha_i, \alpha_j]$, these elements taking on all values of $i, j = 1, 2, \ldots, 6$. This matrix is invertible, as it is just formed by the multiplication of non-singular matrices. Carrying out this inversion then yields:

$$\dot{\boldsymbol{\alpha}} = [[\alpha_i, \alpha_j]]^{-1} R_{\boldsymbol{\alpha}}^T \tag{4.61}$$

Before the inversion of this matrix is discussed we point out a crucial property: each Lagrange bracket is constant in time. To show this, take the total time derivative of an arbitrary term in the matrix:

$$\frac{d}{dt}[\alpha_i, \alpha_j] = \boldsymbol{v}_{\alpha_i} \cdot \boldsymbol{v}_{\alpha_j} + \boldsymbol{r}_{\alpha_i} \cdot \dot{\boldsymbol{v}}_{\alpha_j} - \boldsymbol{v}_{\alpha_i} \cdot \boldsymbol{v}_{\alpha_j} - \dot{\boldsymbol{v}}_{\alpha_i} \cdot \boldsymbol{r}_{\alpha_j} \tag{4.62}$$

$$= \boldsymbol{r}_{\alpha_i} \cdot \frac{\partial}{\partial \alpha_j} [U_{\boldsymbol{r}} + R_{\boldsymbol{r}}] - \boldsymbol{r}_{\alpha_j} \cdot \frac{\partial}{\partial \alpha_i} [U_{\boldsymbol{r}} + R_{\boldsymbol{r}}] \tag{4.63}$$

$$= \boldsymbol{r}_{\alpha_i} \cdot [U_{\boldsymbol{rr}} + R_{\boldsymbol{rr}}] \cdot \boldsymbol{r}_{\alpha_j} - \boldsymbol{r}_{\alpha_j} \cdot [U_{\boldsymbol{rr}} + R_{\boldsymbol{rr}}] \cdot \boldsymbol{r}_{\alpha_i} \tag{4.64}$$

$$= 0 \tag{4.65}$$

as the dyadics $U_{\boldsymbol{rr}}$ and $R_{\boldsymbol{rr}}$ are symmetric. This is an important fact, as it implies that the matrix of Lagrange brackets need only be evaluated functionally only once, and that the resulting inversion that occurs is applicable anywhere during the subsequent perturbed motion. It will also have important implications for averaging, as it allows us to pull the averaging operator across this term.

Lagrange Planetary Equations: Standard Form

Even though the derivation has been simple up to this point, the actual computation of the Lagrange brackets for a particular choice of constants is quite onerous. For the details we refer the reader to [19]. If the $\boldsymbol{\alpha}$ are chosen to be the standard orbital elements: $(a, e, i, \omega, \Omega, \sigma)$, where $\sigma = -n\tau_o$ and τ_o is the time of periapsis passage, the following form for the equations are found [19]

$$\dot{a} = \frac{2}{na} \frac{\partial R}{\partial \sigma} \tag{4.66}$$

$$\dot{e} = \frac{1}{na^2 e} \left[(1 - e^2) \frac{\partial R}{\partial \sigma} - \sqrt{1 - e^2} \frac{\partial R}{\partial \omega} \right] \tag{4.67}$$

$$\dot{i} = \frac{1}{na^2 \sqrt{1 - e^2}} \left[\cot i \frac{\partial R}{\partial \omega} - \csc i \frac{\partial R}{\partial \Omega} \right] \tag{4.68}$$

$$\dot{\omega} = \frac{\sqrt{1 - e^2}}{na^2 e} \frac{\partial R}{\partial e} - \frac{\cot i}{na^2 \sqrt{1 - e^2}} \frac{\partial R}{\partial i} \tag{4.69}$$

$$\dot{\Omega} = \frac{\csc i}{na^2 \sqrt{1 - e^2}} \frac{\partial R}{\partial i} \tag{4.70}$$

$$\dot{\sigma} = -\frac{(1 - e^2)}{na^2 e} \frac{\partial R}{\partial e} - \frac{2}{na} \frac{\partial R}{\partial a} \tag{4.71}$$

where $n = \sqrt{\mu / a^3}$ is the mean motion. While complicated, all of the coefficients in front of the partials of R are time-invariant, even though their values change over time, driven by these ordinary differential equations.

There are many variations on these equations. Common changes are to replace the argument of periapsis with the longitude of periapsis, $\varpi = \omega + \Omega$, to replace the expression σ with some a term that is a function of the mean anomaly, and to redefine the orbit elements to eliminate the terms e and $\sin i$ in the denominators.

Delaunay Variables

There is freedom in choosing the specific constants of motion used in the variation of parameters. Thus, one can ask if there are particular choices which can simplify the equations, or place them into a standard form. In fact, it is possible to do so, and in particular it is possible to choose the $\boldsymbol{\alpha}$ so that the transformation $\boldsymbol{r}(t,\boldsymbol{\alpha}), \boldsymbol{v}(t,\boldsymbol{\alpha})$ is canonical, as the Newtonian form of the equations of motion are themselves in Hamiltonian form. Then the matrix $[\boldsymbol{r}_{\boldsymbol{\alpha}}^T; \boldsymbol{v}_{\boldsymbol{\alpha}}^T]$ is symplectic by definition and $[[\alpha_i, \alpha_j]] = J$.

One particular choice of constants that enable such a transformation are called Delaunay variables. The resulting equation of motion can be stated as

$$\dot{\sigma} = -\frac{\partial R}{\partial L} \tag{4.72}$$

$$\dot{g} = -\frac{\partial R}{\partial H} \tag{4.73}$$

$$\dot{h} = -\frac{\partial R}{\partial G} \tag{4.74}$$

$$\dot{L} = \frac{\partial R}{\partial \sigma} \tag{4.75}$$

$$\dot{H} = \frac{\partial R}{\partial h} \tag{4.76}$$

$$\dot{G} = \frac{\partial R}{\partial g} \tag{4.77}$$

where the symbols G and H and g and h are traditionally switched in the Delaunay notation, respectively, σ is defined as before, $h = \omega$, the argument of periapsis, and $g = \Omega$, the longitude of the ascending node. These are usually termed the angles or the coordinates. The conjugate momenta are all functions of the orbital elements a, e and i as follows:

$$L = \sqrt{\mu a} \tag{4.78}$$

$$H = L\sqrt{1 - e^2} \tag{4.79}$$

$$G = H \cos i \tag{4.80}$$

where L is a direct function of the energy, H equals the angular momentum magnitude, and G is the angular momentum projected on the z-axis. To cast these variables into proper Hamiltonian form, define the system Hamiltonian as $\mathcal{H} = -R$, with the implicit assumption that the two-body potential acts on the original system.

A frequent change in these variables is to introduce a modified Hamiltonian:

$$\mathcal{H} = -\frac{\mu^2}{2L^2} - R \tag{4.81}$$

and replace the epoch σ term with the mean anomaly, $l = \int n\, dt + \sigma$. The new Hamiltonian system is then:

$$\dot{l} = \frac{\partial \mathcal{H}}{\partial L} \tag{4.82}$$

$$\dot{g} = \frac{\partial \mathcal{H}}{\partial H} \tag{4.83}$$

$$\dot{h} = \frac{\partial \mathcal{H}}{\partial G} \tag{4.84}$$

$$\dot{L} = -\frac{\partial \mathcal{H}}{\partial l} \tag{4.85}$$

$$\dot{H} = -\frac{\partial \mathcal{H}}{\partial h} \tag{4.86}$$

$$\dot{G} = -\frac{\partial \mathcal{H}}{\partial g} \tag{4.87}$$

4.4.2 Gauss Equations

The Gauss equations are usually derived either from the Lagrange equations or in a direct manner. An alternate approach is given here that is sometimes useful when deriving similar equations for nonstandard coordinates or problems. Now the perturbing acceleration is represented as an explicit vector $\boldsymbol{a}$. Thus, the equations of motion become:

$$\ddot{\boldsymbol{r}} = U_{\boldsymbol{r}} + \boldsymbol{a} \tag{4.88}$$

Again, the equations with $\boldsymbol{a} \equiv 0$ have a solution expressed as a function of a set of independent constants: $\boldsymbol{r}(t, \boldsymbol{\alpha})$ and $\boldsymbol{v}(t, \boldsymbol{\alpha})$.

Now assume that these functions have been inverted to find the constants of motion as a function of the particle position and velocity at a particular time:

$$\boldsymbol{\alpha} = \boldsymbol{\alpha}(\boldsymbol{r}, \boldsymbol{v}, t) \tag{4.89}$$

When the acceleration $\boldsymbol{a} \equiv 0$ the $\boldsymbol{\alpha}$ are constant and thus $\dot{\boldsymbol{\alpha}} = \boldsymbol{\alpha}_{\boldsymbol{r}} \cdot \boldsymbol{v} + \boldsymbol{\alpha}_{\boldsymbol{v}} \cdot U_{\boldsymbol{r}} + \boldsymbol{\alpha}_t$ $= 0$, where the substitutions $\dot{\boldsymbol{r}} = \boldsymbol{v}$ and $\dot{\boldsymbol{v}} = U_{\boldsymbol{r}}$ are used. The $\boldsymbol{\alpha}$ are again osculating elements.

Take the total time derivative of $\boldsymbol{\alpha}$ for the general case with a non-zero perturbing acceleration to find

$$\dot{\boldsymbol{\alpha}} = \frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{r}} \cdot \dot{\boldsymbol{r}} + \frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{v}} \cdot \dot{\boldsymbol{v}} + \frac{\partial \boldsymbol{\alpha}}{\partial t} \tag{4.90}$$

$$= \frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{r}} \cdot \dot{\boldsymbol{r}} + \frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{v}} \cdot [U_{\boldsymbol{r}} + \boldsymbol{a}] + \frac{\partial \boldsymbol{\alpha}}{\partial t} \tag{4.91}$$

However, the terms $\boldsymbol{\alpha}_{\boldsymbol{r}} \cdot \boldsymbol{v} + \boldsymbol{\alpha}_{\boldsymbol{v}} \cdot U_{\boldsymbol{r}} + \boldsymbol{\alpha}_t = 0$ by definition. Thus this yields the very simple, and general, form of the Gauss equations:

$$\dot{\boldsymbol{\alpha}} = \boldsymbol{\alpha}_{\boldsymbol{v}} \cdot \boldsymbol{a} \tag{4.92}$$

Thus, to compute these all one must do is define an appropriate set of integrals of motion, take their partials with respect to the velocity, and dot them with the perturbing acceleration vector.

Phrasing the Gauss equations in this form allows us to perform some derivations very easily. Consider the two-body problem with an applied acceleration and use the following integrals of motion, the two-body energy $\mathcal{E}$, the angular momentum vector $\boldsymbol{H}$, and the eccentricity (or Laplace) vector $\boldsymbol{e}$. These are defined in terms of the spacecraft state as:

$$\mathcal{E} = \frac{1}{2}\boldsymbol{v} \cdot \boldsymbol{v} - \frac{\mu}{|\boldsymbol{r}|} \tag{4.93}$$

$$\boldsymbol{H} = \boldsymbol{r} \times \boldsymbol{v} \tag{4.94}$$

$$\boldsymbol{e} = \frac{1}{\mu}\boldsymbol{v} \times \boldsymbol{r} \times \boldsymbol{v} - \frac{\boldsymbol{r}}{|\boldsymbol{r}|} \tag{4.95}$$

To apply this result one only needs take the partial of each of these with respect to the velocity vector and then dot the resulting quantity with the acceleration vector, yielding

$$\dot{\mathcal{E}} = \boldsymbol{v} \cdot \boldsymbol{a} \tag{4.96}$$

$$\dot{\boldsymbol{H}} = \tilde{\boldsymbol{r}} \cdot \boldsymbol{a} \tag{4.97}$$

$$\dot{\boldsymbol{e}} = \frac{1}{\mu}\left[\boldsymbol{a} \cdot \widetilde{(\boldsymbol{r} \times \boldsymbol{v})} + \tilde{\boldsymbol{v}} \cdot \tilde{\boldsymbol{r}} \cdot \boldsymbol{a}\right] \tag{4.98}$$

$$= \frac{1}{\mu}\left[2\boldsymbol{r}\boldsymbol{v} - \boldsymbol{v}\boldsymbol{r} - (\boldsymbol{r} \cdot \boldsymbol{v})\boldsymbol{U}\right] \cdot \boldsymbol{a} \tag{4.99}$$

Decomposing the acceleration vector into orthogonal components along the radius vector, S, along the angular momentum vector, W, and along the normal to these in the direction of travel, T, yields

$$\boldsymbol{a} = S\hat{\boldsymbol{r}} + T\hat{\boldsymbol{\theta}} + W\hat{\boldsymbol{H}} \tag{4.100}$$

Using this decomposition an explicit form for the time derivatives of these quantities is

$$\dot{\mathcal{E}} = \frac{H}{r}(\tan\gamma S + T) \tag{4.101}$$

$$\dot{\boldsymbol{H}} = r(T\hat{\boldsymbol{H}} - W\hat{\boldsymbol{\theta}}) \tag{4.102}$$

$$\dot{\boldsymbol{e}} = \frac{H}{\mu}\left[2T\hat{\boldsymbol{r}} - (S + \tan\gamma T)\hat{\boldsymbol{\theta}} - \tan\gamma W\hat{\boldsymbol{H}}\right] \tag{4.103}$$

In terms of orbital elements and integrals of motion the unit vectors have the following definitions:

$$\hat{\boldsymbol{r}} = \cos f \hat{\boldsymbol{e}} + \sin f \hat{\boldsymbol{e}}_\perp \tag{4.104}$$

$$\hat{\boldsymbol{\theta}} = -\sin f \hat{\boldsymbol{e}} + \cos f \hat{\boldsymbol{e}}_\perp \tag{4.105}$$

$$\hat{\boldsymbol{H}} = \frac{\boldsymbol{H}}{H} \tag{4.106}$$

$$\hat{\boldsymbol{e}} = \frac{\boldsymbol{e}}{e} \tag{4.107}$$

$$\hat{\boldsymbol{e}}_\perp = \hat{\boldsymbol{H}} \times \hat{\boldsymbol{e}} \tag{4.108}$$

while the radius r and $\tan\gamma$ have the usual two-body relations and are themselves functions of true anomaly f. It is interesting to note that when $e = 0$, where the Gauss equations are usually singular in argument of periapsis, that the equation for the Laplace vector is well defined, $\dot{\boldsymbol{e}}|_{e=0} = \frac{H}{\mu}\left[2T\hat{\boldsymbol{r}} - S\hat{\boldsymbol{\theta}}\right]$, and thus can serve as a non-singular coordinate. Similarly for the angular momentum in terms of inclination and longitude of the ascending node.

To derive the traditional Gauss equations from the above one only needs to express the integrals as functions of orbital elements and take time derivatives. The simplest example is for the energy, where $\mathcal{E} = -\frac{\mu}{2a}$ and $\dot{\mathcal{E}} = \frac{\mu}{2a^2}\dot{a}$ leading to

$$\dot{a} = \frac{2a^2}{\mu}\frac{H}{p}\left[e\sin f S + (1 + e\cos f)T\right] \tag{4.109}$$

which can be further simplified by substituting for H and p. Carrying out these evaluations for the classical orbital elements yields:

$$\dot{a} = \frac{2}{n\sqrt{1-e^2}}\left(e\sin f S + (1 + e\cos f)T\right) \tag{4.110}$$

$$\dot{e} = \frac{H}{\mu}\left[\sin f S + \left(\frac{e + \cos f}{1 + e\cos f} + \cos f\right)T\right] \tag{4.111}$$

$$\dot{i} = \frac{H}{\mu}\frac{\cos(\omega + f)}{1 + e\cos f}W \tag{4.112}$$

$$\dot{\omega} = \frac{H}{\mu e}\left[\frac{(2 + e\cos f)\sin f}{1 + e\cos f}T - \cos f S\right] - \cos i\dot{\Omega} \tag{4.113}$$

$$\dot{\Omega} = \frac{H}{\mu\sin i}\frac{\sin(\omega + f)}{1 + e\cos f}W \tag{4.114}$$

$$\dot{\sigma} = \frac{r}{na^2e}\left[(\cos f + e\cos^2 f - 2e)S - \sin f(2 + e\cos f)T\right] \tag{4.115}$$

A key item of contrast between the Lagrange and Gauss equations is in their coefficients. The Gauss equations have time-varying terms multiplying the accelerations involving the true anomaly, thus they must each be averaged separately.

5. Properties of Solution

This chapter discusses the properties of solutions to dynamical systems, and in particular focuses on Hamiltonian dynamical systems. The justification for this restriction is given below, but essentially arises from the fact that almost all of the relevant problems of interest to us can be transformed into this general form. We take advantage of this, as many of the fundamental properties that are of interest to us can be easily derived when the system is stated in Hamiltonian form.

5.1 Reduction to Time Invariant Hamiltonian Dynamical Systems

The reader is reminded that all of the generic equations of motion studied in the previous chapter can be transformed into a Hamiltonian formulation. Thus to study the general properties of these solutions it suffices to study the properties of Hamiltonian systems. First, note that the different formulations can all be transformed into a time-varying Hamiltonian system. Starting with the Newtonian formulation, the Hamiltonian function can be defined as the specific energy, independent of whether this is a conserved quantity. Thus $H = \frac{1}{2}\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} - \mathcal{U}(\boldsymbol{r}, t)$ with the coordinates being the inertial position vectors $\boldsymbol{r}$ and the momenta being the inertial velocities $\dot{\boldsymbol{r}}$. It is simple to verify that Hamilton's equations recover the Newtonian equations of motion. Thus the Newtonian formulation is trivially in Hamiltonian form.

Next consider the Lagrangian form of the dynamics, along with a Lagrangian function $L(\boldsymbol{q}, \dot{\boldsymbol{q}}, t)$. As explicitly derived, one can transform these directly into a Hamiltonian form by a simple transformation in which the coordinates remain fixed and the new momenta are $\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}}$. The Hamiltonian function is then constructed as $H(\boldsymbol{q}, \boldsymbol{p}, t) = \dot{\boldsymbol{q}} \cdot \boldsymbol{p} - L$, where the $\dot{\boldsymbol{q}}$ must be explicitly solved for in terms of the coordinates and momenta.

Finally, note that the Lagrange Planetary Equations can be rewritten into a canonical form, such as the Delaunay variables, as explicitly discussed in the previous chapter. Similar transformations can also be applied to the Gauss equations, although it is not always possible to recast the perturbing accelerations as a perturbing potential.

Once a system is placed into a Hamiltonian form, then any application of a canonical transformation will preserve the structure of its equations of motion. Thus, in a very real sense, the specific formulation of the problem does not matter as much, especially as it relates to generic properties of these systems.

Of specific interest are dynamical systems whose Hamiltonian function can be made time invariant by transformation to a special set of coordinates. If this can be done via a canonical transformation, then the system will have an integral of motion, called the Jacobi integral, associated with it. For a time invariant Hamiltonian the Jacobi integral is just the Hamiltonian function. The number of systems for which such a transformation exists is quite limited, however, so that for most general problems of interest the Hamiltonian will always be time varying or time periodic at best.

However, any time varying Hamiltonian can always be related to a time invariant Hamiltonian system, at the cost of raising the dimension of the problem by 2. The procedure for this is quite simple. Assume a Hamiltonian system with $\boldsymbol{q} \in \mathbf{R}^n$ and $\boldsymbol{p} \in \mathbf{R}^n$, and a time varying Hamiltonian $H(\boldsymbol{q}, \boldsymbol{p}, t)$. Introduce a new independent coordinate $\tau(t)$ with its inverse $t(\tau)$, so that $dt/d\tau = t'$, where the $'$ denotes differentiation with respect to the new independent parameter τ. It is possible to take τ identically equal to the time ($t' = 1$), or one can introduce some other scaling, but the resulting expression t' should be independent of τ in order to develop a time invariant system. Formally define the new time coordinate as $q_t = t$ and the new time-momentum as p_t. The new Hamiltonian can then be constructed as $\mathcal{H}(\boldsymbol{q}, q_t, \boldsymbol{p}, p_t) = (H(\boldsymbol{q}, \boldsymbol{p}, q_t) + p_t)\, t'$. Note that this now provides the equations of motion with τ as the new independent variable. Computing Hamilton's equations of motion

$$\boldsymbol{q}' = \frac{\partial H}{\partial \boldsymbol{p}} t' \tag{5.1}$$

$$q_t' = t' \tag{5.2}$$

$$\boldsymbol{p}' = -\frac{\partial H}{\partial \boldsymbol{q}} t' \tag{5.3}$$

$$p_t' = -\frac{\partial H}{\partial q_t} t' \tag{5.4}$$

The original equations of motion for $\boldsymbol{q}$ and $\boldsymbol{p}$ are recovered with the new independent parameter. The time equation is trivially satisfied given the specified relation between τ and t. Finally, note that the time variation of the time-momentum equals the negative time partial of the Hamiltonian. Writing this out in differential form yields $dp_t = -H_t\, dt$ which can be integrated to find

$$p_t(t) - p_t(t_o) = -H(\boldsymbol{q}(t), \boldsymbol{p}(t), t) + H(\boldsymbol{q}(t_o), \boldsymbol{p}(t_o), t_o) \tag{5.5}$$

where we have shifted back to the original time notation. Rearranging exposes the new conservation relation

$$p_t(t) + H(t) = p_t(t_o) + H(t_o) \tag{5.6}$$

using an abbreviated notation. Thus, the new Hamiltonian $\mathcal{H} = p_t + H$ is formally time invariant and has a conservation relation defined for it. The time-momentum equals the negative of the original time-varying Hamiltonian plus an arbitrary constant.

Due to the existence of this result, one can formally treat any Hamiltonian system as a time invariant system. This allows us to only consider this case in these detailed discussions, although the time varying form will still be used where appropriate.

5.2 Properties of General Trajectories

Given a general solution to a Hamiltonian dynamical system there are a number of fundamental results that can be ascribed to the solution flow. While some of these properties also exist for arbitrary dynamical systems, Hamiltonian systems have additional properties that make them distinguished.

Denote the solution to a Hamiltonian system that passes through a point of phase space $\boldsymbol{x}_o$ at an epoch t_o to be $\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}_o, t_o)$. Assume the ordering $\boldsymbol{x} = [\mathbf{q}, \mathbf{p}]$. Thus, the solution flow $\phi(t)$ satisfies the Hamiltonian equations of motion:

$$\frac{\partial \phi}{\partial t} = J \left. \frac{\partial H}{\partial \boldsymbol{x}} \right|_{x(t)=\phi(t)} \tag{5.7}$$

By definition, a trajectory is a canonical transformation between the state $\boldsymbol{x}_o$ and the state $\boldsymbol{x}$, with time t being an arbitrary parameter of the transformation. Thus, the Jacobian of the solution must satisfy the property that $\Phi(t, t_o; \boldsymbol{x}_o) = \partial \boldsymbol{x}(t)/\partial \boldsymbol{x}_o$ be a symplectic matrix (the functional dependence of Φ on the state will often be suppressed). This provides a few important technical results which lead to additional functional results. The basic technical result is that $|\Phi| = 1$, namely that the solution flow is never singular and can always be solved for the inverse of the flow. Denote this inverse using the solution flow description as $\boldsymbol{x}_o = \phi(t_o; \boldsymbol{x}, t)$. Further, for a linear dynamical system where the solution has the form $\boldsymbol{x}(t) = \Phi(t, t_o)\boldsymbol{x}_o$ this inversion can be carried out in closed form as

$$\Phi(t_o, t) = \Phi^{-1}(t, t_o) \tag{5.8}$$

$$\Phi^{-1}(t, t_o) = -J\Phi(t, t_o)^T J \tag{5.9}$$

from the properties of symplectic matrices (Appendix D).

5.2.1 Initial Conditions and Higher-Order Expansions

Solutions of Hamilton dynamical systems are analytic in initial conditions over finite timespans, even for chaotic systems. Thus, one can expand a nominal solution into a higher-order series in order to generate a local characterization of the solution flow. This is a familiar technique used for linear variations about a nominal trajectory, but can be expanded to higher orders. The main issue is to develop a notation that supports such higher-order expansions, as a matrix formulation cannot be unambiguously applied beyond the first order. Thus for this section an index notation is introduced, representing the state at a time t by

$$x_i(t) = \phi_i(t; t_o, \boldsymbol{x}_o) \tag{5.10}$$

$$i = 1, 2, \ldots, 2n \tag{5.11}$$

Using this notation, the partial of x_i with respect to the initial conditions $x_j(t_o)$ will be denoted as $x_{i,j}$, with the implication that the sub j index after the comma implies a partial with respect to the initial state. Specifically:

$$\frac{\partial \phi_i(t; t_o, \boldsymbol{x}_o)}{\partial x_j(t_o))} = \phi_{i,j}(t; t_o, \boldsymbol{x}_o) \tag{5.12}$$

with its natural generalizations to higher orders. For the first partial, note the correspondence between $\Phi = \phi_{i,j}$, where i denotes the row and j denotes the column.

Using this notation the function $\phi(t; t_o, \boldsymbol{x}_o + \delta\boldsymbol{x}_o)$ can be expanded about the nominal initial condition $\boldsymbol{x}_o$ to find:

$$\begin{aligned} x_i(t) + \delta x_i(t) = &\phi_i(t; t_o, \boldsymbol{x}_o) + \phi_{i,j}(t; t_o, \boldsymbol{x}_o)\delta x_j(t_o) \\ &+ \frac{1}{2!}\phi_{i,jk}(t; t_o, \boldsymbol{x}_o)\delta x_j(t_o)\delta x_k(t_o) + \ldots \end{aligned} \tag{5.13}$$

where the Einstein summation convention is used. The higher-order partials $\phi_{i,jk\ldots n}$ are called “state transition tensors” in [124] and can be computed analytically if a closed-form solution exists to the dynamical system, or numerically as discussed in the next chapter.

Given a nominal trajectory $x_i(t)$ these results allow us to describe motion in the neighborhood of that trajectory both forward and backwards in time. The forward solution is trivially inferred from the above and equals:

$$\delta x_i(t) = \phi_{i,j}(t; t_o, \boldsymbol{x}_o)\delta x_j(t_o) + \frac{1}{2!}\phi_{i,jk}(t; t_o, \boldsymbol{x}_o)\delta x_j(t_o)\delta x_k(t_o) + \ldots \tag{5.14}$$

With such a series it is possible to perform a reversion of series (i.e., invert the series to solve for $\boldsymbol{x}_o$), so long as the leading linear term is invertible. This is guaranteed by the symplectic nature of the state transition matrix. Carrying out this reversion to the first two orders yields:

$$\delta x_i(t_o) = \psi_{i,j}\delta x_j(t) + \frac{1}{2!}\psi_{i,jk}\delta x_j(t)\delta x_k(t) + \dots \tag{5.15}$$

where $\psi_{i,j} = \phi_{i,j}^{-1} = -J_{il}\phi_{k,l}J_{kj}$, and $\psi_{i,jk} = -\phi_{il}^{-1}\psi_{l,mn}\phi_{mo}^{-1}\phi_{np}^{-1}$, with the higher-order terms being more complex (see [124] for a more complete discussion).

5.2.2 Solutions Analytic in a Parameter

Solutions can also be viewed as functions of free parameters in the system. Given a dynamical system defined as

$$\dot{\boldsymbol{x}} = JH_{\boldsymbol{x}}(\boldsymbol{x}, t; \mu), \tag{5.16}$$

where μ represents a parameter of the system, one can define a family of solutions as a function of this independent parameter μ. The main assumption is that the Hamiltonian function is analytic in the parameter μ, which is generally the case for physically defined problems. Examples of such parameters include masses or reduced masses of attracting bodies, physically defined parameters of a spacecraft or system that can take on different values, or even values of integrals of motion of the system. One common property that parameters have is that they are constant in time. Time-varying parameters can also be studied, under some assumptions, using the theory of adiabatic invariants (see [4]), but are not considered here.

Generally, for a specific dynamical system these parameters would not be varied, but instead would just be chosen and assigned a specific value. However, it is often convenient to allow these parameters to vary in order to understand the robustness of a particular solution to slightly different models, or to understand the global stability properties of a dynamical system across all values of a given parameter. An example of the latter would be varying the mass fraction in the restricted three-body problem from 0 to 1/2 in order to determine the stability properties of a certain class of motion.

One specific application is to the continuation of a family of periodic orbits as a parameter is varied. Consider a solution to Eq. 5.16, denoted as $\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}_o, t_o, \mu)$. Such a solution can be found by numerical integration or by analytic solution. If they are obtained analytically, one can immediately describe the properties of the solution as the parameter is varied. For a numerically computed solution this description is not as easy, but can be derived directly from the equations of motion combined with a numerical integration. Consider a variation in the parameter μ:

$$\boldsymbol{x}(t) + \delta\boldsymbol{x}_\mu(t) = \phi(t; \boldsymbol{x}_o, t_o, \mu + \delta\mu) \tag{5.17}$$

$$= \phi(t; \boldsymbol{x}_o, t_o, \mu) + \frac{\partial \phi}{\partial \mu}\delta\mu + \dots \tag{5.18}$$

At the leading order the desired variation in the solution can be found

$$\delta \boldsymbol{x}_\mu(t) = \frac{\partial \phi}{\partial \mu}\delta\mu + \dots \tag{5.19}$$

and note that this variation is controlled by the partial $\phi_\mu = \frac{\partial \phi}{\partial \mu}$. Computation of this partial is most precise if the variational equation is integrated along the solution. Applying standard linearization approaches yields

$$\frac{d}{dt}\phi_\mu = JH_{xx}\phi_\mu + JH_{\boldsymbol{x}\mu} \tag{5.20}$$

where the $H_{\boldsymbol{xx}}$ and $H_{\boldsymbol{x}\mu}$ terms are evaluated along the nominal solution contemporaneously with ϕ_μ. The initial conditions for this solution is simply $\phi_\mu = 0$.

These equations are, in general, non-homogenous, time-varying differential equations. Given a state transition matrix for the system, $\Phi(t, t_o)$, a general solution for the partial derivative can be written out as

$$\phi_\mu(t; \boldsymbol{x}_o, t_o, \mu) = \int_{t_o}^{t} \Phi(t,\tau) H_{\boldsymbol{x}\mu}(\boldsymbol{x}(\tau), \tau, \mu)\, d\tau \tag{5.21}$$

From a practical perspective, it is often easiest to directly solve the differential equation Eq. 5.20 for ϕ_μ along with the nominal solution and the state transition matrix.

These computations can be generalized to higher-order systems of equations following the approaches previously outlined.

5.2.3 Eigenstructure of the State Transition Matrix

The symplecticity of $\Phi \in \mathbf{R}^{6\times 6}$ has important consequences for the structure of this matrix, which in turn have important implications for Hamiltonian dynamical systems. A matrix is defined, most generally, in terms of its eigenstructure, which can be considered to consist of its eigenvalues, right eigenvectors and left eigenvectors, and potentially generalized eigenvectors. The right eigenvector of an eigenvalue λ_i is denoted as $\boldsymbol{u}_i \in \mathbf{R}^6$ and satisfies $\Phi \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i$. A left eigenvector of the same eigenvalue is denoted as $\boldsymbol{v}_i \in \mathbf{R}^6$ and satisfies $\Phi^T \boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$. In general, it is possible to represent a matrix without repeated eigenvalues in terms of its eigenvalues and eigenvectors as:

$$\Phi = \sum_{i=1}^{2n} \lambda_i \boldsymbol{u}_i \boldsymbol{v}_i^T \tag{5.22}$$

$$= [U][\lambda][V]^T \tag{5.23}$$

where $[U]$ and $[V]$ are the right and left eigenvectors arranged as columns, respectively, and $[\lambda]$ is a diagonal matrix with the eigenvalues along its diagonal. The more general case of repeated eigenvalues is discussed later.

Eigenvalues

The eigenvalues of Φ can be discussed using the defining equation

$$\|\lambda I - \Phi\| = 0 \tag{5.24}$$

where $\| - \|$ denotes the determinant and I is the 6×6 identity matrix. Given that Φ is symplectic, and thus has non-zero eigenvalues, it is easy to show, using the inverse form of Φ (see Appendix D), that

$$\left\|\lambda^{-1} I - \Phi\right\| = 0 \tag{5.25}$$

where $\lambda\lambda^{-1} = 1$. Thus, every eigenvalue will have a matching inverse. For a general complex eigenvalue we can describe it as $\lambda = \rho\, e^{i\theta}$. Then its inverse has the description $\lambda^{-1} = \frac{1}{\rho}\, e^{-i\theta}$, where ρ is a real number, i is the imaginary number and e^x is the exponential function. One fundamental implication of this is that the characteristic polynomial for any symplectic Φ is symmetric, i.e., has the form:

$$\lambda^{2n} + a_1\lambda^{2n-1} + \ldots + a_m\lambda^{2n-m} + \ldots + a_m\lambda^m + \ldots + a_1\lambda + 1 = 0 \tag{5.26}$$

Thus every entry in the polynomial of form $a_m\lambda^{2n-m}$ has a counterpart $a_m\lambda^m$. Since this reciprocal relationship holds for arbitrary values of the a_i, it follows that when a symplectic matrix Φ has a unity eigenvalue, that these must also appear in pairs. In this case the eigenvectors can become more difficult to compute, and will be discussed later. Another constraint arises in the fact that Φ is a real matrix, and thus all complex eigenvalues must have a complex conjugate counterpart. Specifically, if there exists an eigenvalue λ_i which has a complex part, then $\bar{\lambda}_i$, its complex conjugate, must also be an eigenvalue.

The most general situation occurs for an eigenvalue that is complex and not of unit magnitude, $\lambda = \rho\, e^{i\theta}$, where ρ and θ are real. Then the inverse eigenvalue is $\frac{1}{\rho}\, e^{-i\theta}$, the complex conjugate of the original eigenvalue is $\rho\, e^{-i\theta}$, and the complex conjugate of the inverse is $\frac{1}{\rho}\, e^{i\theta}$, forming a set of four eigenvalues. If an eigenvalue is real, then $\lambda = \rho$ and identically equals its complex conjugate and its inverse is just the reciprocal, $\lambda^{-1} = 1/\rho$, forming a pair. If an eigenvalue has a unit magnitude, then $\lambda = e^{i\theta}$ and its complex conjugate is equal to its inverse, or $\lambda^{-1} = \bar{\lambda} = e^{-i\theta}$, again forming a pair.

The usual approach to representing the eigenvalues of a symplectic matrix is on the complex plane, with the unit circle serving to orient the values. As any free parameter of the dynamical system is varied, the eigenvalues of the associated state transition matrix will evolve on the complex plane. Due to their strict constraints, the eigenvalues can only transition between different classes by intersecting each other either on the unit circle or on the real axis.

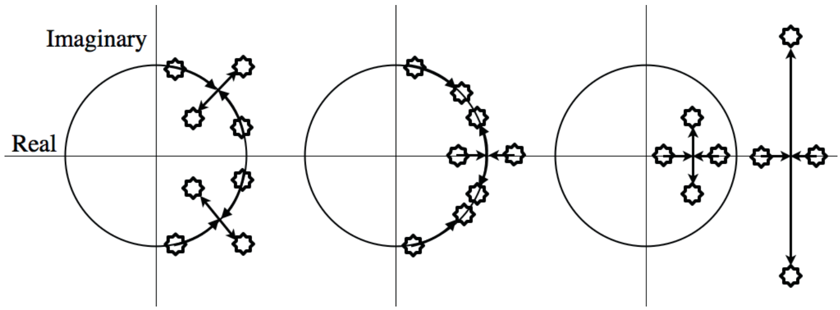

Fig. 5.1 Eigenvalues of the state transition matrix plotted on the complex plane, relative to the unit circle. The three generic types of transitions for a four-dimensional symplectic matrix are shown.

Eigenvectors

Next, given an eigenvalue of Φ, λ_i, consider its right eigenvector $\boldsymbol{u}_i$, defined by:

$$\Phi \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i \tag{5.27}$$

Multiplying each side of the equation by Φ^{-1}, substituting the form for the inverse and simplifying yields:

$$\lambda_i^{-1} J \boldsymbol{u}_i = \Phi^T J \boldsymbol{u}_i \tag{5.28}$$

Thus, the left eigenvector of the inverse eigenvalue to λ_i is equal to the right eigenvalue of λ_i times the symplectic matrix J, or $\boldsymbol{v}_{-i} = J\boldsymbol{u}_i$, where we introduce a temporary subscript notation where $\lambda_i^{-1} = \lambda_{-i}$. Similarly the right eigenvector to λ_i^{-1}, denoted as $\boldsymbol{u}_{-i}$, serves as the generator to the left eigenvector of λ_i, or $\boldsymbol{v}_i = J\boldsymbol{u}_{-i}$.

These associated left and right eigenvectors are orthogonal to each other as $\boldsymbol{u}_i \cdot \boldsymbol{v}_{-i} = \boldsymbol{u}_i J \boldsymbol{u}_i \equiv 0$ from the skew-symmetric property of J. This can be generalized even further, borrowing from [5]. Consider an arbitrary left and right eigenvalue and eigenvector, λ_i and $\boldsymbol{u}_i$ and λ_j and $\boldsymbol{v}_j$, assuming Φ does not have repeated eigenvalues. Taking the right eigenvector definition $\Phi\boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i$, multiplying on the left by $\boldsymbol{v}_j$, and bringing all terms to one side yields the relationship:

$$(\lambda_j - \lambda_i)\boldsymbol{v}_j \cdot \boldsymbol{u}_i = 0 \tag{5.29}$$

If $i \neq j$ then $\lambda_j \neq \lambda_i$ and the left and right eigenvectors must be orthogonal to each other. For the case $i = j$ there is no explicit constraint from this result. However, in that case one knows from linear independence of the eigenvectors (for a non-repeated root) that $\boldsymbol{v}_i$ and $\boldsymbol{u}_i$ cannot be orthogonal to each other and can be properly scaled so that $\boldsymbol{v}_i \cdot \boldsymbol{u}_i = 1$. This provides the fundamental result:

$$\boldsymbol{v}_j \cdot \boldsymbol{u}_i = \delta_{ij} \tag{5.30}$$

where $\delta_{ij} = 1$ if $i = j$ and $= 0$ of $i \neq j$. This fits with the decomposition of the matrix given above, as $[U][\lambda][V]^T[U] = [U][\lambda]$ and $[V]^T[U][\lambda][V]^T = [\lambda][V]^T$.

The form is not so simple if there is a repeated eigenvalue without two linearly independent eigenvectors. For Hamiltonian systems this situation occurs generically for the state transition matrix evaluated over one period of a periodic orbit in a system which can be canonically transformed into a time-invariant system. This is discussed explicitly later when periodic orbits are considered.

5.3 Conservation Principles

5.3.1 Nonlinear Integrals

An integral of motion is any function of the state at a given epoch, and perhaps of the epoch itself, such that its value is constant as the state moves along the solution flow. Specifically, given some function $h(\boldsymbol{x}, t)$, this is an integral of motion if it is constant along the flow. For this to hold, the total time derivative of this quantity must equal zero, or

$$\frac{dh}{dt} = \frac{\partial h}{\partial \boldsymbol{x}} \cdot \dot{\boldsymbol{x}} + \frac{\partial h}{\partial t} \tag{5.31}$$

$$= \frac{\partial h}{\partial \boldsymbol{x}} \cdot J \cdot \frac{\partial H}{\partial \boldsymbol{x}} + \frac{\partial h}{\partial t} \tag{5.32}$$

$$= 0 \tag{5.33}$$

Since the first term is independent of the specific functional form in which the state is specified, assuming a canonical transformation, the time derivative can be rewritten entirely in terms of the generic Hamiltonian of the system

$$\dot{h} = \{h, H\} + \frac{\partial h}{\partial t} \tag{5.34}$$

where $\{.,.\}$ denote Poisson brackets and are defined as

$$\{h, H\} = \frac{\partial h}{\partial \boldsymbol{x}} \cdot J \cdot \frac{\partial H}{\partial \boldsymbol{x}} \tag{5.35}$$

For h to be an integral of motion the following must be true

$$\{h, H\} + \frac{\partial h}{\partial t} = 0 \tag{5.36}$$

The simplest example of this is often the Hamiltonian function itself. If a transformation can be found that places the Hamiltonian into a time invariant form, then the system in question has an integral of motion in the Hamiltonian function. This is clear as the quantity $\{H, H\}$ is a scalar and hence must be symmetric, yet its transpose must also equal its negative as $J^T = -J$, making $\{H, H\} = 0$. Thus

$$\dot{H} = \frac{\partial H}{\partial t} \tag{5.37}$$

and only equals zero if the Hamiltonian is independent of time. If true, this is called a Jacobi integral, although we will also use the more colloquial term of "energy integral." It is not always apparent that a system has such an integral, although when it does this has important implications for the solution of the system. For instance, if the circular restricted three-body problem is posed in a Newtonian formulation it is not evident from the Hamiltonian that the system has an energy integral. It is only when a transformation into a uniformly rotating frame of the proper period is enacted that it becomes obvious. Despite this, the solutions of the Newtonian formulation have constraints on them, arising from the integral of motion, just as solutions of the rotating formulation do. It is important to note that a Jacobi integral exists for any Lagrangian system that can be transformed into a frame where the Lagrangian is time invariant. The form of this integral is just the Hamiltonian from the Legendre transformation: $\dot{\boldsymbol{q}} \cdot L_{\dot{\boldsymbol{q}}} - L$.

Another important form of a conservation principle occurs when the system can be transformed into a Hamiltonian in which some coordinates or momenta are missing. This usually only occurs for a system with a physical symmetry or one with an unreduced conservation principle. The simplest manner in which these conservation laws appear is if a specific coordinate is absent from the Hamiltonian, say q_I. Then $\dot{p}_I = -\partial H/\partial q_I = 0$, and p_I is a constant. The coordinate is not necessarily constant in this case, as H may depend on p_I. The converse case is if the Hamiltonian does not contain a certain momentum p_I, as then $\dot{q}_I = \partial H/\partial p_I = 0$, and the coordinate itself is a constant. As coordinates and momenta can always be swapped via a canonical transformation (see [56]), one can always change this situation into one where the momenta is conserved. In all cases, conservation of linear or angular momentum can generally be placed into this form.

This definition of an integral can even be applied to the Jacobi integral described above. Specifically, this tells us that if the Hamiltonian is taken as a momentum of the system, the conjugate coordinate to it will be the time. Thus, when a form of the Hamiltonian can be found that is independent of time, the Jacobi integral is recovered. The time-invariance transformation also falls into place, as in the absence of a transformation that makes the current Hamiltonian time-invariant, one can just define this time-varying quantity as the new momentum, define a new time, and trivially construct a new Hamiltonian that has a conservation law defined for it.

5.3.2 Initial Conditions as Local Integrals of Motion

Every trajectory has a full set of integrals of motion defined for it, which is just the state of the system at a specified epoch. Consider the general solution function, $\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}_o, t_o)$ and its inverse $\boldsymbol{x}_o = \phi(t_o; \boldsymbol{x}, t)$. Taking the total time derivative of $\boldsymbol{x}_o$ with respect to t yields

$$\frac{d\boldsymbol{x}_o}{dt} = \frac{\partial\phi(t_o;\boldsymbol{x}(t),t)}{\partial\boldsymbol{x}(\boldsymbol{t})} \cdot \frac{d\boldsymbol{x}}{dt} + \frac{\partial\phi(t_o;\boldsymbol{x}(t),t)}{\partial t} \tag{5.38}$$

For a time invariant system it can be shown that $\frac{\partial\phi(t_o;\boldsymbol{x}(t),t)}{\partial t} = -\dot{\boldsymbol{x}}(t_o)$, and substituting the Hamiltonian equations for the state time derivatives defines the identity

$$\Phi(t_o,t)JH_{\boldsymbol{x}}|_t - JH_{\boldsymbol{x}}|_o = 0 \tag{5.39}$$

Introducing the inverse identity $\Phi(t_o,t) = \Phi^{-1}(t,t_o) = -J\Phi(t,t_o)^T J$ the relation can be reduced to

$$-H_{\boldsymbol{x}}|_o + \Phi^T(t,t_o)H_{\boldsymbol{x}}|_t = 0 \tag{5.40}$$

which is identically satisfied. Thus, the total time derivative of the initial conditions is identically zero and every trajectory formally has a complete set of integrals defined for it in any given state defined at an epoch. This is a fundamental result for any dynamical system. When combined with an expansion of a solution in terms of its initial conditions in the vicinity of a nominal solution, this implies that one can construct local "integrable" solutions for motion over finite timespans in the vicinity of a known solution.

However, while each set of initial conditions defines a set of integrals, the result is only local and does not translate to a global characterization in general. To do that would require the convergence of these local solutions over a large, if not global, domain. In the absence of such convergence, this result does not provide us the global characterization associated with integrable problems. Despite this limitation, such expansions and time-limited solutions are frequently sufficient for practical purposes [124].

5.3.3 Linear Integrals

The existence of an integral of motion places some additional constraints on the linearized solution flow. Assuming that $h(\boldsymbol{x},t) = C$ is an integral of the motion, note the following results and identities:

$$h(\boldsymbol{x}_o,t_o) = h(\phi(t;\boldsymbol{x}_o,t_o),t) = C \tag{5.41}$$

$$h(\boldsymbol{x}_o+\delta\boldsymbol{x},t_o) = h(\phi(t;\boldsymbol{x}_o+\delta\boldsymbol{x},t_o),t) \tag{5.42}$$

$$C + \left.\frac{\partial h}{\partial\boldsymbol{x}}\right|_{t_o} \cdot \delta\boldsymbol{x} = C + \left.\frac{\partial h}{\partial\boldsymbol{x}}\right|_t \cdot \Phi(t,t_o;\boldsymbol{x}_o)\cdot\delta\boldsymbol{x} \tag{5.43}$$

where higher-order terms are ignored. As this result must hold for all possible values of $\delta\boldsymbol{x}$, it follows that

$$\left.\frac{\partial h}{\partial\boldsymbol{x}}\right|_{t_o} = \left.\frac{\partial h}{\partial\boldsymbol{x}}\right|_t \cdot \Phi(t,t_o;\boldsymbol{x}_o) \tag{5.44}$$

The left-hand side is constant by definition, as it is evaluated at the specified epoch of the solution, and thus the right-hand side defines an associated integral of motion for the state transition matrix Φ, and was originally discussed by Poincaré [131]. This will be of particular interest when special solutions to the equations of motion are considered, such as equilibrium points and periodic orbits. It is also in similar in form to Eq. 5.40.

5.4 Equilibrium Points and Stability

Now the simplest solutions to the equations of motion are considered, those in which the dynamics at a particular point in phase space are identically zero, or $\dot{\boldsymbol{x}} = 0$. The existence of an equilibrium point often implies that the Hamiltonian function is time-invariant, although exceptions do exist, as will be seen in Chapter 12. The following will only consider time-invariant systems.

5.4.1 Form of the State Transition Matrix

Suppose that a Hamiltonian function $H(\boldsymbol{x})$ has an equilibrium point, i.e., there exists a point $\boldsymbol{x}^*$ such that $\partial H/\partial \boldsymbol{x}|_{\boldsymbol{x}^*} = 0$. Then the dynamics at this point in phase space are null, or $\dot{\boldsymbol{x}}^* = 0$, and the point ideally remains stationary, or $\phi(t; t_o, \boldsymbol{x}^*) = \boldsymbol{x}^*$. The main question then progresses to what the motion of the system is when slightly perturbed from this point in phase space. A related question is whether a given equilibrium point is stable in the sense of Lyapunov or not, and can be answered if the motion in the vicinity of the equilibrium point can be described. Formally, the solution can be characterized as $\boldsymbol{x}^* + \delta\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}^* + \delta\boldsymbol{x}_o, t_o)$. Assuming that $\delta\boldsymbol{x}_o$ is an infinitesimal quantity, a Taylor Series expansion of the solution function can be performed to find $\delta\boldsymbol{x}(t) = \Phi(t, t_o) \cdot \delta\boldsymbol{x}_o$. The associated equation of motion for the state transition matrix is found to be:

$$\dot{\Phi} = J\, H_{\boldsymbol{xx}}|_{\boldsymbol{x}^*}\, \Phi \tag{5.45}$$

where the matrix $JH_{\boldsymbol{xx}}$ is constant as it is evaluated at the equilibrium point. Thus the solution can be expressed in closed form using the exponential matrix

$$\begin{aligned} \Phi(t, t_o) &= \sum_{m=0}^{\infty} \frac{1}{m!}(t - t_o)^m [JH_{\boldsymbol{xx}}]^m && (5.46) \\ &= e^{(t-t_o)JH_{\boldsymbol{xx}}} && (5.47) \end{aligned}$$

Application of this result relies on the dynamics matrix being constant.

5.4.2 Eigenstructure of the State Transition Matrix

A few properties of the eigenvectors and eigenvalues of the state transition matrix Φ are now established. First note that if a vector $\boldsymbol{u}$ is an eigenvector of the dynamics matrix $JH_{\boldsymbol{xx}}$, then it is an eigenvector of the state transition matrix Φ. Specifically, let $JH_{\boldsymbol{xx}}\boldsymbol{u} = \sigma\boldsymbol{u}$ where σ is a complex number in general. Then, multiplying the state transition matrix by $\boldsymbol{u}$ and using the exponential equation yields

$$\Phi\boldsymbol{u} = \sum_{m=0}^{\infty} \frac{1}{m!}(t - t_o)^m [JH_{\boldsymbol{xx}}]^m \boldsymbol{u} \tag{5.48}$$

$$= \left(\sum_{m=0}^{\infty} \frac{1}{m!}(t - t_o)^m \sigma^m \right) \boldsymbol{u} \tag{5.49}$$

$$= e^{(t-t_o)\sigma}\boldsymbol{u} \tag{5.50}$$

and thus the associated eigenvalue to an eigenvector $\boldsymbol{u}$ has the form $\lambda = e^{(t-t_o)\sigma}$, linking the eigenstructure of $JH_{\boldsymbol{xx}}$ with that of Φ. From the properties of eigenvalues of a symplectic matrix, one can also infer the properties of the eigenvalues of the dynamics matrix.

Let us write $\sigma = \alpha + i\beta$, where α is the real part of the eigenvalue and β is the imaginary part, with $i = \sqrt{-1}$. Then, from the symplectic nature of Φ it is known that if $e^{(t-t_o)(\alpha+i\beta)}$ is an eigenvalue then so must its inverse, $e^{(t-t_o)(-\alpha-i\beta)}$, its complex conjugate, $e^{(t-t_o)(\alpha-i\beta)}$, and the inverse of its complex conjugate, $e^{(t-t_o)(-\alpha+i\beta)}$. Thus if $\sigma = \alpha + i\beta$ is an eigenvalue of $JH_{\boldsymbol{xx}}$ then so must $-\alpha - i\beta$, $\alpha - i\beta$, and $-\alpha + i\beta$. Note that if either α or β are zero, than the eigenvalues only come in pairs, as $\pm\alpha$ or $\pm i\beta$. For a 1-DOF system $\Phi \in \mathbf{R}^{2x2}$ and it can only have two eigenvalues, meaning that either $\alpha = 0$ or $\beta = 0$. For a 2-DOF or higher system $\Phi \in \mathbf{R}^{2n\times 2n}$, $n \geq 2$, and the more general case can occur as well. However, if the total state $2n$ is not divisible by 4, i.e., if n is odd, then at least one pair of eigenvalues must be pure real or pure imaginary.

There are, generically, three different possibilities that can occur for the eigenvalues, each of which has specific consequences for motion close to an equilibrium point. Each of these are reviewed in turn in the following.

Real Eigenvalues of $JH_{\boldsymbol{xx}}$

If σ is a real eigenvalue then there are two eigenvalues with equal magnitude and opposite signs, $\pm\alpha$. The corresponding eigenvalues of the state transition matrix are then $e^{\pm\alpha(t-t_o)}$, and there are both asymptotically stable and unstable motions in the vicinity of the equilibrium point. This is a hallmark of Hamiltonian systems and means that they cannot exhibit generic asymptotic stability. The associated eigenvectors, $\boldsymbol{u}_\pm$, are real and each eigenvalue/eigenvector pair defines a one-dimensional manifold for motion close to the equilibrium point:

$$\delta\boldsymbol{x}_\pm(t) = a\, e^{\pm\alpha(t-t_o)}\boldsymbol{u}_\pm \tag{5.51}$$

where a is an arbitrary amplitude, positive or negative, on these manifolds. The positive solution $\delta\boldsymbol{x}_+$ is unstable and will depart the equilibrium point at an exponentially increasing rate in time. This is often characterized in terms of the characteristic time of the exponential expansion, $\tau = 1/\alpha$. For example, in the Earth–Sun system the $L_{1,2}$ equilibrium points have a characteristic instability time on the order of 23 days, while in the Earth–Moon system the same points have an instability time on the order of 2 days. If the linear unstable manifold is propagated forwards in time using a numerical integration scheme, the manifold continues into a nonlinear trajectory that can stray far from the equilibrium point. Conversely, the negative solution $\delta\boldsymbol{x}_-$ is asymptotically stable as time increases and will approach the equilibrium point with the same characteristic time. To compute the nonlinear stable manifold, one starts a point on the linear stable manifold and integrates backwards in time to find where the trajectory has come from, which again can stray far from the equilibrium point. These properties of the stable and unstable manifolds, that they stray far from the equilibrium points, have been used in practical applications for transferring spacecraft to and from libration points in the Earth–Sun system and, more recently, in the Earth–Moon system. They have even been characterized as "super-highways" in the solar system, although that terminology is perhaps a bit overreaching.

Imaginary Eigenvalues of $\boldsymbol{J}\boldsymbol{H}_{\boldsymbol{x}\boldsymbol{x}}$

If σ is an imaginary eigenvalue then there are two eigenvalues with equal magnitudes and opposite signs, $\pm i\beta$ (where i denotes the imaginary unit in the next few paragraphs). The eigenvalues of the state transition matrix are then $e^{\pm i\beta(t-t_o)}$ and both lie on the unit circle. From Euler's formula note that the eigenvalues can also be expressed as $\cos(\beta(t-t_o)) \pm i\sin(\beta(t-t_o))$, and thus that these eigenvalues represent solutions that oscillate about the equilibrium point. This is a weak form of Lyapunov stability for a linear dynamical system, with no guarantees that when nonlinear dynamics are taken into account that the the solution will remain bounded. The question of nonlinear stability is much more difficult and few results exist. The fundamental result in this area is the Kolmogorov–Arnold–Moser Theorem [4] which places conditions on an equilibrium point, or periodic orbit, under which it can be nonlinearly stable. Conversely, Marchal has shown that a linearly stable point can at most have a nonlinear instability that is polynomial in time, and not exponential [97]. Since practical applications are generally only interested in finite time solutions, Marchal's result is taken as a motivation for accepting linear stability as practically sufficient for stability.

Since the eigenvalues are imaginary, the eigenvectors must be complex with $\boldsymbol{u}_\pm = \boldsymbol{u}_R \pm i\boldsymbol{u}_I$. Taken together, these eigenvectors define a 2-D surface around the equilibrium point filled with closed trajectories, called a center manifold. The center manifold can be expressed in terms of real numbers as

$$\delta\boldsymbol{x}_C(t) = 2a\left[\cos(\beta(t-t_o)+\varphi)\boldsymbol{u}_R - \sin(\beta(t-t_o)+\varphi)\boldsymbol{u}_I\right] \tag{5.52}$$

where a is again an arbitrary amplitude (now chosen to be positive in general) and φ is an initial phase angle. Together these two parameters trace out the two-dimensional center manifold. This can be seen more clearly if the arbitrary phase angle is extracted to find

$$\begin{aligned}\delta \boldsymbol{x}_C(t) = 2a\cos\varphi\,[\cos(\beta(t-t_o))\boldsymbol{u}_R - \sin(\beta(t-t_o))\boldsymbol{u}_I] \\ -2a\sin\varphi\,[\sin(\beta(t-t_o))\boldsymbol{u}_R + \cos(\beta(t-t_o))\boldsymbol{u}_I]\end{aligned} \tag{5.53}$$

with the coefficients $a\cos\varphi$ and $a\sin\varphi$ serving as the two-dimensional generator of the center manifold.

The linear center manifold has a constant period of motion $2\pi/\beta$ that is independent of amplitude. In the Earth–Sun system the $L_{1,2}$ center manifolds have a period of approximately half a year, while in the Earth–Moon system they have a period of approximately 2 weeks. Nonlinear continuation of the center manifolds is more difficult than for the stable and unstable manifolds. In general, families must be computed at increasing amplitudes away from the equilibrium point using a differential correction approach. Such procedures are discussed in the following chapter when computation of periodic orbit families are covered. Using the techniques of analysis and perturbation theory it is also possible to analytically construct the center manifolds related to an equilibrium point, as has been developed rigorously to a high level of sophistication by Simo's Barcelona group (see [51]).

Complex Eigenvalues of $\boldsymbol{J}\boldsymbol{H}_{\boldsymbol{xx}}$

If the dynamical system has two or more degrees of freedom, it is also possible for σ to be a complex eigenvalue with non-zero α and β. Then there are four eigenvalues associated with the system, $\pm\alpha \pm i\beta$. The eigenvalues of the state transition matrix are then $e^{\pm\alpha(t-t_o)}e^{\pm i\beta(t-t_o)}$ and trajectories consist of exponential growth and contraction in addition to oscillation, forming unstable and stable spirals emanating from the equilibrium point.

The eigenvectors are complex in general where the overbar denotes complex conjugation and with the form $\boldsymbol{u}_\pm$ and $\bar{\boldsymbol{u}}_\pm$, with the $\pm$ being associated with the sign of the real part of the eigenvalue. Each complex conjugate pair of eigenvectors define a 2-D surface around the equilibrium point filled with exponentially increasing or decreasing spirals. These manifolds can be expressed as

$$\begin{aligned}\delta \boldsymbol{x}_{\pm S}(t) = 2a\,e^{\pm\alpha(t-t_o)}\,[\cos(\beta(t-t_o)+\varphi)\,(\boldsymbol{u}_\pm + \bar{\boldsymbol{u}}_\pm) \\ +i\sin(\beta(t-t_o)+\varphi)\,(\boldsymbol{u}_\pm - \bar{\boldsymbol{u}}_\pm)]\end{aligned} \tag{5.54}$$

where again a is an arbitrary positive amplitude and φ is an initial phase angle. Each spiral traces out a two-dimensional manifold, creating a four-dimensional manifold overall when both are considered.

5.4.3 General Motion in the Vicinity of an Equilibrium Point

The generic motion in the vicinity of an equilibrium point will have components that lie on all of the available manifolds in the vicinity of the equilibrium point.

For definiteness, let us consider a 2-DOF system with a pair of stable/unstable eigenvalues and a center manifold. A general expression for motion relative to the equilibrium point is specified as

$$\begin{aligned}\delta\boldsymbol{x}(t) = a_- \, e^{-\alpha t}\boldsymbol{u}_- + a_+ \, e^{\alpha t}\boldsymbol{u}_+ + 2a_{cc}\left[\cos(\beta t)\boldsymbol{u}_R - \sin(\beta t)\boldsymbol{u}_I\right] \\ -2a_{sc}\left[\sin(\beta t)\boldsymbol{u}_R + \cos(\beta t)\boldsymbol{u}_I\right]\end{aligned} \tag{5.55}$$

As all of these eigenvalues are distinct, note the existence of a left eigenvector $\boldsymbol{v}_i$ for each eigenvalue with the property that $\boldsymbol{v}_i \cdot \boldsymbol{u}_j = \delta_{ij}$. Thus, these eigenvectors can be used to select what component of an arbitrary initial vector, $\delta\boldsymbol{x}_o$ lies on each of the manifolds. Specifically, $a_- = \boldsymbol{v}_- \cdot \delta\boldsymbol{x}_o$ and $a_+ = \boldsymbol{v}_+ \cdot \delta\boldsymbol{x}_o$. Projections onto the center manifold become more algebraically complex, and thus are more simply stated as $\boldsymbol{v}_c \cdot \delta\boldsymbol{x}_o$ and $\bar{\boldsymbol{v}}_c \cdot \delta\boldsymbol{x}_o$. Introduction of these projections allows us to take the state transition matrix definition full circle, and rewrite the motion in the vicinity of the equilibrium point in terms of the eigenstructure of that point

$$\begin{aligned}\delta\boldsymbol{x}(t) = \big[e^{-\alpha t}\boldsymbol{u}_-\boldsymbol{v}_- + e^{\alpha t}\boldsymbol{u}_+\boldsymbol{v}_+ + 2\cos(\beta t)\left(\boldsymbol{u}_R\boldsymbol{v}_R - \boldsymbol{u}_I\boldsymbol{v}_I\right) \\ -2\sin(\beta t)\left(\boldsymbol{u}_R\boldsymbol{v}_I + \boldsymbol{u}_I\boldsymbol{v}_R\right)\big] \cdot \delta\boldsymbol{x}_o\end{aligned} \tag{5.56}$$

where the products of eigenvectors are all outer, or dyad, products. This formula provides an explicit decomposition of the state transition matrix into its fundamental components and is equivalent to Eq. 5.22.

5.4.4 Constraints Due to Integrals

For the particular case of a time-invariant system and an equilibrium point, the constraint on the state transition matrix shown in Eq. 5.44 does not apply. To understand this, note that the partial of the integral, in this case $H(\boldsymbol{x})$, is identically zero when evaluated at the point $\boldsymbol{x}^*$. Thus, there is no constraint on the matrix Φ in Eq. 5.44.

If the system has an additional integral of motion, denoted as $h(\boldsymbol{x}) = C$, where C is a constant and h a scalar function (assumed time-invariant), there are additional constraints. Note the usual relationship $\dot{h} = \{h, H\} = 0$. If h is not the Hamiltonian and is a linearly independent integral, its gradient will in general not equal zero. Thus in this case the neighborhood of the equilibrium point $\boldsymbol{x}^*$ can be evaluated to find the result:

$$\left.\frac{\partial h}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}^*} \cdot \left[I - \Phi(t - t_o; \boldsymbol{x}^*)\right] = 0 \tag{5.57}$$

where the fact that the value of $h_{\boldsymbol{x}}$ is constant when evaluated at an equilibrium point is used. From this one can immediately note that Φ has a unity eigenvalue, and that the quantity $h_{\boldsymbol{x}}$ is a left-eigenvector of the matrix. This implies the presence of a second unity eigenvalue for the matrix with the right-eigenvector $Jh_{\boldsymbol{x}}$. Thus, for every additional integral of motion for a given dynamical system beyond the Hamiltonian, there exist two unity eigenvalues for the state transition matrix evaluated in the vicinity of the equilibrium point. The presence of such additional integrals of motion can complicate the numerical search for equilibrium points, as discussed in Section 6.2, due mainly to the fact that they imply that the dynamics matrix JH_{xx} has a pair of zero eigenvalues.

5.5 Periodic Orbits and Stability

5.5.1 Definition of a Periodic Orbit

Assume a Hamiltonian system with state $\boldsymbol{x} = [q, p]$, Hamiltonian function $H(\boldsymbol{x}, t)$, and attendant equations of motion $\dot{\boldsymbol{x}} = JH_{\boldsymbol{x}}$ with solutions $\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}_o, t_o)$. A solution to this set of equations is said to be periodic with period T if the state of the system exactly repeats itself after a timespan T and the equations of motion evaluated at this state repeat themselves. As with all trajectories, a periodic orbit can be uniquely defined by an initial state, here denoted as $\boldsymbol{x}_t^*$, where the t subscript indicates that it can be defined at any point along the trajectory. Thus the condition for a periodic orbit is

$$\boldsymbol{x}_t^* = \phi(t + T; \boldsymbol{x}_t^*, t) \tag{5.58}$$

$$JH_{\boldsymbol{x}}(\boldsymbol{x}_t^*, t + T) = JH_{\boldsymbol{x}}(\boldsymbol{x}_t^*, t) \tag{5.59}$$

and it can be immediately seen that for a time invariant Hamiltonian a sufficient condition for a solution to be periodic is that it repeats itself, as then the dynamics will also repeat. An equilibrium point is a special case of a periodic solution where the period of motion equals zero. For a time-varying Hamiltonian there is a more stringent condition on the dynamical system itself, namely that the gradient of the Hamiltonian repeat itself over the same timespan T. This is usually relaxed to the stipulation that the Hamiltonian itself is time periodic, meaning that there exists some constant time T such that $H(\boldsymbol{x}, t + T) = H(\boldsymbol{x}, t)$, for all time t.

For time-varying systems that do not repeat themselves, i.e., there exists no time interval T such that $H(\boldsymbol{x}, t + T) = H(\boldsymbol{x}, t)$ for all t, such systems cannot have periodic solutions in general. This is the situation for "real world" motion, as the orbits of the planets never repeat themselves exactly, etc. Still, the concept of periodic orbits in these systems is important, as planetary systems can generally be reduced to simplified models that are periodic or time invariant. If these simplified systems are "close" to the real systems, the influence of the periodic orbits often persists in the perturbed problem, meaning that trajectories in the real systems still feel the influence of the periodic orbits and their stability properties, even if those solutions do not exist in the equations at hand. The simplest way to think of their influence is that the periodic solutions exist in a lower-order model of the system, and that the non-periodic terms represent higher-order perturbations that destroy the periodicity, yet which may not entirely eliminate the influence of the first-order periodic solutions.

A motivating interest in periodic solutions is that they are defined for all time. Specifically, once $\boldsymbol{x}(t+T) = \boldsymbol{x}(t)$ is established, this solution can be generalized to

$\boldsymbol{x}(t+mT) = \boldsymbol{x}(t)$ for $m = \pm 1, \pm 2, \ldots$. Thus, knowledge of the solution over a time interval $t \in [0, T)$ is sufficient information to know the solution for all time. Except for integrable systems, this is generally the only time one can have such complete knowledge of the solution of the equations of motion.

As should be apparent from the conditions for periodicity, there is a fundamental difference between periodic orbits in time-invariant systems and in time periodic systems. For time invariant systems there are no *a priori* constraints on what possible periods of the solutions there may be. Indeed, for time-invariant systems there are usually connected families of periodic solutions that have periods which continuously cover some range of values. For every periodic orbit with a given period, there will in general always exist a continuous family of periodic orbits with different periods in its neighborhood.

For time-periodic systems, there can only be periodic orbits with periods that are multiples of the smallest period of the Hamiltonian. Thus, periodic orbits are isolated in terms of period, and as will be shown, are also isolated in initial conditions.

5.5.2 Floquet Theory

The motion in the vicinity of a periodic orbit in phase space has a special structure associated with it. In particular, while the state transition matrix expanded about a periodic orbit cannot be expressed in closed form in general (unlike for an equilibrium point), the functional form of its general solution can be found. Floquet's theorem provides such an explicit functional form for the state transition matrix relative to a periodic orbit.

The state transition matrix follows the equation:

$$\dot{\Phi}(t, t_o) = J H_{\boldsymbol{x}\boldsymbol{x}}(\boldsymbol{x}(t)) \Phi(t, t_o) \tag{5.60}$$

where $\boldsymbol{x}(t) = \phi(t + T; \boldsymbol{x}_o, t_o)$. Since the nominal solution is periodic with an assumed period of T, the dynamics matrix $JH_{\boldsymbol{x}\boldsymbol{x}}$ is periodic with the same period. Based on this additional constraint Floquet's Theorem can be proved. The simplest statement of the theorem is that the state transition matrix evaluated relative to a periodic orbit has the standard form:

$$\Phi(t, t_o) = P(t, t_o)\, e^{M(t - t_o)} \tag{5.61}$$

where P is a non-singular, periodic matrix function of time with period T and M is a constant matrix.

Consider the state transition matrix solution over one period T, $\Phi(T)$ (where we will take $t_o = 0$ and stop carrying this term). The matrix $\Phi(T)$ is a symplectic matrix and thus its eigenvalues follow the fundamental rules for these systems. The state transition matrix of a periodic orbit taken over one period will always have two eigenvalues equal to 1, and cannot be reduced to a diagonal matrix of eigenvalues. This will be considered in more detail later, yet this fact is addressed in the following derivation. The remaining part of the matrix can be decomposed into its left and right eigenvectors and its eigenvalues. The unity eigenvalues can at least be isolated into a Jordan block form, and will have associated left and right eigenvectors and generalized eigenvectors. This allows us to generically write the state transition matrix into a decomposed form:

$$\Phi(T) = U\Lambda V^T \tag{5.62}$$

where V is the matrix of left eigenvectors and generalized eigenvectors, U is the matrix of right eigenvectors and generalized eigenvectors, and Λ is a matrix in Jordan form with distinct eigenvalues of the state transition matrix along the diagonals and the one Jordan block for the repeated unity eigenvalues (for a time-invariant system), with the structure

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \tag{5.63}$$

The matrix Λ can be written in exponential form as $\Lambda = e^{[\alpha]T}$. The matrix $[\alpha]$ is again in Jordan form with diagonal entries associated with all the non-repeated eigvenvalues, equal to $\alpha_i = \ln(\lambda_i)/T$, and one Jordan block with zeros along the diagonal and one above the diagonal. Thus, the state transition matrix can be written as

$$\Phi(T) = U\, e^{[\alpha]T} V^T \tag{5.64}$$

This decomposition of the state transition matrix is used to motivate a new matrix function $\psi(t)$ defined as

$$\psi(t) = U\, e^{[\alpha]t} V^T \tag{5.65}$$

Note that $\psi(t)$ is the matrix solution to the time invariant matrix equation

$$\dot{\psi} = U[\alpha]V^T\psi \tag{5.66}$$

To show this recall the property of the left and right eigenvectors, namely that $V^TU = U^TV = I$. As the solution to a time invariant matrix equation, the function $\psi(t)$ has a few key properties. Of specific interest to us is the inverse relation $\psi^{-1}(t) = \psi(-t)$.

Using this new function $\psi(t)$, a decomposition of the state transition matrix can be defined

$$\Phi(t, t_o) = P(t, t_o)\psi(t - t_o) \tag{5.67}$$

or conversely a new function $P(t, t_o)$ is defined by:

$$P(t, t_o) = \Phi(t, t_o)\psi^{-1}(t - t_o) \tag{5.68}$$

For this definition it can be shown that P is a periodic function of time with period T. The proof is simple. First, note that, by definition, $P(0, 0) = I$, and furthermore that $P(T,0) = I$, the latter due to the definition of ψ in Eq. 5.65. Next, consider the equation of motion of P,

$$\begin{aligned}\dot{P}(t,t_o) &= JH_{\boldsymbol{xx}}(t)\Phi(t,t_o)\psi^{-1}(t-t_o) \\ &\quad -\Phi(t,t_o)\psi^{-1}(t-t_o)U[\alpha]V^T\end{aligned} \tag{5.69}$$

To establish periodicity, it is just needed to show that $\dot{P}(0,0) = \dot{P}(T,0)$, as then the solution P must be periodic since its value is equal at these two epochs. Substituting these time limits into the above expression yields the following:

$$\dot{P}(0,0) = JH_{\boldsymbol{xx}}(0) - U[\alpha]V^T \tag{5.70}$$

$$\dot{P}(T,0) = JH_{\boldsymbol{xx}}(T)\Phi(T,0)\psi^{-1}(T) - \Phi(T,0)\psi^{-1}(T)U[\alpha]V^T \tag{5.71}$$

but $H_{\boldsymbol{xx}}(T) = H_{\boldsymbol{xx}}(0)$ along a periodic orbit and $\Phi(T,0)\psi^{-1}(T) = I$, making the two expressions equal and finishing the proof.

In final form, note that the solution to the state transition matrix in the vicinity of a periodic orbit has the general form:

$$\Phi(t,t_o) = P(t,t_o)U\,e^{[\alpha](t-t_o)}V^T \tag{5.72}$$

where U and V^T are the right and left eigenvector (and generalized eigenvector) matrices of $\Phi(T)$, respectively, and $[\alpha] = \ln(\Lambda)/T$.

5.5.3 Stability of Periodic Orbits

A particular application of Floquet's Theorem is to the stability of a periodic orbit, as their stability properties provide an immediate description of the phase space flow in the vicinity of that solution. Specifically, if an orbit is stable this implies that phase flows in the vicinity of the orbit will stay close to the periodic orbit. Conversely, if an orbit is unstable, this implies that there are pathways that flow from close to the solution to "far" from the solution, and allows for the phase space to be traversed to either approach or depart from the periodic orbit.

Stability analysis is made particularly simple by using the Floquet decomposition. First, consider the propagation of the state transition matrix to an arbitrary epoch in the future, denoted as $t + mT$, assuming an initial epoch of $t_o = 0$:

$$\begin{aligned}\Phi(t+mT) &= P(t)U\,e^{[\alpha](t+mT)}V^T &(5.73)\\ &= P(t)U\,e^{[\alpha]t}V^TU\,e^{[\alpha]mT}V^T &(5.74)\\ &= \Phi(t)U\Lambda^mV^T &(5.75)\\ &= \Phi(t)\Phi(T)^m &(5.76)\end{aligned}$$

The stability of the solution is clearly dominated by the eigenvalues of the state transition matrix evaluated over one orbit period. If all of the eigenvalues Λ have unity magnitude, i.e., are of the form $\lambda_j = e^{i\theta_j}$, then the matrix Λ^m will not grow under iteration and the orbit is considered to be stable. Note that this does not constitute nonlinear stability, a much more difficult topic for which few sharp results exist, the most important being the Kolgomorov–Arnold–Moser Theorem [4]. In general, stability does imply lack of hyperbolic instability, with any instabilities that may exist only being polynomial in time at most [97]. Conversely, the existence of one eigenvalue with a magnitude not equal to unity implies instability, due to the existence of its inverse pair with magnitude less than or greater than unity, respectively.

Similar to equilibrium points, periodic orbits will also have stable, unstable and center manifolds associated with them, depending on the stability of the system. The fundamental stability result for a periodic orbit reduces to the computation of the eigenvalues and eigenvectors of the state transition matrix over one orbit period, $\Phi(T)$, which can be represented again as a set of eigenvalues $\lambda_i = e^{\alpha_i T}$ and corresponding eigenvectors, $\boldsymbol{u}_i$. While the eigenvalues of the state transition matrix, when mapped over one orbit period, are invariant with where they start within one periodic orbit, the eigenvectors will, in general, depend on where they start within one the periodic orbit. Specifically, the eigenvectors for $\Phi(T, 0)\boldsymbol{u}(0) = \lambda\boldsymbol{u}(0)$ and $\Phi(T + t, t)\boldsymbol{u}(t) = \lambda\boldsymbol{u}(t)$ are different, even though the eigenvalues are the same. This arises from the relationship

$$\Phi(t + T, t) = \Phi(t, 0)\Phi(T, 0)\Phi^{-1}(t, 0) \tag{5.77}$$

thus leading to the relationship $\boldsymbol{u}(t) = \Phi(t, 0)\boldsymbol{u}(0)$ [58]. Thus, after every orbit period the eigenvectors repeat their direction, and expand or contract according to whether $|\lambda|$ is greater or less than one. For an eigenvector associated with an eigenvalue with unity magnitude, $\lambda = e^{i\theta T}$, the eigenvector will in general be rotated by a total net angle θ over one orbit period, tracing out lines on a torus surrounding the periodic orbit.

The analogy between manifolds of equilibrium points and of periodic orbits have many similarities, most of which are clearly exposed when the concept of surface of section and Poincaré map is introduced. The main difference is the dimension of the manifolds, however. For the stable and unstable eigenvalues (i.e., when λ is real and either greater or less than unity) the manifolds are now two-dimensional surfaces, essentially one-dimensional manifolds emanating from every point along a closed periodic orbit and thus forming a two-dimensional sheet. The center manifold, for a generic periodic orbit, consists of sets of quasi-periodic tori that surround the periodic orbit, forming a two-dimensional surface for a given offset amplitude. Thus, as the amplitude is varied these define three-dimensional objects. All of these manifolds can also be extended beyond the linear regime, as described previously for the equilibrium point manifolds. The procedure for extending these manifolds becomes more difficult, due to their higher dimensions. This problem has been extensively studied in the literature, using numerical techniques [118], semi-analytical techniques [109] and analytical techniques [40, 51].

5.5.4 Unity Eigenvalues for Time Invariant Systems

For periodic orbits in time-invariant systems, i.e., systems that conserve their Hamiltonians, there are always a pair of unity eigenvalues in addition to the eigenvalues discussed above, and were already alluded to in the section on Floquet Theory.

Assume that we are studying a periodic orbit in a time-invariant system, thus the Hamiltonian is a constant of motion. Again consider Eq. 5.44, now applied to the Hamiltonian of the system. Due to the periodicity of the underlying motion the partial $H_{\boldsymbol{x}}(0) = H_{\boldsymbol{x}}(T)$ and is non-zero, leading to:

$$H_{\boldsymbol{x}}(0)\left[I - \Phi(T,0)\right] = 0 \tag{5.78}$$

meaning that Φ has a unity eigenvalue and that $H_{\boldsymbol{x}}$ is a left eigenvector. By symmetry there is a second unity eigenvalue with $JH_{\boldsymbol{x}}$ as its right eigenvector. This makes sense, as a displacement along the dynamics of the initial point should just correspond to a point displaced along the periodic orbit, or $\delta\boldsymbol{x} = \dot{\boldsymbol{x}}\delta t = JH_{\boldsymbol{x}}\delta t$. Since the orbit repeats itself, this displaced point will repeat itself with the same period. The presence of these unity eigenvalues for periodic orbits in time-invariant systems complicate the solution process for finding periodic orbits.

Whenever a repeating eigenvalue is found for a matrix, it is important to understand whether the associated eigenvectors with these repeated roots are linearly independent. To evaluate this the concept of a generalized eigenvector must be introduced, discussed in some detail in [58] for astrodynamical systems. A generalized eigenvector of grade k for an eigenvalue with multiplicity m satisfies

$$(\Phi - \lambda I)^k \boldsymbol{u}_k = 0 \tag{5.79}$$

$$(\Phi - \lambda I)^{k-1} \boldsymbol{u}_k \neq 0 \tag{5.80}$$

For a periodic orbit with repeated unity eigenvalues, one can always find the generalized eigenvector of rank 1, $JH_{\boldsymbol{x}}$. Thus, the generalized eigenvector of rank 2 must satisfy

$$(\Phi - I)\,\boldsymbol{u}_2 = JH_{\boldsymbol{x}} \tag{5.81}$$

as $(\Phi - I)JH_{\boldsymbol{x}} = 0$. Multiplying by $\Phi^{-1} = -J\Phi^T J$ and noting that $H_{\boldsymbol{x}}$ is a left eigenvector of Φ yields

$$\left[I - \Phi^{-1}\right]\boldsymbol{u}_2 = JH_{\boldsymbol{x}} \tag{5.82}$$

Thus, subtracting the two leads to the condition that

$$\left[\frac{1}{2}\left(\Phi + \Phi^{-1}\right) - I\right]\boldsymbol{u}_2 = 0 \tag{5.83}$$

Thus, for a non-zero generalized eigenvector of rank 2 to exist the matrix $\frac{1}{2}(\Phi + \Phi^{-1}) - I$ must be singular, and $\boldsymbol{u}_2$ must lie in its null space. This matrix can be shown to always be singular by choosing $\boldsymbol{u}_2 = JH_{\boldsymbol{x}}$ and noting that this is also a right eigenvector of Φ^{-1}, which can be shown from the symplectic property. Then $\left[\frac{1}{2}(\Phi + \Phi^{-1}) - I\right] JH_{\boldsymbol{x}} = JH_{\boldsymbol{x}} - JH_{\boldsymbol{x}} \equiv 0$, proving that the matrix must have at least one zero eigenvalue when $JH_{\boldsymbol{x}}$ is a unity eigenvector, which always occurs for time-invariant systems evaluated at a periodic orbit. Since Φ and Φ^{-1} have the same set of eigenvalues, due to the inverse property of their eigenvalues, this implies that there exists a second zero eigenvalue of this matrix as well. This generalized eigenvector must include a component along the gradient of the Hamiltonian, $H_{\boldsymbol{x}}$, however in general it also includes components along the other eigenvectors of the system [58]. That it must contain a component along $H_{\boldsymbol{x}}$ arises from the fact that this vector is orthogonal to all other eigenvectors, including $JH_{\boldsymbol{x}}$. Thus, since the remaining generalized eigenvector must complete the linear space, it must also contain a component of $H_{\boldsymbol{x}}$.

For a time-varying system, such a displacement along the orbit without modifying the time changes the dynamics and yields a solution in the vicinity of the periodic orbit and not on the periodic orbit. Thus, periodic orbits in time-periodic systems, which do not conserve the Hamiltonian, generally do not have unity eigenvalues. If they do, these only occur at specific stability bifurcation points and do not arise generically.

5.5.5 Periodic Orbit Families

A fundamental aspect of periodic orbits in time-invariant systems is that they can be continued into families by varying a parameter such as the Jacobi constant or a parameter of the dynamical system. The discussion of such families is postponed until the concept of a surface of section and Poincaré map are introduced. Once these ideas are presented, the discussion for continuation of a family becomes clearer.

5.6 General Trajectories and Stability

The two most important classes of solutions have now been discussed, equilibrium points and periodic orbits. There are important higher-dimensional generalizations of periodic orbits called quasi-periodic orbits which are an important class of solution, but are in general difficult to compute and characterize. We do not consider these classes of solutions, although they will be of future interest to the study of small body orbiters. The importance of solutions such as equilibria, periodic orbits, and quasi-periodic orbits is mainly attached to the fact that they can be specified for all time and that their stability can be unambiguously computed. Additionally, through their stable and unstable manifolds they influence wider regions of phase space. Despite all of these things, the number of such special trajectories are vanishingly small with respect to more general trajectories without such special initial conditions. Thus, it is relevant to discuss these other solutions. However, due to our not having a clear expression of these trajectories valid over all time (an exception being for integrable systems), there are only limited things that can be stated about such trajectories.

First, recall our notation for a solution, $\boldsymbol{x}(t) = \phi(t; \boldsymbol{x}_o, t_o)$. In some special cases, namely if the state $\boldsymbol{x}_o$ lies on a manifold to an equilibrium point or a periodic orbit, then the solution may have a well-defined limit. If it lies on a stable manifold, then the trajectory is defined as $t \rightarrow \infty$, if on an unstable manifold it is defined as $t \rightarrow -\infty$. It is significant to note that the dimension of an object's manifolds will always be greater than the object in question itself. Thus, one can immediately note the outsized influence which equilibria and periodic orbits have.

If one moves away from these special cases, or has a trajectory that moves away from a special orbit either forwards or backwards in time, this leads to a general solution without a known limiting structure. The solution may still be subject to other constraints, such as having to lie on an energy surface or remain within a zero-velocity surface, but it is not necessary to assume these phase space constraints in general. Instead, all one knows is that the solution will either continue to evolve over time or may run into a physical constraint, such as impacting the surface of a small body, a planet, or the Sun. Beyond the specific path of the trajectory, which can only be ascertained by some solution procedure, the other main item of interest is determining how neighboring trajectories evolve. In fact, from a practical point of view, knowing how neighboring trajectories evolve can be more helpful in understanding a general trajectory than where the trajectory itself goes. This is primarily due to the fact that spacecraft locations in phase space are never known precisely, and thus knowledge of how the local phase volume evolves informs us as to whether neighboring solutions will remain close to each other or diverge.

The standard approach to characterizing the local dynamics about a nominal trajectory is via Lyapunov Characteristic Exponents (LCE). These are, in some sense, a generalization of the characteristic exponents found for an equilibrium point and a periodic orbit. Instead of measuring how rapidly a neighboring trajectory expands along a specified direction, an LCE instead measures the maximum rate of expansion in a region of the neighboring phase space. There are several techniques that are used to compute LCEs, or to ascertain specific properties related to them. The following is the simplest and most direct definition.

First, assume a nominal trajectory defined by an epoch state and time, $\boldsymbol{x}_o, t_o$. Linearizing about the nominal trajectory yields the state transition matrix, which takes a small variation $\delta\boldsymbol{x}_o$ into a later variation relative to the nominal trajectory: $\delta\boldsymbol{x}(t) = \Phi(t, t_o)\delta\boldsymbol{x}_o$. In principle it is always possible to choose the norm $|\delta\boldsymbol{x}_o|$ small enough to satisfy linearity assumptions that affect the deviation of $\delta\boldsymbol{x}(t)$ for any timespan $t - t_o$. Given these steps the finite time LCE is computed as

$$\chi(t) = \max_{\delta\boldsymbol{x}_o} \frac{1}{t} \ln\left(\frac{|\Phi(t, t_o)\delta\boldsymbol{x}_o|}{|\delta\boldsymbol{x}_o|}\right) \tag{5.84}$$

It is instructive to look at this function more closely.

As the logarithm is a definite function, its maximum value will occur at the maximum value of its argument. This can be rewritten by squaring the norm and dividing through by the magnitude of $\delta\boldsymbol{x}_o$ and is the same as finding $\max_{|\boldsymbol{u}|=1} \boldsymbol{u} \cdot$

$\Phi^T\Phi \cdot \boldsymbol{u}$. To solve this problem, write out a constrained Lagrangian function to find the extremal values, $\Xi = \boldsymbol{u} \cdot \Phi^T\Phi \cdot \boldsymbol{u} - \xi(\boldsymbol{u} \cdot \boldsymbol{u} - 1)$, where Ξ is a scalar function and ξ is a Lagrange multiplier. Extrema are found by solving the equations $\frac{\partial \Xi}{\partial \boldsymbol{u}} = 0$ and $\frac{\partial \Xi}{\partial \xi} = 0$. The second equation gives us $\boldsymbol{u} \cdot \boldsymbol{u} = 1$ while the first gives us $\left[\Phi^T\Phi - \xi I\right] \boldsymbol{u} = 0$. Thus, one only needs to find eigenvalues and eigenvectors of the matrix $\Phi^T\Phi$ in order to compute the extremal values. The eigenvalues of $\Phi^T\Phi$ can be computed from the eigenvalues of Φ, denoted here as $\lambda_i, i = 1, 2, \ldots, 2n$ and have the usual structure for symplectic matrices. Specifically, the eigenvalues of $\Phi^T\Phi$ equal $\xi_i = \lambda_i\bar{\lambda}_i, i = 1, 2, \ldots, 2n$. Thus, there may exist repeated unity eigenvalues if $\lambda_i = e^{i\theta}$, but these will not be defective in general and thus will have linearly independent eigenvectors (see [100] for a discussion of the general properties of these eigenvalues and eigenvectors). Evaluating the cost function Ξ at a general extremum yields $\Xi = \lambda_i\bar{\lambda}_i, i = 1, 2, \ldots, 2n$ and thus that the maximum occurs for $\max_{i\in[1,2n]} \lambda_i\bar{\lambda}_i$. Carrying this back to the original definition of $\chi(t)$,

$$\chi(t) = \frac{1}{t} \ln \max_i \sqrt{\lambda_i\bar{\lambda}_i} \tag{5.85}$$

It can shown in general that the limit of this quantity converges to a constant value as $t \to \infty$, and that it can be considered to be an integral of motion for any particular trajectory [119]. This provides the formal definition of an LCE:

$$\chi_\infty = \lim_{t\to\infty} \chi(t) \tag{5.86}$$

If $\chi_\infty > 0$, the trajectory is considered to be unstable as neighboring trajectories will deviate exponentially from the given trajectory. Conversely, if $\chi_\infty = 0$ this implies that neighboring solutions are bounded, in a linear sense, although it does not guarantee nonlinear boundedness. The practical computation of the LCE is difficult, it being impossible to determine a general trajectory for all time. There are a wealth of practical computation schemes and related definitions which are discussed in the reference [42].

It is instructive to apply this definition to an equilibrium point. For the state transition matrix evaluated at an equilibrium point, the maximum eigenvalue $\lambda_i\bar{\lambda}_i$ will have the generic value $e^{2\alpha t}$ where α is the largest real part of the eigenvalues of $JH_{\boldsymbol{xx}}$ evaluated at the equilibrium point. Thus, it can be immediately seen that $\chi_\infty = \alpha$, which fits precisely with one's expectations. In particular, if the equilibrium point is stable then all of the eigenvalues satisfy $\lambda_i\bar{\lambda}_i = 1$ and $\chi(t) \equiv 0$. Similar results are found if applied to a periodic orbit.

5.7 Surfaces of Section and Poincaré Maps

Finally, the important concepts of surface of section maps and Poincaré maps are introduced. Although these two are usually related to each other, it instructive to separate our discussion of them. Following their appropriate definition families of periodic orbits and additional properties of these solutions can be discussed.

5.7.1 Surface of Section Maps

For time-invariant dynamical systems it is often convenient to reduce the dimensionality of the system via a "surface of section." A surface of section is some geometrically defined surface or condition in phase space which, when a trajectory crosses this surface or passes through this condition, the precise point of intersection is obtained and reported in some coordinate frame. Usually the trajectory must cross the surface or condition in a proscribed direction. The simplest example of a surface of section is some coordinate axis in the configuration space, say the x-axis. Then, whenever a trajectory crosses through the value $y = 0$ the values of all of its other coordinates and momenta (or velocities) are reported. For a simple orbit traveling about the origin in the x–y plane there should be at least two x-crossings per orbit – however, by restricting the surface of section to only be reported (to only exist) when the trajectory crosses in a certain direction, say for $\dot{y} > 0$, then there could only be one surface of section crossing per orbit. A surface of section need not be defined only in the configuration space, although these are often the most convenient and commonly used surfaces. For example, periapsis passages can be used as a surface of section, defined as the condition $\dot{r} = \boldsymbol{r} \cdot \dot{\boldsymbol{r}}/r = 0$ with the additional condition that $\ddot{r} > 0$.

To set this up in a more generic fashion, let the surface of section be defined by a scalar equation $S(\boldsymbol{x}) = 0$, with $\boldsymbol{x} \in \mathbf{R}^{2n}$ (see Fig. 5.2). In the above examples this function would equal x_2 or $\dot{\boldsymbol{r}} \cdot \boldsymbol{r}/r$. To be well-defined, the solution flow to be observed using the surface of section must be transversal to the surface of section, meaning that the trajectory is non-tangent to the surface. This can be enforced by requiring $\frac{\partial S}{\partial \boldsymbol{x}} \cdot \dot{\boldsymbol{x}} > 0$, where a preferred direction of crossing has been defined. The solution flow relative to a well-defined surface of section will repeatedly cross through the surface after each orbit and will not have a grazing, or tangential, intersection with it. A given trajectory being transverse to the surface of section at one crossing is no guarantee that it will be transversal to the surface at a future crossing, or even that it will ever cross the surface again. It is possible for a trajectory to initially have transversal crossings of a surface of section yet later evolve to the point where there are grazing, or perhaps no, additional crossings. An example of the latter is if a trajectory with a periapsis surface of section escapes from the system in question – it will never have another periapsis crossing and hence will never appear on the surface again. Due to this, the choice of surface of section is often very important in analyzing a dynamical system, as a poor choice may lead to no intersections. Thus, the selection of surfaces of section is highly dependent on the problem and the particular type of flow that is being investigated.

Surfaces of section are usually only set up in time invariant dynamical systems, as then, whenever a trajectory crosses through the same surface of section coordinates again, the motion, by definition, defines a periodic orbit. For a time-varying dynamical system, the surface of section would have time-varying dynamics across its surface and does not yield any systematic simplification. It is interesting to consider a time periodic dynamical system, i.e, one with a Hamiltonian where $H(\boldsymbol{x}, t) = H(\boldsymbol{x}, t + T)$, that is transformed into its time invariant form with its attendant increase in dimensions. For such a time invariant system the natural surface

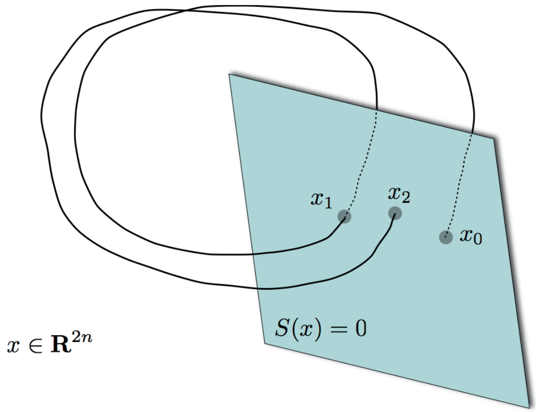

Fig. 5.2 Illustration of a surface of section.

of section to choose is the time coordinate. Such a surface would consist of reporting the dynamics whenever the time coordinate equals $t_o + nT$, $n = 0, \pm 1, \pm 2, \ldots$. By definition, the solution flow will always be transverse to this surface of section. This particular application of a surface of section is called a stroboscopic map and has been widely applied to time-periodic systems.

Let us take the periapsis passage in a two-body orbit as an example, using a Lagrangian formulation. Then $S(\boldsymbol{x}) = \dot{r} = \frac{\boldsymbol{r}\cdot\dot{\boldsymbol{r}}}{r}$ and $S_{\boldsymbol{x}} = \left[\frac{\dot{\boldsymbol{r}}}{r}; \frac{\boldsymbol{r}}{r}\right]$, where application of the result $\dot{r} = 0$ simplifies the partial with respect to position. The transversality condition then becomes $S_{\boldsymbol{x}} \cdot \dot{\boldsymbol{x}} = \frac{\dot{\boldsymbol{r}}\cdot\dot{\boldsymbol{r}}}{r} + \frac{\boldsymbol{r}\cdot\ddot{\boldsymbol{r}}}{r} > 0$. If the acceleration in the two-body problem is substituted as an example, $\ddot{\boldsymbol{r}} = -\mu\boldsymbol{r}/r^3$, this leads to $S_{\boldsymbol{x}} \cdot \dot{\boldsymbol{x}} = \frac{1}{r}\left[\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} - \frac{\mu}{r}\right]$. Thus the surface is only crossed when this is greater than zero, implying that the local speed is greater than local circular speed, a prerequisite to have a periapsis passage.

The main advantage of the surface of section is that it allows us to replace a continuous time solution with a discrete map, effectively removing the time from the system. Reducing the trajectory to these surface crossings allows us to define a mapping function $G(\boldsymbol{x})$. Assume an initial point on the surface of section is chosen, $\boldsymbol{x}_0$, at a time t_0, such that $S(\boldsymbol{x}_0) = 0$ and the transversality condition is satisfied. Then the mapping to the next surface crossing can be represented as $\boldsymbol{x}_1 = G(\boldsymbol{x}_0) = \left\{\boldsymbol{x}(t_1) | S(\phi(t_1, t_0, \boldsymbol{x}_0)) = 0,\, S_{\boldsymbol{x}} \cdot \dot{\boldsymbol{x}}|_{t_1} > 0, t_1 > t_0\right\}$. This mapping implicitly defines a passage time t_1, constrained in general to be the first passage of the surface of section after t_0, although the exact value of this time is not of specific interest in terms of the surface of section map. Repeating this procedure, it is possible to propagate a single trajectory through many surface crossings, denoted as:

$$
\begin{aligned}
\boldsymbol{x}_{i+1} &= G(\boldsymbol{x}_i) && (5.87)\\
&= G(G(\boldsymbol{x}_{i-1})) && (5.88)\\
&= G^2(\boldsymbol{x}_{i-1}) && (5.89)\\
&= G^{i+1}(\boldsymbol{x}_0) && (5.90)
\end{aligned}
$$

This notation still carries around a full $2n$ state vector, although technically this can be reduced by one dimension to a $2n-1$-dimensional state vector through the constraint $S(\boldsymbol{x}_i) = 0$. It is important to note that in the computation of a surface of section, to compute subsequent passages it is necessary to integrate the entire trajectory forward (or backwards) in time. For a stroboscopic map in a time periodic dynamical system the definition of the surface of section is $\boldsymbol{x}_n = \phi(t_o + nT; \boldsymbol{x}_o, t_o)$.

5.7.2 Poincaré Maps

As the surface of section map was only defined for a time invariant system, the discrete surface of section mapping defined above has a Jacobi integral of motion. Specifically,

$$H(\boldsymbol{x}_0) = H(G^i(\boldsymbol{x}_0)) = C_0 \tag{5.91}$$

As the Hamiltonian is a relatively simple algebraic function it is also possible to eliminate an additional coordinate or momenta from consideration, reducing the dimensionality of the state by two (one for the surface of section and the other for the energy) to a $2n-2$-dimensional system. Generally, one removes a velocity or momentum component using the constant Hamiltonian, while one removes a coordinate using a surface of section. This is not a fundamental restriction, however. Once the additional state is removed from the map, the value of the Hamiltonian along a specific trajectory becomes a new parameter of the system and the resulting system is defined as a Poincaré map.

To formalize this, assume that two states are removed from the original $2n$-dimensional state vector $\boldsymbol{x}$, one each from the conditions $S(\boldsymbol{x}) = 0$ and $H(\boldsymbol{x}) = C$. For a given dynamical system it is important, of course, to systematically remove the same states at each crossing. Let us define the so-reduced state vector as $\boldsymbol{y} \in \mathbf{R}^{2n-2}$. It is important to remember that there is a unique state $\boldsymbol{x}$ associated with each reduced state $\boldsymbol{y}$, surface of section function S, and energy value C. Redefine the mapping function, now explicitly specifying the energy level that is being considered, with the form $g(\boldsymbol{y}; C)$. This discrete mapping function defines a Poincaré map, a mapping function that returns a trajectory back to a surface of section and maps every initial point chosen on that section with the same value of the Hamiltonian. Our dynamics are then:

$$\begin{aligned} \boldsymbol{y}_{i+1} &= g(\boldsymbol{y}_i; C) && (5.92)\\ &= g^{i+1}(\boldsymbol{y}_0; C) && (5.93) \end{aligned}$$

The properties of the Poincaré map reduction have been studied extensively in the literature for a variety of applications. Our main application of this map will be in the computation of periodic orbits in time-invariant systems and in understanding the stability of such periodic motions. The map has also been extensively applied to the study of dynamical systems, an application not explored here.

It should be noted that if a dynamical system has additional integrals of motion, it is possible to further reduce the dimensionality of the Poincaré map. Perhaps the main benefit of the map is that it allows one to reduce the dimensionality of the system without having to go through the laborious process of eliminating integrals of motion from the full dynamical system. It can be assumed the reduction of the dynamical system to a Poincaré map has explicitly removed a coordinate–momentum conjugate pair. If dealing with a Lagrangian system, it suffices to remove a coordinate value and its time derivative, as when transformed into a Hamiltonian system this will also correspond to a coordinate–momentum pair being removed.

Systematic application of a Poincaré map to a collection of initial conditions implies that these conditions are chosen with the same Hamiltonian value, although the trajectories will be distinct in general. Mapping such collections repeatedly through the map then allows for the exploration of a dynamical system through an inspection of $2n-2$ dimensions, instead of the usual $2n$ dimensions. For 2-DOF systems this is tractable, as the resulting map is only two-dimensional. For higher-degree-of-freedom systems the utility of this approach is not as great, as the map itself lives in a four-dimensional or higher space.

5.7.3 Linearized Poincaré Map

Notions of local motion can be carried over to the Poincaré map. Assume a nonlinear map between points $\boldsymbol{y}_0$ and $\boldsymbol{y}_1 = g(\boldsymbol{y}_0; C)$. Consider an initial variation of $\boldsymbol{y}_0$ such that $\delta\boldsymbol{y}_0$ lies in the surface of section and conserves the Hamiltonian. Specifically, this means that $\frac{\partial S}{\partial \boldsymbol{y}_0} \cdot \delta\boldsymbol{y}_0 = 0$ and $\frac{\partial H}{\partial \boldsymbol{y}_0} \cdot \delta\boldsymbol{y}_0 = 0$. This small variation will also be mapped back to the surface of section and will naturally conserve energy, defining $\boldsymbol{y}_1 + \delta\boldsymbol{y}_1 = g(\boldsymbol{y}_0; C) + \Phi_{0,1}\delta\boldsymbol{y}_0$, where the reduced state transition matrix $\Phi_{0,1}$ is defined as the $(2n-2)\times(2n-2)$ matrix that linearly maps the Poincaré map back to itself in the vicinity of a nominal trajectory. Note the following identities $\frac{\partial S}{\partial \boldsymbol{y}_1} \cdot \delta\boldsymbol{y}_1 = 0$ and $\frac{\partial H}{\partial \boldsymbol{y}_1} \cdot \delta\boldsymbol{y}_1 = 0$. The practical computation of this linearized map is discussed later, at this point one only needs to note its existence.

This has defined a mapping of local variations about a nominal trajectory

$$\delta\boldsymbol{y}_1 = \Phi_{1,0}\delta\boldsymbol{y}_0 \tag{5.94}$$

which naturally generalizes to the iterated map

$$\begin{aligned}\delta\boldsymbol{y}_{n+1} &= \Phi_{n+1,n}\delta\boldsymbol{y}_n &(5.95)\\ &= \left(\Pi_{i=0}^{n}\Phi_{i+1,i}\right)\delta\boldsymbol{y}_0 &(5.96)\end{aligned}$$

where the state transition matrix $\Phi_{i+1,i}$ depends on the associated nominal trajectory between the points $\boldsymbol{y}_i$ and $\boldsymbol{y}_{i+1} = g(\boldsymbol{y}_i; C)$, and is defined for a specific surface of section and energy value. If the reduction to the nonlinear Poincaré map is performed by systematic reduction of a coordinate–momentum pair, the resulting system is Hamiltonian and hence the linearized Poincaré map will be a symplectic matrix, and thus will inherit all the relevant properties defined for these maps.

5.7.4 Periodic Orbits and Stability

Now, let us reconsider periodic orbits in light of the Poincaré map. Note that a periodic orbit will correspond to a fixed point in a Poincaré map, denoted as $\boldsymbol{y}^*$. It is defined as

$$\boldsymbol{y}^* = g(\boldsymbol{y}^*; C) \tag{5.97}$$

To evaluate the stability of this fixed point, consider a small variation in the neighborhood of its initial conditions and its expansion through the linearized map

$$\begin{aligned} \boldsymbol{y}^* + \delta\boldsymbol{y}_1 &= g(\boldsymbol{y}^* + \delta\boldsymbol{y}_0; C) && (5.98)\\ &= g(\boldsymbol{y}^*; C) + \Phi^*_{1,0}\delta\boldsymbol{y}_0 + \dots && (5.99)\\ \delta\boldsymbol{y}_1 &= \Phi^*_{1,0}\delta\boldsymbol{y}_0 + \dots && (5.100) \end{aligned}$$

The linear map about a fixed point is independent of which iteration it is being applied to, and in particular $\Phi^*_{1,0} = \Phi^*_{i+1,i} = \Phi_M$. The linearized Poincaré map Φ_M is defined as the monodromy matrix, and it describes the stability of the periodic orbit in question. To note this, recall from Floquet theory that the eigenvalues of the state transition matrix mapped over one period were independent of where the matrix was evaluated along the orbit. Thus, the eigenvalues that the monodromy marix inherits from the full state transition matrix are independent of where the surface of section is chosen along the periodic orbit, and iterates of the monodromy matrix will then describe local motion on the surface of section from crossing to crossing

$$\delta\boldsymbol{y}_n = \Phi^n_M\delta\boldsymbol{y}_0 \tag{5.101}$$

Before the eigenstructure of Φ_M is discussed, note that the reduction process to a constant energy surface and to a surface of section will remove the two generic unity eigenvalues present in the full state transition matrix. From Floquet theory, the monodromy matrix has the general form $e^{M'T}$, where M' is now a constant matrix of size $(2n-2)\times(2n-2)$ and generically does not have zero eigenvalues.

Consider the eigenvalue and right eigenvector pairs of Φ_M, $(\lambda_i, \boldsymbol{u}_i, i = 1, 2, \dots, 2n-2)$. Again, associated with each λ_i is its complex conjugate, its inverse, and the inverse of its complex conjugate. If λ_i is real, then only its inverse must also exist, and if the magnitude of λ_i is unity, then only its complex conjugate must exist. The interpretation and analysis of the stable, unstable and center manifolds associated with the fixed point are all analogous to the discussion for the structure in the vicinity of an equilibrium point, except now instead of being continuous in time the dynamics occur at discrete mappings.

Stability of Periodic Orbits in a 2-DOF problem

Consider the monodromy matrix computed for a periodic orbit in a 2-DOF problem. In this case it is a simple 2×2 matrix, and thus it is possible to provide a tractable

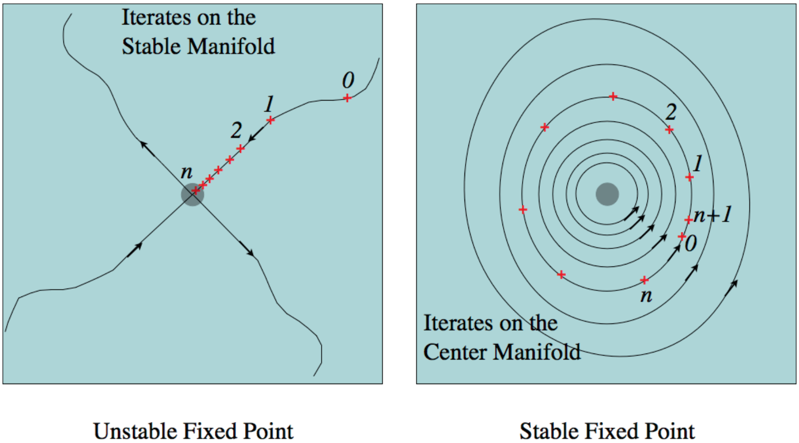

Fig. 5.3 Illustration of a stable and unstable fixed point, showing the manifolds and iterates.

discussion of its stability. The general form of the monodromy matrix will be

$$\Phi_M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{5.102}$$

with the constraint that $ad - bc = 1$, which for 2×2 matrices is the necessary and sufficient condition for them to be symplectic. The eigenvalues of this matrix can be specifically computed from

$$\lambda_{\pm} = \frac{1}{2}(a+d) \pm \frac{1}{2}\sqrt{(a+d)^2 - 4} \tag{5.103}$$

and $\lambda_{\pm}\lambda_{\mp} = 1$. If $|a+d| < 2$ the periodic orbit is linearly stable and $\bar{\lambda}_{\pm} = \lambda_{\mp}$. If instead $|a+d| > 2$ the eigenvalues are both real with one greater than 1 in magnitude and the other less than 1 in magnitude, and $\lambda_{\pm} = 1/\lambda_{\mp}$. Thus, in this case the orbit is unstable in general, and has one stable and one unstable manifold. In the limiting case when $a+d=2$, the eigenvalues equal 1 and are repeated, and usually represent a transition of the eigenvalues from being on the unit circle to being off the unit circle, or vice versa. If $a+d=-2$, the eigenvalues equal -1 and are repeated. This also represents a possible transition of the eigenvalues from stability to instability, or vice versa. The implications of these different transitions are discussed below. Note that periodic orbits in 2-DOF systems cannot exhibit complex instability, unlike equilibrium points in 2-DOF systems.

Stability of Periodic Orbits in a 3-DOF problem

Consider the monodromy matrix computed for a periodic orbit in a 3-DOF problem. In this case it is a 4×4 matrix, and again it is possible to provide a tractable discussion of its stability. The general form of the monodromy matrix will be

$$\Phi_M = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ j & k & l & m \\ n & p & q & r \end{bmatrix} \tag{5.104}$$

with the symbols i and o being reserved. The characteristic equation can be expressed as

$$0 = \lambda^4 + \alpha\lambda^3 + \beta\lambda^2 + \alpha\lambda + 1 \tag{5.105}$$

$$\alpha = -(a + f + l + r) \tag{5.106}$$

$$\begin{aligned} \beta = &\, af - be + al - cj + ar - dn \\ &+ fl - gk + fr - hp + lr - mq \end{aligned} \tag{5.107}$$

Defining $\gamma = \lambda + \lambda^{-1}$ it is possible to show that γ satisfies the equivalent characteristic equation

$$\gamma^2 + \alpha\gamma + \beta - 2 = 0 \tag{5.108}$$

For stability it can be shown that γ must be a real number and satisfy $-2 \le \gamma \le 2$. These conditions will hold if and only if the following inequalities hold

$$8|\alpha| - 16 \le 4\beta - 8 \le \alpha^2 \le 16 \tag{5.109}$$

This defines a stability plot in the α, β space, showing the different possible stability states for a periodic orbit (see Fig. 5.4).

5.8 Periodic Orbit Families

Given the Poincaré map definition and the identification of the monodromy matrix it becomes tractable to discuss the continuation of periodic orbits into families in a more complete fashion. A number of different topics are discussed in the following, including the generic isolation of periodic orbits at fixed values of Jacobi energy, the possible stability transitions along a period orbit family as its energy changes, and the intersections of periodic orbits with other families of the same and higher periods.

5.8.1 Isolation of Periodic Orbits at Constant Jacobi Values

First determine the conditions for another periodic orbit to exist in the vicinity of a given periodic orbit at a fixed value of energy. Given a fixed point $\boldsymbol{y}^*$ of a Poincaré map

$$\boldsymbol{y}^* = g(\boldsymbol{y}^*; C) \tag{5.110}$$

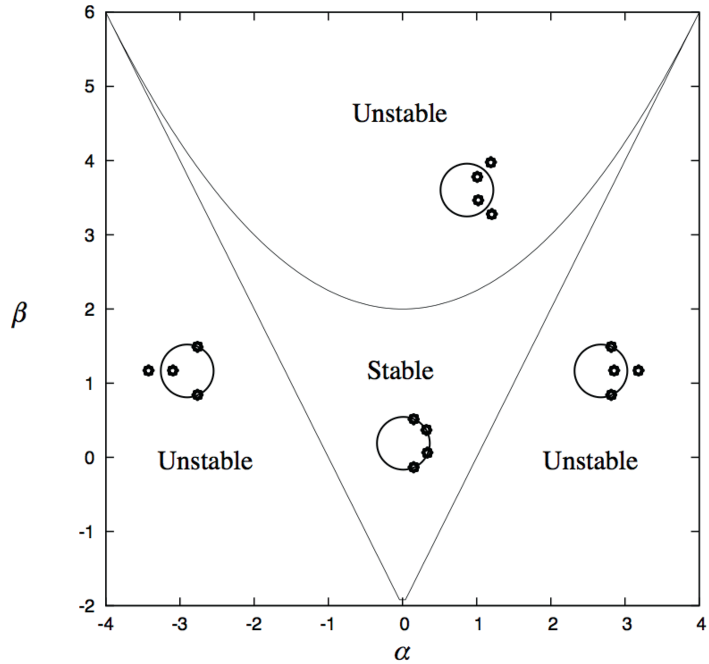

Fig. 5.4 Stability diagram for periodic orbits in a time invariant 3-DOF system as a function of α vs. β.

consider a variation in the state while keeping the energy fixed. Assume the variation is small enough to expand the fixed point using the monodromy matrix, and tacitly assume that the new point maps back into itself

$$\boldsymbol{y}^* + \delta\boldsymbol{y}^* = g(\boldsymbol{y}^*; C) + \Phi_M \delta\boldsymbol{y}^* \tag{5.111}$$

Simplifying the condition becomes

$$[\Phi_M - I]\,\delta\boldsymbol{y}^* = 0 \tag{5.112}$$

Recall that Φ_M has had its unity eigenvalues removed and thus it generically will not have any unity eigenvalues except at exceptional points. If there are unity eigenvalues, then there will be neighboring periodic orbits with their relative location to $\boldsymbol{y}^*$ located along the null space of $\Phi_M - I$. If, as is expected for the generic case, there are no unity eigenvalues of Φ_M, then the periodic orbit is isolated at the fixed value of energy.

This is generally the case for the stroboscopic Poincaré map defined for time-periodic systems. In fact, for these systems where the conserved Hamiltonian equals the time-momentum plus the Hamiltonian, one can show that the value of this Hamiltonian is in general invariant with the state and is always equal to the same constant. This implies that periodic orbits in stroboscopic systems are generically isolated, unless there exists another integral of motion that introduces additional unity eigenvalues.

5.8.2 Continuation of Periodic Orbits with Jacobi Energy

Given the generic isolation of periodic orbits at a given energy, their continuation is explored with a variation in the energy. Now in addition to varying from the fixed point we also allow the Hamiltonian value C to vary by δC, leading to the condition

$$[\Phi_M - I]\,\delta\boldsymbol{y}^* = \left.\frac{\partial g}{\partial C}\right|_* \delta C \tag{5.113}$$

where the partial of the Poincaré map with respect to energy can be computed relatively easily, as is discussed in the next chapter. Assuming that Φ_M has no unity eigenvalues the new, neighboring periodic orbit is defined by

$$\boldsymbol{y}^* + [\Phi_M - I]^{-1}\, g_C^* \delta C \tag{5.114}$$

and maps into itself at the linear level. An algorithm for using the linearized Poincaré map to nonlinearly converge on a fixed point is discussed in the next chapter. If, instead, Φ_M has unity eigenvalues then this implies that there exists a neighboring periodic orbit with the same value of energy. This is not a generic situation and will generally only apply for an exceptional periodic orbit.

Thus, a family of periodic orbits can be defined that passes through $\boldsymbol{y}^*$ at a value of C and generally continues for larger and smaller values of C. This can be represented as a mapping from C to $\boldsymbol{y}$ as $\boldsymbol{y} = \boldsymbol{h}(C)$, where $\boldsymbol{h} \in \mathbf{R}^{2(n-1)}$. The graph of $\boldsymbol{h}$ then defines the family, where the tangent vector to the curve is defined by $[\Phi_M - I]^{-1}\, g_C$.

5.8.3 Stability Transitions along a Periodic Orbit Family

As a family of periodic orbits is traced out, the eigenvalues of the associated monodromy matrix are expected to vary as well. To emphasize this we now show the monodromy matrix as a function of energy, $\Phi_M(C)$, assuming that there exists a family of fixed points defined by $\boldsymbol{y} = \boldsymbol{h}(C)$.

Let us consider a 2-DOF problem first, as there are only two eigenvalues for such a system. If the family is stable at a point $\boldsymbol{h}(C_0)$, with eigenvalues $e^{\pm i\theta}$ (where i is the imaginary unit) where the rotation angle $\theta \neq 0, \pi$, then the local family will always be a stable fixed point as well. The only way the system can transition to an unstable orbit is for the eigenvalues to meet at $\theta = 0, \pi$ and transition to the positive or negative real axis, respectively. The converse situation holds if the fixed point is unstable at a given value of C_0. The two different possible transitions have significantly different implications.

Intersection along $\theta = 0$

First consider the implications if $\theta \to 0$. Drawing on an analysis of such stability transitions for symmetric 2-DOF periodic orbits by Hénon [66], one can distinguish two cases. For both cases it can be assumed that Φ_M has unity eigenvalues so $\|\Phi_M - I\| = 0$ and the matrix $[\Phi_M - I]$ cannot be uniquely inverted. Then the solutions that are consistent with this condition can be explored

$$[\Phi_M - I]\,\delta\boldsymbol{y} = \boldsymbol{g}_C \delta C \tag{5.115}$$

In this case one can always choose a $\delta\boldsymbol{y}$ along an eigenvector of unity such that the left-hand side equals zero.

Local Extremum If either term of $\boldsymbol{g}_C \neq 0$, then the right-hand side can only be made to equal zero if $\delta C = 0$. This implies that the current family is forced to change along a direction defined by the null space of $\Phi_M - I$ with no change in energy. This generically corresponds to the periodic family curve $\boldsymbol{h}(C)$ being at an extremum with respect to C and, in some sense, turning back on itself in phase space. See Fig. 5.5 left for a graphical representation. Following such a transition there will exist two branches of the periodic orbit family at a given value of energy, one will be stable and the other unstable. Such reversals in a periodic orbit family are relatively common, and essentially represent a bifurcation with the family appearing or disappearing as the energy C is varied.

Family Intersection If, on the other hand, $|\boldsymbol{g}_C| \to 0$ as $\theta \to 0$, then this represents an intersection of the current family with another family with the same value of energy, C. For this to be defined these two fixed points must physically cross through the same point in phase space at the same value of C. At such a crossing of families, they will generally exchange stability, as shown in Fig. 5.5 right. The vanishing of the $\boldsymbol{g}_C$ terms is a hallmark of such an intersection, as in the vicinity of the intersection there are multiple solutions which the linear conditions cannot accommodate. If the continuation conditions were developed to second order or higher (depending the the number of intersections), then there should be a definitive set conditions that can be solved for.

Intersection along $\theta = \pi$

In this case the eigenvalues collide along the negative real axis at $\lambda = -1$. At such a bifurcation the matrix $\Phi_M - I$ has full rank and is invertible. Thus, the family seems to seamlessly transition from stable to unstable, or vice versa, at such a transition. What is really going on here is a bit more interesting, however. If the double-iterate of the Poincaré map is considered, $\boldsymbol{g}^2(\boldsymbol{y}^*)$, this is still a fixed point but now its monodromy matrix is Φ_M^2 and, at the stability transition point, will have a pair of unity eigenvalues. This is an example of a period-doubling bifurcation, where a

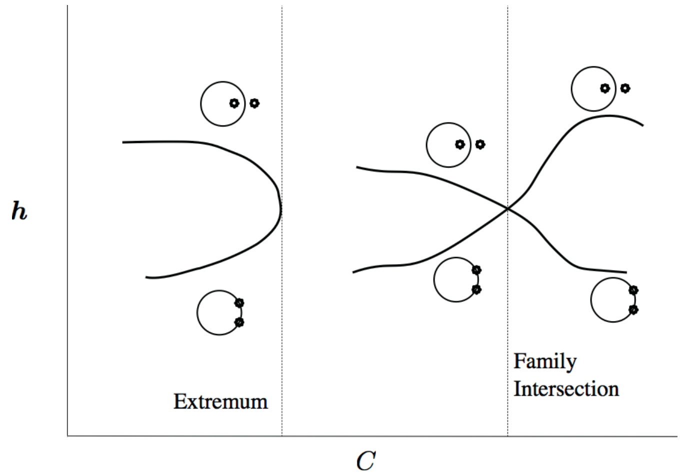

Fig. 5.5 Local extremum of a periodic orbit family (left) and an intersection of two periodic orbit families (right).

pair of fixed points of twice the period are born, inheriting the stability type of the single-period orbit that undergoes its transition. The resulting family may either persist or may intersect with the single-period family again at a different value of C, providing the single family with its same initial stability type. Figure 5.6 shows the generic situation.

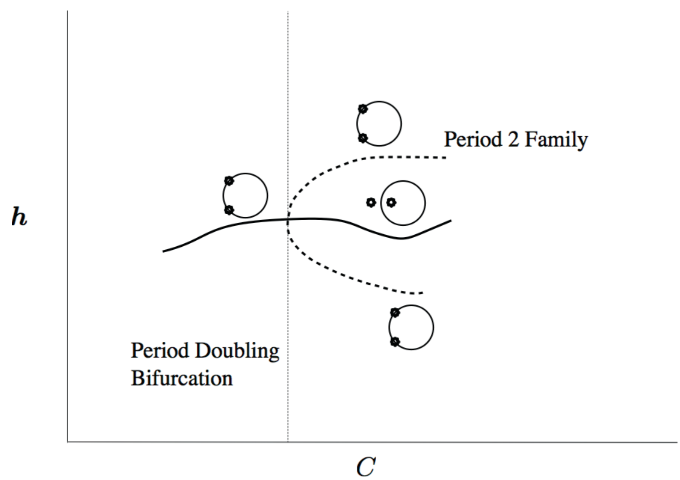

Fig. 5.6 Period doubling stability bifurcation.

Higher-period Intersections

Passage of θ through π corresponds to the birth of a period-2 family. Similarly passage through special values of $\theta = 2\pi m/n$ may result in n-period families to bifurcate. Whether such higher-period fixed points bifurcate or not, it is important to note that the base fixed point will not undergo any stability transitions for these bifurcations, and indeed will seemingly remain unaware of whether such bifurcations are occurring.

Higher-dimension Systems

Similar reasoning applies to systems with three or more degrees of freedom, except there exists one additional type of intersection. Here, it is possible for two sets of eigenvalues to intersect along the unit circle and bifurcate into an unstable periodic orbit with a set of two-dimensional stable and unstable spiral manifolds. Since these stability bifurcations do not occur along the real axis, continuation of the family is not a problem. It is interesting to note that there are rigorous conditions for when such an intersection will result in a stability bifurcation. Specifically, Krein's Theorem states that if the pair of colliding eigenvalues on the unit circle are traveling in the same direction, then a stability bifurcation will not occur. If they are traveling in the opposite direction, then a bifurcation may occur [5].

6. Solution and Characterization Methods

In the following we discuss various methods by which the equations of motion can be solved to find the trajectory of a particle or a solution flow in the neighborhood of a trajectory. We make an implicit assumption that the equations we wish to solve are non-integrable, meaning that there exists no closed-from solution. In many situations we will rely on a closed-form solution of a simpler, non-perturbed system to help generate a solution to a more general problem. As the given problems are non-integrable, we are always limited in that our solutions and solution methods are approximate, and only represent the true solution up to some degree of precision over a finite timespan. Some exceptions to these occur for the special solutions of dynamical systems, namely (relative) equilibria and periodic orbits. A relative equilibria is an exact solution to a system, and in general does not rely on a solution of the simpler, non-perturbed problem. A periodic orbit is not exact, as it has a limit on the precision (except in the rare cases where a completely convergent series is found in closed form), although there is no limit on the timespan for the solution, that being infinite. All other solutions have limitations on both precision and time.

Our goal in constructing solutions is not to find single trajectories of arbitrary precision – that can already be done using numerical techniques. Rather, our goal is to represent a solution and the neighborhood about that solution, since that is the most important aspect of our dynamical understanding of motion.

In the following we will sometimes not specifically assume any structure for the equations of motion, due to the generality of many solution techniques, but can assume a generic formulation of the equations of motion as $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \mathbf{R}^{2n}$ and $\boldsymbol{f} : \mathbf{R}^{2n} \times \mathbf{R} \to \mathbf{R}^{2n}$. In other instances we will include an added term representing perturbations from an integrable problem. Exceptions to this convention will be noted. We will also use the Hamiltonian formalism when appropriate.

6.1 Numerical Integration

The most direct, and in many ways simplest and most precise, method to solve for the motion of a particle given the equations of motion is to numerically integrate the trajectory. A numerical integration takes a current value of the state and the dynamics function, which must be computable as a function of the state, and generates a discrete map in time to a subsequent (or previous) state. For a given numerical integrator there is always an error associated with this map; however, this error can be driven to a small value by appropriate choice of the time step and algorithm. The field of numerical analysis is far too comprehensive to be easily summarized, but for astrodynamics applications there are a few main approaches that are used, described in brief later. The main point is that the field is sufficiently advanced so that there is no reason to doubt the stated numerical precision of a solution, given the caveats associated with the particular numerical scheme. Thus, one can consider the output of a numerical integrator to be a valid solution to the differential equations, to within some degree of precision. It should not be mistaken with the actual solution, which will have additional properties that the numerical solution may not enjoy, such as symplecticity and analyticity, yet for a given initial state it is an accurate representation of the solution. As such, these solutions can be used to test analytical results, to provide precise predictions or reconstructions of a particular trajectory, and can be used as a tool for analysis.

Technically, the solution that comes out of a numerical integrator is a series of discrete states $\boldsymbol{x}_i$ at given time steps, t_i, where $i = 0, 1, \ldots, N$, t_0 is the initial time, and t_N is the final time (where $t_N < t_0$ can occur without ambiguity). Most numerical routines also provide a means by which to interpolate between the states $\boldsymbol{x}_i$ and $\boldsymbol{x}_{i+1}$, sometimes using neighboring states to improve the interpolation. This provides the illusion of a continuous solution, whereas the interpolated points will in general have lower precision than the discrete points output by the system, and may not share the fundamental properties of the solutions as they are not directly constructed from the equations of motion. It is important to note that the total timespan of a given map, $t_N - t_0$, need not be an integer number of some fundamental time step, as almost all integrators have the capability to adjust the time step to achieve any desired end point in time. Because of this a numerical solution can be represented as if it were a continuous time solution, $\boldsymbol{x}(t) = \boldsymbol{\phi}(t; \boldsymbol{x}_o, t_o)$, where the solution is also subject to the chosen parameters of the system. This allows us to represent a numerical solution precisely in the same way as an analytical solution would be represented.

6.1.1 Lagrangian, Hamiltonian and Orbit Element Computations

If the system is defined by a Lagrangian in terms of coordinates and their time rate of change, $L(\boldsymbol{r}, \dot{\boldsymbol{r}}, t)$, then the corresponding equations of motion are found by solving Lagrange's equation $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{r}}}\right) - \frac{\partial L}{\partial \boldsymbol{r}} = 0$ for the acceleration, resulting in a set of equations $\ddot{\boldsymbol{r}} = \boldsymbol{F}(\boldsymbol{r}, \dot{\boldsymbol{r}}, t)$. Then make the usual transformation of a second-order differential equation into a first-order by defining $\boldsymbol{x} = [\boldsymbol{r}, \boldsymbol{v}]$, where $\dot{\boldsymbol{r}} = \boldsymbol{v}$, and

$\boldsymbol{f} = [\boldsymbol{v}, \boldsymbol{F}(\boldsymbol{r}, \boldsymbol{v}, t)]$. Hamilton's equations with a Hamiltonian function $H(\boldsymbol{q}, \boldsymbol{p}, t)$ are already in the appropriate form, as if given $\boldsymbol{x} = [\boldsymbol{q}; \boldsymbol{p}]$ the equations $\dot{\boldsymbol{x}} = JH_{\boldsymbol{x}}$ follow, where $\boldsymbol{f}(\boldsymbol{x}, t) = JH_{\boldsymbol{x}}$.

We also consider the orbit element forms of the equations that arise from the Lagrange and Gauss equations. In these one can generically assemble the orbit elements into a single state vector, for the classical orbit elements this could be $\boldsymbol{x} = [a, e, i, \omega, \Omega, \tau_o]$. Then both sets of equations can be immediately written into general form. In fact, there is a bit more structure to the equations than that. Lagrange's planetary equations have the form $\dot{\boldsymbol{x}} = F_L(\boldsymbol{x})\frac{\partial R}{\partial \boldsymbol{x}}$, where F_L is a 6×6 matrix that is a nonlinear function of the orbit elements but not time, and $R(\boldsymbol{x})$ is the perturbing potential. Similarly, the Gauss equations take the form $\dot{\boldsymbol{x}} = F_G(\boldsymbol{x}, t)\boldsymbol{a}$, where F_G is now a 6×3 matrix that is a function of the state $\boldsymbol{x}$ and time, while $\boldsymbol{a} \in \mathbf{R}^3$ is the applied acceleration. Once in these forms the equations of motion can be directly passed to a generic numerical integration scheme.

6.1.2 State Transition Matrix Computations

Note that the state transition matrix is a crucial component of the theory. The state transition matrix can be analytically solved for only in the case where the nominal solution is an equilibrium point of a time-invariant system. For any other solution one must numerically integrate to find $\Phi(t, t_o)$. Already discussed is the differential equation which Φ satisfies, which is always derived from the fundamental equations of motion as

$$\dot{\Phi}(t, t_o) = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_t \Phi(t, t_o) \tag{6.1}$$

$$\Phi(t_o, t_o) = I \tag{6.2}$$

where the solution of the nominal trajectory $\boldsymbol{\phi}(t; \boldsymbol{x}_o, t_o)$ must be substituted into the dynamics matrix

$$A(t) = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_t$$

at each point of time. It is often convenient to integrate both the state $\boldsymbol{x}(t)$ and the state transition matrix simultaneously, thus evolving the dynamics associated with a point in phase space and the linear description of the dynamics about that point.

Depending on what form of the equations of motion are being integrated, the state transition matrix may not be symplectic. However, for many cases the state transition matrix can be simply transformed into a symplectic form. For example consider a Lagrangian framework, where

$$\Phi(t, t_o) = \frac{\partial(\boldsymbol{r}(t), \boldsymbol{v}(t))}{\partial(\boldsymbol{r}_o, \boldsymbol{v}_o)}$$

Unless the position and velocity are relative to an inertial frame, this will not be a symplectic matrix. Recall the simple Legendre Transformation, where the current position vector can be taken as the Hamiltonian coordinate $\boldsymbol{q}(t) = \boldsymbol{r}(t)$ and the momentum is defined as the partial of the Lagrangian with respect to the velocity, $\boldsymbol{p}(t) = \frac{\partial L}{\partial \boldsymbol{v}}$. Using these definitions one can express a symplectic state transition matrix in terms of the Lagrangian variables as

$$\frac{\partial(\boldsymbol{q}(t), \boldsymbol{p}(t))}{(\boldsymbol{q}_o, \boldsymbol{p}_o)} = \begin{bmatrix} I & 0 \\ \left.\dfrac{\partial^2 L}{\partial \boldsymbol{v} \partial \boldsymbol{r}}\right|_t & \left.\dfrac{\partial^2 L}{\partial \boldsymbol{v}^2}\right|_t \end{bmatrix} \Phi(t, t_o) \begin{bmatrix} I & 0 \\ \left.\dfrac{\partial^2 L}{\partial \boldsymbol{v} \partial \boldsymbol{r}}\right|_{t_o} & \left.\dfrac{\partial^2 L}{\partial \boldsymbol{v}^2}\right|_{t_o} \end{bmatrix}^{-1} \tag{6.3}$$

where the state transition matrix Φ is computed for the Lagrangian variables.

6.1.3 Common Integrators

The field of numerical integration continues to be an area of innovation and research advancement. Thus in the following only a few different types of numerical integration routines are mentioned, leaving the interested reader to decide for themselves which one to use. Each tends to have advantages that suit themselves for different applications. The day when one single approach to numerical integration will be used is still apparently in the distant future.

Variable Step Predictor–Corrector Routines

Perhaps the most widely used numerical integrators are those that simultaneously develop a prediction and an estimate of the current error in the numerical integration. Based on the predicted error and the desired tolerance in the computation the time step can be adjusted to be longer or shorter, varying this as the solution is being computed. A common formulation for these integrators are the Runga–Kutta methods, and they are widely available in pre-packaged computational environments. These methods are often quite useful for astrodynamics problems where there can be relatively long periods of small perturbations interspersed with short periods of more intense interactions between bodies. When numerically integrating high-eccentricity orbits this occurs quite frequently, with longer time steps being acceptable through apoapsis and short time steps through periapsis.

Fixed-Step Integrators

Another school of thought is willing to forgo speed for greater precision in computation. Along these lines are the extremely accurate fixed-step integration methods such as Burlisch–Stoer. Use of these algorithms in the literature is usually associated with computations that strive to achieve machine precision over long timespans [52, 53].

Symplectic or Variational Integrators

More recently, the development and application of symplectic or variational integrators has gained much attention. These are integration methods where the discrete time steps are chosen so that they conserve some fundamental property of the dynamical system. For instance, symplectic integrators have been developed for the integration of Hamiltonian systems and preserve the symplectic structure of the underlying equations of motion. These approaches have been further generalized recently to conserve other quantities that might be associated with the geometry of motion or the underlying dynamical system. When symplecticity of the state transition matrix is important it is possible to implement an integration scheme for just that matrix which will ensure symplecticity independent of how the nominal trajectory is integrated [184]. A useful review of numerical integrations, with a definite bias toward variational integrators, is provided in [60].

6.1.4 Verification of a Numerical Integrator's Performance

More important than what numerical integration routine is used is that the researcher performs verification and validation tests of their numerical integrations to ensure that the algorithms have been implemented correctly and that adequate error tolerances have been chosen. For astrodynamical problems, which usually are dealing with a system perturbed from two-body motion, an effective way to verify integration performance is to directly integrate the two-body problem. As this problem is integrable, it is possible to transform the system at each point into orbit elements and verify that they remain conserved for stressful integration cases such as high-eccentricity orbits. For systems that have a Jacobi integral defined for the full system, it is usually also good practice to confirm that the integration scheme keeps the integral constant. These are not proofs of the integrator's accuracy, but are effective ways to verify performance.

For integrations of the state transition matrix it is also useful to perform validations of the numerical accuracy of its computation. For all dynamical systems without dissipation, the determinant of the state transition matrix should generically equal 1, and this can be used as a check. Additionally, for a nominally symplectic matrix it is possible to check whether the matrix verifies the symplectic condition, and view deviation from this either as an indication of poor error tolerances or a mistake in the implementation.

6.2 Computation of Equilibrium Solutions

6.2.1 General Algorithm

Given a dynamical system reduced to Hamiltonian form, $\dot{\boldsymbol{x}} = JH_{\boldsymbol{x}}$, an equilibrium solution is a particular state of the system, $\boldsymbol{x}^*$, which causes the dynamics function to be identically zero. The defining equation is:

$$H_{\boldsymbol{x}}(\boldsymbol{x}^*) = \boldsymbol{0} \tag{6.4}$$

where the Hamiltonian is assumed to be time-invariant and hence is a constant of the motion. If the equation is written as a Lagrangian system, the equilibrium condition can be reduced to a set of algebraic equations of the form

$$\frac{\partial L(\boldsymbol{r}^*, \mathbf{0})}{\partial \boldsymbol{r}} = \mathbf{0}, \tag{6.5}$$

where this is a three-dimensional equation instead of the six-dimensional Hamiltonian form. Thus, for a specific problem, it is often easier to solve the Lagrangian form for zero accelerations in the configuration space, as the equilibrium condition for the velocities is trivially satisfied.

When the dynamical system has no integrals of motion other than the Jacobi integral (for a time invariant Hamiltonian or Lagrangian), then equilibrium points are in general non-degenerate, or $|H^*_{\boldsymbol{xx}}| \neq 0$. This implies that a Newton–Raphson method can be applied to compute the equilibrium points. Specifically, given a test value for an equilibrium point $\boldsymbol{x}_i$, search for the correction $\delta\boldsymbol{x}_i$ that will satisfy the condition: $H_{\boldsymbol{x}}(\boldsymbol{x}_i + \delta\boldsymbol{x}_i) = \mathbf{0}$. If "close" to the solution (i.e., if a reasonable guess for the point is provided) it is possible to expand the function in a Taylor Series: $H_{\boldsymbol{x}}(\boldsymbol{x}_i) + H_{\boldsymbol{xx}}(\boldsymbol{x}_i) \cdot \delta\boldsymbol{x}_i + \ldots = \mathbf{0}$. Solution of this equation at the linear order provides the update:

$$\delta\boldsymbol{x}_i = -H_{\boldsymbol{xx}}(\boldsymbol{x}_i)^{-1} \cdot H_{\boldsymbol{x}}(\boldsymbol{x}_i) \tag{6.6}$$

and a new test solution is computed as $\boldsymbol{x}_{i+1} = \boldsymbol{x}_i + \delta\boldsymbol{x}_i$ and the procedure is iterated until the equilibrium condition is satisfied to an acceptable degree of precision.

6.2.2 Complications from Additional Integrals

The presence of an integral of motion in addition to the Jacobi integral complicates this procedure and ensures that the above algorithm will fail. As seen earlier in Chapter 5 the state transition matrix expanded about an equilibrium solution has a pair of unity eigenvalues when an additional integral of motion exists. Specifically, for an integral of motion linearly independent from the Hamiltonian, $h(\boldsymbol{x}) = C_h$, the state transition matrix evaluated at an equilibrium $\boldsymbol{x}^*$ satisfies the following condition: $h_{\boldsymbol{x}}(\boldsymbol{x}^*) \cdot [I - \Phi(t, t_o; \boldsymbol{x}^*)] = 0$. Thus, $h_{\boldsymbol{x}}$ is a left eigenvector of Φ with a unity eigenvalue. The symmetry of Hamiltonian systems ensures that there exists another unity eigenvalue associated with the matrix with a right eigenvector $Jh_{\boldsymbol{x}}(\boldsymbol{x}^*)$. As the associated linear system is time-invariant the explicit solution exists

$$\Phi(t, t_o; \boldsymbol{x}^*) = e^{JH^*_{\boldsymbol{xx}}(t-t_o)} \tag{6.7}$$

$$= I + (t - t_o)JH^*_{\boldsymbol{xx}} + \frac{1}{2}(t - t_o)^2 \left[JH^*_{\boldsymbol{xx}}\right]^2 + \ldots \tag{6.8}$$

Substitution of this form of the solution into the eigenvalue equations yields the following conditions:

$$H^*_{\boldsymbol{xx}} \cdot J \cdot h^*_{\boldsymbol{x}} = 0 \tag{6.9}$$

Thus, the Hamiltonian is degenerate at an equilibrium with two zero eigenvalues and any numerical technique using the Newton–Raphson method to compute the point will become ill-defined as the test solution comes close to the equilibrium.

This happens as associated with every equilibrium point of a system with an additional integral is a family of neighboring equilibrium points at the same value of energy and same value of the integral $h(\boldsymbol{x})$. Specifically, suppose an equilibrium solution is given such that $H^*_{\boldsymbol{x}} = \mathbf{0}$ and $h(\boldsymbol{x}^*) = C_h$. Consider an arbitrary variation from this point $\delta\boldsymbol{x}$, the question is whether there exists a neighboring point that will satisfy both the equilibrium condition and have the same value of the integral h. If so, then there should be a solution satisfying the two conditions $H^*_{\boldsymbol{xx}} \cdot \delta\boldsymbol{x} = 0$ and $h^*_{\boldsymbol{x}} \cdot \delta\boldsymbol{x} = 0$. As the Hamiltonian is degenerate at the equilibrium point the first condition is satisfied by the eigenvector $\delta\boldsymbol{x} = Jh^*_{\boldsymbol{x}}$, meaning that there is another equilibrium point lying in this direction. This vector direction also satisfies $h^*_{\boldsymbol{x}} \cdot Jh^*_{\boldsymbol{x}} = 0$ trivially. Thus, there is not a unique point to which the algorithm can converge, and hence the procedure becomes ill-defined.

This demonstration also suggests a specific approach that can be used to construct a more robust computation algorithm. Instead of searching over the full state direction $\delta\boldsymbol{x}$, one can restrict the search to the direction $\delta\boldsymbol{x}_\perp$ defined such that $\delta\boldsymbol{x}_\perp \cdot Jh_{\boldsymbol{x}}(\boldsymbol{x}) = 0$, i.e., not allowing variations along the line of continuous equilibria. Practically speaking, this equation can be solved in terms of one variation direction to define a $2n-1$ space in which to search. It is also possible to make a further reduction, by choosing to eliminate a second direction by further constraining $\delta\boldsymbol{x}_\perp$ by the condition $h_{\boldsymbol{x}} \cdot \delta\boldsymbol{x}_\perp = 0$, maintaining a constant value of h. However, this presupposes that a value of h has been chosen that allows an equilibrium to exist, something that is not necessarily true.

An explicit example of this form of degeneracy can be given. Consider the relative two-body problem in a rotating reference frame with the Hamiltonian function $H = \frac{1}{2}\left(P_r^2 + \frac{1}{r^2}P_\theta^2\right) - \omega P_\theta - \frac{\mu}{r}$, where ω is the rotation rate. Due to the absence of θ the momentum P_θ is an integral of motion. The second, associated integral of motion for θ exists by quadratures once the solution for r and P_r is found. Applying the equilibrium condition yields

$$H_{\boldsymbol{x}} = \begin{bmatrix} -\frac{1}{r^3}P_\theta^2 + \frac{\mu}{r^2} \\ 0 \\ P_r \\ \frac{1}{r^2}P_\theta - \omega \end{bmatrix} \tag{6.10}$$

$$H_{\boldsymbol{xx}} = \begin{bmatrix} \frac{3}{r^4}P_\theta^2 - \frac{2\mu}{r^3} & 0 & 0 & -\frac{2}{r^3}P_\theta \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\frac{2}{r^3}P_\theta & 0 & 0 & \frac{1}{r^2} \end{bmatrix} \tag{6.11}$$

It can be verified that the matrix $H_{\boldsymbol{xx}}$ has two zero eigenvalues. The gradient of h is $h_{\boldsymbol{x}} = [0, 0, 0, 1]$ and $Jh_{\boldsymbol{x}} = [0, 1, 0, 0]$. If the search is constrained in the direction normal to $Jh_{\boldsymbol{x}}$ this implies that θ is fixed. If the search in the direction $h_{\boldsymbol{x}}$ is constrained, this implies that P_θ is fixed. Doing so provides us with two equations which can be solved to find $P_r = 0$ and $r = P_\theta^2/\mu$, with an ancillary condition

$\frac{1}{r^2}P_\theta - \omega = 0$. Note that an improper choice of P_θ may yield an inconsistency with one of these equations. One can either use the final equation to define the proper value of P_θ, something which is trivial for the given simple problem, or not be restricted to searching for it at a specified value of P_θ to begin with, which allows one to simultaneously solve for r, P_r and P_θ. For more general systems where the solutions are not as easily found, it is preferred to solve for the value of the integral simultaneously with the solution of the other states. Then the full solution is found to be: $r^* = \left(\mu/\omega^2\right)^{1/3}$, $P_r^* = 0$, $P_\theta^* = \left(\mu^2/\omega\right)^{1/3}$.

6.2.3 Stability Computation

Once an equilibrium point of the system is found, then the stability of that point can be determined by evaluating the eigenvalues of the matrix $JH^*_{\boldsymbol{xx}}$, evaluated at the equilibrium point. Only in the simplest cases can the eigenvalues and eigenvectors of this matrix be determined analytically. It is noted that even in the restricted three-body problem, the Euler equilibrium solutions must be found numerically for a precise evaluation of them. Thus, it is usually easiest to use a computational toolbox to evaluate the eigenvalues and eigenvectors of the dynamics matrix at the equilibrium point. Once these are found the methods outlined in the previous chapter for describing motion in the vicinity of the equilibrium point can be applied.

6.3 Computation of Periodic Orbits

6.3.1 General Algorithm

The traditional approach to analytical or numerical computation of periodic orbits relies heavily on the symmetric properties of the forces in the system [70, 65, 18]. Examples of this include the computation of periodic orbits in the restricted three-body problem and the Hill problem, best exemplified by the seminal work of Hénon. As a direct application to asteroids, this approach has been used to compute symmetric periodic orbits about tri-axial ellipsoids [153]. For application to "real" objects, however, there are no symmetries in the force fields and the periodic orbit computation must be generalized to a form that does not rely on such symmetries. The algorithm of Deprit and Henrard [31] views periodic orbits in terms of their intrinsic differential geometric properties and enables such computations, as applied by Lara [86] and others. In the following we present a more direct approach that we find to be more intuitive and which only requires the computation of the state transition matrix.

For the computation of a periodic orbit it is necessary to solve the equation:

$$\boldsymbol{x} = \phi(T; \boldsymbol{x}, 0) \tag{6.12}$$

for a given period T. The standard approach is to assume an initial state $\boldsymbol{x}_0$ that is sufficiently close to, but does not solve, the periodicity condition. Often this initial estimate can be found by applying simple two-body theory to the perturbed problem. Then assume that a small correction, $\delta \boldsymbol{x}$, may solve the problem

$$\boldsymbol{x}_0 + \delta \boldsymbol{x} = \phi(T; \boldsymbol{x}_0 + \delta \boldsymbol{x}, 0) \tag{6.13}$$
$$= \phi(T; \boldsymbol{x}_0, 0) + \Phi(T, 0)\delta \boldsymbol{x} + \dots \tag{6.14}$$

If the higher-order terms are ignored this equation can be rearranged to

$$[I - \Phi(T, 0)]\, \delta \boldsymbol{x} = \phi(T; \boldsymbol{x}_0, 0) - \boldsymbol{x}_0 \tag{6.15}$$

where the right-hand side is not exactly zero. At this point, prior to a periodic orbit having been converged on, the matrix on the left is generally non-singular and can be inverted to yield an estimate for the correction. Then the trial solution can be updated as $\boldsymbol{x}_1 = \boldsymbol{x}_0 + \delta \boldsymbol{x}$.

Normally this process would be repeated until the mismatch $|\phi(T, 0, \boldsymbol{x}_i) - \boldsymbol{x}_i| < \epsilon$. If the system is time periodic, then the matrix $\Phi(T, 0)$ will, generically, not have unity eigenvalues, even when the periodic orbit has been converged upon, and is itself the monodromy matrix for that orbit. Thus, in this sense, computation of periodic orbits in time periodic systems are much easier than in time-invariant systems, although generation of the appropriate initial conditions can be difficult. First, the period of the motion must be equal to an integer multiple of the system period and thus is immediately specified. Second, a Newton–Raphson method using the full state transition matrix works well to converge upon a periodic orbit, assuming the initial estimate is chosen sufficiently close to a periodic motion. For perturbed orbital problems, it is often sufficient to choose the motion that would be periodic in the two-body problem as a first estimate for the algorithm.

For a time-invariant system there are additional challenges. First, the period of the orbit is a free parameter. Second, and more importantly, when the solution tends towards a periodic orbit, two eigenvalues of Φ tend towards unity and, if at a periodic orbit, would equal unity. Thus, as the target state is approached the matrix $[I - \Phi]$ becomes singular and the algorithm diverges. One remedy would be to choose the pseudo-inverse of the singular matrix. An alternate, and constructive approach, is to remove the offending unity eigenvalues by restricting the linear variation to a Poincaré map. The advantage of the latter approach is that it defines the monodromy matrix of the periodic orbit once computed. This approach, first detailed in [175], is presented in the following.

6.3.2 Reduction of the State Transition Matrix to a Linear Poincaré Map

The reduction is easiest to derive for a Lagrangian system, but can be generalized for a Hamiltonian system as well. In the following assume that the full Lagrangian state is $\boldsymbol{x} = [\boldsymbol{r}, \dot{\boldsymbol{r}}]$, that the Lagrangian is time-invariant, $L(\boldsymbol{r}, \dot{\boldsymbol{r}})$, and thus a Jacobi integral exists which can be evaluated as the Hamiltonian of the system, $H(\boldsymbol{x}) = \dot{\boldsymbol{r}} \cdot \frac{\partial L}{\partial \dot{\boldsymbol{r}}} - L$, albeit with positions and velocities instead of coordinates and momenta. Assume a simple surface of section, chosen to equal a coordinate value $S(\boldsymbol{x}) = r_I = 0$, where $I = 1, 2,$ or 3. Transversality in this case means that $S_{\boldsymbol{x}} \cdot \dot{\boldsymbol{x}} = \dot{r}_I \neq 0$. Using the simple example of the surface of section being $r_I = 0$, its associated speed v_I is then eliminated from the Jacobi integral. Note that for most Lagrangian systems encountered in astrodynamics $\frac{\partial H}{\partial v_I} = v_I$ and is non-zero at the surface of section by definition, and thus it is usually simple to eliminate this term. Then the reduced state is $\boldsymbol{y} = [r_J, r_K, \dot{r}_J, \dot{r}_K]$ where $J, K \neq I$ and do not equal each other.

The procedure starts with choosing an initial state $\boldsymbol{x}_0$ such that $S(\boldsymbol{x}_0) = 0$ and $H(\boldsymbol{x}_0) = C$, a specified value of energy. The trajectory is then propagated until a time t_1 when $S(\boldsymbol{x}(t_1)) = 0$. The state transition matrix is propagated simultaneously to find $\Phi(t_1, t_0)$, where this will map deviations in the initial state to deviations in the final state at time t_1, or $\delta\boldsymbol{x}(t_1) = \Phi(t_1, t_0)\delta\boldsymbol{x}_0$. The initial deviation is $\delta\boldsymbol{x}_0 = [\delta r_J, \delta r_K, \delta r_I, \delta v_J, \delta v_K, \delta v_I]$. It is trivial to choose $\delta r_I = 0$, thus keeping the initial state on the surface of section. The energy of the deviated trajectory must also be restricted to a constant value, or $H(\boldsymbol{x}_0 + \delta\boldsymbol{x}_0) = H(\boldsymbol{x}_0)$. Expanding the Hamiltonian and simplifying yields the condition at the linear order $H_{\boldsymbol{x}}|_0 \cdot \delta\boldsymbol{x}_0 = 0$. Expanding this in terms of the reduced state results in $H_{\boldsymbol{y}}\big|_0 \cdot \delta\boldsymbol{y}_0 + H_{v_I}|_0\, \delta v_I = 0$, which can be solved for the necessary variation in the initial velocity v_I to maintain a constant value of energy

$$\delta v_I = -\frac{1}{H_{v_I}|_0}\, H_{\boldsymbol{y}}\big|_0 \cdot \delta\boldsymbol{y}_0 \tag{6.16}$$

Now a linear mapping can be constructed that takes a variation in the reduced state, $\delta\boldsymbol{y}_0$, and maps it into a variation in the initial state $\delta\boldsymbol{x}_0$ that can be mapped forward using the full state transition matrix

$$\delta\boldsymbol{x}_0 = (P_0 + P_H)\, \delta\boldsymbol{y}_0 \tag{6.17}$$

where P_0 and P_H are $2n \times (2n-2)$ matrices in general that enforce the initial projection onto the surface of section and constant energy surface. Specifically the matrices have the following form for the assumed configuration

$$P_0 = \left[\begin{array}{c|c} I_{n-1,n-1} & 0_{n-1,n-1} \\ \hline 0_{1,n-1} & 0_{1,n-1} \\ \hline 0_{n-1,n-1} & I_{n-1,n-1} \\ \hline 0_{1,n-1} & 0_{1,n-1} \end{array}\right] \tag{6.18}$$

$$P_H = -\hat{\boldsymbol{w}}_{I+n}\, H_{\boldsymbol{y}}\big|_0 \tag{6.19}$$

Note that the matrices in P_0 all have their subscripts defined for both the identity matrices and the zero matrices. In the equation for P_H for the example considered, $\hat{\boldsymbol{w}}_{I+n}$ is the zero vector with a 1 in the $I+n$th row and in general is the unit vector along the eliminated velocity component. Then P_H is formed by taking the outer product of the two vectors, creating a matrix of size $2n \times 2(n-1)$.

This vector is then applied to the state transition matrix to find

$$\delta\boldsymbol{x}(t_1) = \Phi(t_1, t_0)\, (P_0 + P_H)\, \delta\boldsymbol{y}_0$$

The state deviation at t_1 automatically satisfies energy conservation, or $H_{\boldsymbol{y}}\big|_1 \cdot \delta\boldsymbol{x}(t_1) = 0$, and thus the value of $\delta v_I(t_1)$ can be ignored. The same does not hold for the surface of section constraint, however, and $S(\boldsymbol{x}(t_1)+\delta\boldsymbol{x}(t_1)) = S_{\boldsymbol{x}}|_1 \cdot \delta\boldsymbol{x}(t_1) \neq 0$ in general. In fact, for an arbitrary variation of $\delta\boldsymbol{y}_0$ there will only be a line of initial conditions that have the deviated state at t_1 on the surface of section, with the rest of the states either having already passed through the surface or not having reached it yet (see Fig. 6.1). This occurs generically as the time from one surface of section passage to the next is not constant over linear variations, but in general will change with the initial state.

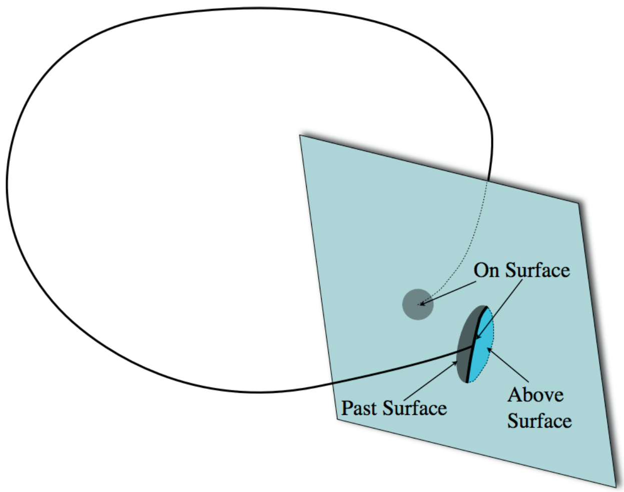

Fig. 6.1 First return map of a solution flow to a surface of section. The initial set is on the surface but the first return at the nominal time only intersects the surface along a line, with the other half of the flow having passed through the surface already and the other not having passed through yet.

To address this allow the time of passage to vary in order to force the deviated trajectory to cross the surface of section

$$S(\boldsymbol{x}(t_1+\delta t)+\delta\boldsymbol{x}(t_1)) = S(\boldsymbol{x}(t_1)) + S_{\boldsymbol{x}}|_1 \cdot \left[\frac{\partial \boldsymbol{x}(t_1)}{\partial t}\delta t + \delta\boldsymbol{x}(t_1)\right] \tag{6.20}$$

Restricting to linear terms, equating this to zero, and noting that $\partial\boldsymbol{x}(t_1)/\partial t = \dot{\boldsymbol{x}}(t_1)$ results in the condition

$$S_{\boldsymbol{x}}|_1 \cdot \dot{\boldsymbol{x}}(t_1)\delta t + S_{\boldsymbol{x}}|_1 \cdot \delta\boldsymbol{x}(t_1) = 0 \tag{6.21}$$

which can be solved for the necessary time variation as a function of the deviated final state

$$\delta t = -\frac{1}{S_{\boldsymbol{x}}|_1 \cdot \dot{\boldsymbol{x}}(t_1)}\, S_{\boldsymbol{x}}|_1 \cdot \delta\boldsymbol{x}(t_1) \tag{6.22}$$

Given the fundamental transversality assumption, $S_{\boldsymbol{x}}|_1 \cdot \dot{\boldsymbol{x}}(t_1) = v_I(t_1) \neq 0$.

Now the final deviated state can be adjusted to ensure that it lies on the surface of section

$$\delta \boldsymbol{x}_1 = \delta \boldsymbol{x}(t_1) + \dot{\boldsymbol{x}}(t_1)\delta t \tag{6.23}$$

$$= \left[I - \frac{1}{S_{\boldsymbol{x}}|_1 \cdot \dot{\boldsymbol{x}}(t_1)} \dot{\boldsymbol{x}}(t_1)\ S_{\boldsymbol{x}}|_1 \right] \cdot \delta \boldsymbol{x}(t_1) \tag{6.24}$$

$$= P_S \cdot \delta \boldsymbol{x}(t_1) \tag{6.25}$$

Note that the final deviation $\delta \boldsymbol{x}_1$ has been defined to denote the deviation that lies on the surface of section. Also, note that the mapping matrix P_S is a $2n \times 2n$ matrix and is full rank in general. Finally, note that the term $\dot{\boldsymbol{x}}(t_1)\ S_{\boldsymbol{x}}|_1$ is an outer product of two vectors.

Thus, the final deviation $\delta \boldsymbol{x}_1$ now both conserves energy and lies completely on the surface of section. Thus, for the final step this vector can be projected onto the reduced state to find the final deviation of the linearized Poincaré map

$$\delta \boldsymbol{y}_1 = P_0^T P_S \Phi(t_1, t_0) \left(P_0 + P_H \right) \delta \boldsymbol{y}_0 \tag{6.26}$$

This has explicitly defined a reduced linear map from the surface of section to the surface of section in the vicinity of the nominal trajectory

$$\Phi_{1,0} = P_0^T P_S \Phi(t_1, t_0) \left(P_0 + P_H \right) \tag{6.27}$$

At this point, if $\Phi_{1,0}$ is required in a symplectic form, the linear transformation discussed previously from a Lagrangian system to a Hamiltonian can be implemented (Eq. 6.3). It is essential to note that, if evaluated at a fixed point, the matrix $\Phi_{1,0} = \Phi_M$ and will have both of the generic unity eigenvalues removed from it.

6.3.3 General Algorithm Revisited

Having a properly reduced linear Poincaré map the computation scheme can be reconsidered, now restricted to the reduced state. Consider the nonlinear Poincaré map from an initial point $\boldsymbol{y}_0$ and fixed energy C to the next crossing

$$\boldsymbol{y}_1 = \boldsymbol{g}(\boldsymbol{y}_0, C) \tag{6.28}$$

Now, consider a deviation in the initial state that makes this into a fixed point map

$$\boldsymbol{y}_0 + \delta \boldsymbol{y}_0 = \boldsymbol{g}(\boldsymbol{y}_0 + \delta \boldsymbol{y}_0, C) \tag{6.29}$$

$$= \boldsymbol{y}_1 + \Phi_{1,0} \delta \boldsymbol{y}_0 + \dots \tag{6.30}$$

assuming the algorithm to compute $\Phi_{1,0}$. The initial deviation is chosen to satisfy this equation, leading to

$$\delta \boldsymbol{y}_0 = [I - \Phi_{1,0}]^{-1} (\boldsymbol{y}_1 - \boldsymbol{y}_0) \tag{6.31}$$

and the test solution is iterated as $\boldsymbol{y}_1 = \boldsymbol{y}_0 + \delta \boldsymbol{y}_0$ and the process repeated. Now, if the initial state is within the basin of attraction of the fixed point, the process will converge and, as discussed previously, the matrix $I - \Phi_{1,0}$ will generically remain non-singular and as $\boldsymbol{y}_0 \to \boldsymbol{y}^*$ then $\Phi_{1,0} \to \Phi_M$, the monodromy matrix.

6.3.4 Stability Computation

Once the fixed point is converged to a specified accuracy the monodromy matrix Φ_M is also automatically defined by reduction of the original state transition matrix. Stability of the resulting fixed point is then determined by computing the eigenvectors and eigenvalues of Φ_M. Again, it is usually easiest to perform such computations using a standard software package.

6.3.5 Families of Periodic Orbits

Having computed a periodic orbit at a particular value of a parameter or energy value, the question arises as to how this single object can be generalized to a larger family of such solutions, generated as the parameter value is changed. From the earlier discussion on family continuation, the first estimate of the new fixed point at energy $C + \delta C$ is given as $\delta \boldsymbol{y}^* + [I - \Phi_M]^{-1} \boldsymbol{g}_C^* \delta C$. The remaining issue is how to compute

$$\boldsymbol{g}_C^* = \frac{\partial \boldsymbol{g}(\boldsymbol{y}^*, C)}{\partial C}$$

For the example system from above, and for the reduced state in general, there is one coordinate which is eliminated and which has a non-zero gradient along the Hamiltonian. Thus

$$\delta C = \frac{\partial H}{\partial v_I} \delta v_I$$

where the velocity coordinate is chosen to ensure that this gradient is non-zero. As the gradient term is a scalar, it can be solved directly for the necessary variation in the velocity component for a given change in energy,

$$\delta v_I = \frac{1}{H_{v_I}} \delta C$$

Next, note that the velocity v_I can be varied independently of the reduced state variation, which can remain zero in general. Thus, the deviation in the final reduced state is equal to

$$\delta \boldsymbol{y}_1 = P_0^T P_S \Phi(t_1, t_0) \hat{\boldsymbol{w}}_{I+3} \frac{\delta C}{H_{v_I}} \tag{6.32}$$

where the unit vector $\hat{\boldsymbol{w}}_{I+3}$ is as described before. From this the ratio $\delta \boldsymbol{y}_1/\delta C$ can be directly computed to find the necessary partial

$$\boldsymbol{g}_C^* = P_0^T P_S \Phi(t_1, t_0) \hat{\boldsymbol{w}}_{I+3} \frac{1}{H_{v_I}} \tag{6.33}$$

It is also possible to continue periodic orbit families in parameters of the equations of motion. The procedure is largely the same as outlined here for tracing out a family as a function of energy, except now the partial of the state with respect to the parameter must be determined. In Chapter 5 the defining differential equation for this partial is described, and following its computation it can also be projected to the surface of section using the same methodology as outlined here.

6.4 Semi-Analytical Solutions: Higher-Order Solution Expansions

An alternate solution approach is to take a nominal, numerical solution of the system as the known solution and expand about it using an assumed form for the relative solution. This is exactly what is done when a linear expansion of an orbit is performed; in the following this approach is generalized and continued to higher orders in a systematic fashion. The following is largely borrowed from the derivation given in [124].

Assume a standard form for the equations of motion, $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, t)$, and that $\boldsymbol{x} \in \boldsymbol{R}^{2n}$ and $\boldsymbol{f} : \mathbf{R}^{2n} \times \mathbf{R} \to \mathbf{R}^{2n}$. Then the solution of these equations can be represented as $\boldsymbol{x}(t) = \boldsymbol{\phi}(t; \boldsymbol{x}_o, t_o)$ and $\boldsymbol{\phi}$ is analytic in the initial conditions for $|t| < \infty$. It is not necessary to assume any special conditions on the solution flow, such that it be a periodic orbit, be an equilibrium point or lie on a manifold. All of these conditions can be modeled by this theory, however.

The goal is to expand the solution as a Taylor Series in terms of its initial conditions, with the coefficients being time-varying functions. Introduce tensor notation to specify the equations of motion and solution as

$$\dot{x}_i = f_i(\boldsymbol{x}, t) \tag{6.34}$$

$$x_i(t) = \phi_i(t; \boldsymbol{x}_o, t_o) \tag{6.35}$$

$$i = 1, 2, \ldots, 2n \tag{6.36}$$

where $\boldsymbol{x} = (x_1, x_2, \ldots, x_{2n})$ is kept as a convenient way to denote the entire state. For the expansion introduce a variation about the initial conditions, $\boldsymbol{x}_o + \delta\boldsymbol{x}_o$ which is introduced in the solution function. Carrying out the expansion for $\boldsymbol{\phi}(t; \boldsymbol{x}_o + \delta\boldsymbol{x}_o, t_o) - \boldsymbol{\phi}(t; \boldsymbol{x}_o)$ leads to

$$\delta x_i(t) = \sum_{p=1}^{\infty} \frac{1}{p!} \phi_{i, j_1 j_2 \cdots j_p}(t, t_o) \delta x_{j_1}^o \delta x_{j_2}^o \cdots \delta x_{j_p}^o \tag{6.37}$$

where the Einstein summation convention is assumed for the repeated indices in the above summation, i.e.,

$$\phi_{i,j}\delta x_j^o = \sum_{j=1}^{2n} \phi_{i,j}\delta x_j^o \quad \text{and} \quad \phi_{i,j_1j_2\ldots j_p} = \frac{\partial^p \phi_i(t;\boldsymbol{x}_o,t_o)}{\partial x_{j_1}^o \partial x_{j_2}^o \cdots \partial x_{j_p}^o}$$

Note that the series is formal, but due to the properties of ordinary differential equations it is expected to converge for small enough $\delta\boldsymbol{x}_o$ when evaluated over a finite time, providing an explicit equation for $\delta x_i(t)$ as a function of δx_j^0.

Similarly, expand the equations of motion evaluated at a time t in terms of the above $\delta x_i(t)$, for $\boldsymbol{f}(\boldsymbol{x}(t)+\delta\boldsymbol{x}(t),t) - \boldsymbol{f}(\boldsymbol{x}(t),t)$ to find

$$\delta\dot{x}_i(t) = \sum_{p=1}^{\infty} \frac{1}{p!} f_{i,j_1j_2\ldots j_p}(\boldsymbol{x}(t),t)\delta x_{j_1}(t)\delta x_{j_2}(t)\cdots\delta x_{j_p}(t) \tag{6.38}$$

In a final step substitute Eq. 6.37 into Eq. 6.38 to derive expressions for the time derivatives of the coefficients $\phi_{i,j_1j_2\ldots j_p}(t,t_o)$ so they can be posed as solutions to a series of equations of motion. Now balance orders of the initial conditions, realizing that they are the constants of motion for this solution procedure and that the resulting expansions must balance at each order. Carrying this out recovers the expressions in [124], which are listed through third order as

$$\dot{\phi}_{i,j} = f_{i,\alpha}(t)\phi_{\alpha,j} \tag{6.39}$$

$$\dot{\phi}_{i,jk} = f_{i,\alpha}(t)\phi_{\alpha,jk} + f_{i,\alpha\beta}(t)\phi_{\alpha,j}\phi_{\beta,k} \tag{6.40}$$

$$\begin{aligned}\dot{\phi}_{i,jkl} &= f_{i,\alpha}(t)\phi_{\alpha,jkl} + f_{i,\alpha\beta}(t)\left(\phi_{\alpha,j}\phi_{\beta,kl} + \phi_{\alpha,jk}\phi_{\beta,l} + \phi_{\alpha,jl}\phi_{\beta,k}\right) \\ &\quad + f_{i,\alpha\beta\gamma}(t)\phi_{\alpha,i}\phi_{\beta,j}\phi_{\gamma,k}\end{aligned} \tag{6.41}$$

The first equation is just the State Transition Matrix, while the higher-order equations are generalizations of this, and thus are called the State Transition Tensors (STTs). The convergence properties of these solutions are tested in [124] and shown to be quite effective in modeling the nonlinear dynamics relative to a nominal trajectory. All of the higher-order partials are symmetric in their indices, i.e., $\phi_{i,\alpha\beta\gamma}(t,t_o)$ and $f_{i,\alpha\beta\gamma}(t)$ are symmetric in α, β and γ in general.

Having the differential equations defined for the STTs all that is needed are their initial conditions to carry out numerical integrations. These are quite trivial

$$\phi_{i,j}(t_o,t_o) = \delta_{ij} \tag{6.42}$$

$$\phi_{i,j_1j_2\ldots j_p}(t_o,t_o) = 0 \text{ for } p \geq 2 \tag{6.43}$$

It is most efficient to integrate the nominal trajectory $\phi_i(t;\boldsymbol{x}_o,t_o)$ in tandem with the STTs, as the entire set of equations then form a closed system. The computational cost of this approach is not trivial. Including the nominal state the number of differential equations to compute for a first-order expansion is $2n(2n+1)$, at second order is $2n\left(2n^2+3n+1\right)$, accounting for the symmetric in the higher-order partials, and continues to scale as $(2n)^4$ at third order, etc.

Depending on the specific form of the equations of motion many of the dynamics terms $f_{i,j_1 j_2 \dots j_p}$ may be identically zero above a certain index value. For example, if the system is modeled using Newton's Equations, $\boldsymbol{f}(\boldsymbol{x}, t) = [\boldsymbol{v}; U_{\boldsymbol{r}}(\boldsymbol{r})]$, the higher-order expansions would follow

$$f_{i,i+3} = 1 \text{ for } i = 1, 2, 3 \tag{6.44}$$

$$f_{i+3,k_1,k_2,\dots,k_p} = U_{x_i,x_{k_1},x_{k_2},\dots,x_{k_p}} \text{ for } i, k_i = 1, 2, 3 \tag{6.45}$$

$$f_{i,j_1 j_2 \dots j_p} = 0 \text{ for all other } j_p \text{ for } i = 1, 2, 3 \tag{6.46}$$

and

$$f_{i,j_1 j_2 \dots j_p} = 0 \text{ for } i, j_i = 4, 5, 6 \tag{6.47}$$

6.5 Analytic Solutions

There is a long history of analytical solution methods that arises out of Celestial Mechanics. By an analytic solution we mean that the solution to the equations of motion can be expressed in terms of analytical functions expanded in some parameter about a known solution. The use of analytical solutions are several. First, an analytical solution evaluated to a high degree of precision essentially replaces a numerical integration with a closed-form function evaluation. This is the fundamental and original motivation behind development of these solutions, and it has been developed to extremely high levels of precision in the study of motions in the restricted three-body problem [51]. The existence of an analytic solution of this form enables the structure of motion in these problems to be studied in detail.

Further motivations include the development of approximate solutions valid over extremely long timespans, and is considered further in the following section when we discuss secular solutions. Development of analytical solutions enable much more difficult problems to be considered, such as mapping orbit uncertainty forward in time. Specifically, given an analytical solution, it becomes possible to solve the diffusionless Fokker–Planck–Kolmogorov Equation to find the probability density function of particle as a function of time. Additionally, it becomes simple to explore the neighborhood of a solution given an analytical approximation to it, enabling applications to navigation and orbit determination techniques.

Two approaches are discussed that develop analytical solutions by very different methods. The first applies analytic continuation and follows the classical development by Moulton. In this approach the solution is built up at each order by solving a quadrature. The other approach is the Lie–Deprit algorithm, which has a basis in von Zeipel's method. This method enforces the condition that the solution is constructed so that it is a canonical solution, and hence preserves the Hamiltonian constraints of the original problem statement. We do not discuss this method in detail, but instead refer to some appropriate texts for its exposition.

6.5.1 Moulton's Method of Analytic Continuation

The following is modified from Moulton [111], consult that text for a more rigorous description of the convergence of the series involved, plus a more in-depth description of this general solution method and its variations.

Assume a general differential equation of the form:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, t) + \epsilon \boldsymbol{g}(\boldsymbol{x}, t) \tag{6.48}$$

where the problem can be completely solved when $\epsilon = 0$. Denote the general solution to the problem as $\boldsymbol{x}(t) = \boldsymbol{\phi}(t; \boldsymbol{x}_o, t_o, \epsilon)$, where ϵ is retained as a parameter. Denote the known, analytical solution when $\epsilon = 0$ as $\boldsymbol{x}^{(0)} = \boldsymbol{\phi}(t; \boldsymbol{x}_o, t_o, 0)$ and note that this is not necessarily a Hamiltonian system.

By hypothesis, the solutions to this full system of equations are analytic in ϵ, and thus the solution can be expanded as an analytic function of ϵ. The term ϵ need not be a physically defined parameter, and can just represent the fact that the order of magnitude of $|\boldsymbol{g}| \ll |\boldsymbol{f}|$ in general. Specifically one can expand the general solution into a Taylor series about the known solution

$$\boldsymbol{x}(t) = \boldsymbol{x}^{(0)} + \epsilon \boldsymbol{x}^{(1)} + \frac{1}{2!}\epsilon^2 \boldsymbol{x}^{(2)} + \ldots \tag{6.49}$$

We do not try to express what the radius of convergence of this series is, but note that for an analytic and well-behaved dynamical solution over a finite timespan that such a radius of convergence will exist.

The derivation of the method is simple, one just substitutes the above general form of the solution into the full equations of motion, expands them in orders of ϵ, and then balances terms at each order of epsilon, which is where analyticity is assumed. Skipping some of the involved details of the expansion, carrying it out and comparing terms at orders ϵ^0, ϵ^1 and ϵ^2 yields the results:

$$\epsilon^0 : \ \dot{\boldsymbol{x}}^{(0)} = \boldsymbol{f}(\boldsymbol{x}^{(0)}, t) \tag{6.50}$$

$$\epsilon^1 : \ \dot{\boldsymbol{x}}^{(1)} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_0 \boldsymbol{x}^{(1)} + \boldsymbol{g}(\boldsymbol{x}^{(0)}, t) \tag{6.51}$$

$$\epsilon^2 : \ \dot{\boldsymbol{x}}^{(2)} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_0 \boldsymbol{x}^{(2)} + 2\left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}}\right|_0 \boldsymbol{x}^{(1)} + \boldsymbol{p}(\boldsymbol{x}^{(0)}, \boldsymbol{x}^{(1)}, t) \tag{6.52}$$

and at the kth level has the same general form:

$$\begin{aligned} \epsilon^k : \ \dot{\boldsymbol{x}}^{(k)} = {} & \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_0 \boldsymbol{x}^{(k)} + k\left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}}\right|_0 \boldsymbol{x}^{(k-1)} \\ & + \boldsymbol{p}(\boldsymbol{x}^{(0)}, \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(k-1)}, t) \end{aligned} \tag{6.53}$$

where the $\boldsymbol{p}$ are functions that arise from the higher-order partials of the equations of motion and are all functions of the expansion at lower orders. These are worked out to the second and sometimes higher order in Moulton, but can also be computed using a symbolic manipulator.

Then, assuming that the solution $\boldsymbol{x}^{(0)}(t)$ is known in closed form, each order equation only involves linear terms at that order with all nonlinear terms appearing at lower order, meaning that the problem can be solved by quadratures once a solution for the homogeneous linear term is found. For the case where the solution $\boldsymbol{x}^{(0)}$ is constant, such as occurs for orbit element formulations, the linear equations can be solved in closed form with the exponential matrix, and if $\boldsymbol{f} \equiv 0$ the entire solution process can be reduced to quadratures.

Let us specifically consider the case when $\boldsymbol{f} \equiv 0$, then the solution to these equations can be expressed in a concise, recursive form. First note that the solution at order 0 is just the initial conditions, $\boldsymbol{x}^{(0)} = \boldsymbol{x}_o$, implying that the initial conditions for all higher-order terms are identically zero, $\boldsymbol{x}^{(k)}(t_o) = 0$, $k \geq 1$. Thus the solutions have the form:

$$\boldsymbol{x}^{(1)}(t, t_o) = \int_{t_o}^{t} \boldsymbol{g}(\boldsymbol{x}_o, \tau)\, d\tau \tag{6.54}$$

$$\boldsymbol{x}^{(2)}(t, t_o) = \int_{t_o}^{t} 2 \left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}}\right|_o \boldsymbol{x}^{(1)}(\tau, t_o)\, d\tau \tag{6.55}$$

$$\begin{aligned} \boldsymbol{x}^{(k)}(t, t_o) = \int_{t_o}^{t} \Bigg[& k \left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}}\right|_0 \boldsymbol{x}^{(k-1)}(\tau, t_o) \\ & + \boldsymbol{p}(\boldsymbol{x}^{(0)}, \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(k-2)}, \tau, t_o) \Bigg] d\tau \end{aligned} \tag{6.56}$$

While implementation of this method can be cumbersome beyond the first order, it provides a systematic method to describe these higher-order dynamics.

6.5.2 Generating Function Approaches

These approaches generally rely on the canonical form of the equations of motion and state the problem as a general expansion of a generating function, defined for all orders of perturbation. At each order of expansion one finds partial differential equations that must be solved. These partial differential equations are usually simplified by eliminating fast variables, retaining only the more slowly varying terms. Then the solution procedure is advanced to the next order, systematically building up a higher-order solution. The von Zeipel method is the first widely applied version of this solution technique, and is detailed in [19]. An important innovation for this theory was introduced by Deprit, through the use of Lie transforms, which eliminated the need to perform analytical operations on the von Zeipel solution and enabled the direct solution of the initial conditions. A detailed discussion of the Lie–Deprit method is given in [13]. The manner in which these theories were applied took fundamental advantage of the periodic motion of the unperturbed two-body system to construct generating functions that eliminated such periodic angles at subsequently higher orders. In this way, these analytical expansion theories essentially perform high-order averaging of a dynamical system and, at leading order, provide similar results to those discussed next for the mean motion derivations.

Despite the rigor and advantages of these methods, we do not present the theory in detail, as this would require significant additional discussions that stray from the core purpose of this text. It must be noted that these expansion techniques have not been widely applied to the problem of small-body orbiters to date, except to the motion about planetary satellites [90], and may represent a new and fruitful application of this theory. Possible drawbacks include the relative strength of the perturbations, which may require high-order expansions in order to fully capture the effects analytically.

6.6 Mean Motion Derivations

A particularly effective method for the approximate evaluation of the dynamics of a system is through the use of averaged equations. Once derived, such averaged equations can themselves be integrated numerically, or can often be solved in closed form. The Lie–Deprit method taken at the first order alone provides what can be considered to be an "averaged" form of the equations of motion, as it provides a systematic way of removing the fast variables while retaining the slower variables of the system. This approach has the advantage of allowing the solutions to be expanded to higher orders, as has been investigated in numerous studies. However, as indicated above this methodology has considerable theoretical overhead and is beyond the scope of the current text, which is focused on more direct and practical approaches for describing orbital motion about small bodies.

The method presented below, in contrast, develops the averaged equations of motion at first order and discusses properties of these solutions. These averaged equations of motion capture the secular evolution of the system and can either be numerically integrated or in some cases solved in closed form. The advantage of this approach is that it is possible to easily capture the qualitative effect of perturbations over long timespans with significantly reduced computational requirements. Similar approaches have been systematically developed to higher orders by Laskar [92] for investigating the long-term stability of the solar system.

These methods are generally applied to orbit element formulations such as the Lagrange Planetary Equations or the Gauss Equations. Thus, the dynamical system of interest for this approach has a simplified form as compared to the general system for Moulton's analytical theory

$$\dot{\boldsymbol{x}} = \epsilon \boldsymbol{g}(\boldsymbol{x}, t) \tag{6.57}$$

where when $\epsilon = 0$ the resulting equation is trivially solved by specifying some set of initial conditions, $\boldsymbol{x}_0$, which one can think of as being the orbit elements of an unperturbed satellite in motion about a point mass body. In developing the acceleration perturbations $\boldsymbol{g}(\boldsymbol{x}_0, t)$, even though the nominal solution consists of constants it is usually necessary to map this solution into a Cartesian frame which specifies the motion of the satellite relative to the system as a function of time.

This is generally where the time-variation in the equations of motion arises from. The following development makes the tacit assumption that, all else being equal, the resulting perturbing acceleration $\boldsymbol{g}(\boldsymbol{x}_0, t)$ will be periodic or quasi-periodic in time t. In the application section there are explicit examples of these perturbations and the relevant assumptions are discussed again at that time.

6.6.1 First-Order Solutions

Now apply Moulton's theory to first order to find the approximate evolution of the system as a function of time. If the perturbation is relatively small, then these changes may not be dramatic, although they will systematically change the state of the system. Writing out the general expression for the first-order solution, ignoring the parameter ϵ, yields

$$\dot{\boldsymbol{x}}_1 = \boldsymbol{g}(\boldsymbol{x}_0, t) \tag{6.58}$$

where the analytic continuation results are used, meaning that $\boldsymbol{x}_0$ is the unperturbed motion. For orbital systems, as discussed above, one can assume that $\boldsymbol{g}(\boldsymbol{x}_0, t + T) = \boldsymbol{g}(\boldsymbol{x}_0, t)$, where T is generally the orbit period of the unperturbed motion or is related to it. There are often other time-varying quantities in these equations, in computing the periodicity one often assumes that these times are held fixed, and may be averaged over later under some mild conditions. The solution for $\boldsymbol{x}_1$ can be explicitly written out as

$$\boldsymbol{x}_1(t) - \boldsymbol{x}_1(0) = \int_0^t \boldsymbol{g}(\boldsymbol{x}_0, \tau)\, d\tau \tag{6.59}$$

where for convenience we take $t_o = 0$. Then over a period $t + T$

$$\boldsymbol{x}_1(t + T) - \boldsymbol{x}_1(0) = \int_0^{t+T} \boldsymbol{g}(\boldsymbol{x}_0, \tau)\, d\tau \tag{6.60}$$

$$= \int_0^{T} \boldsymbol{g}(\boldsymbol{x}_0, \tau) d\tau + \int_T^{T+t} \boldsymbol{g}(\boldsymbol{x}_0, \tau)\, d\tau \tag{6.61}$$

$$= \boldsymbol{x}_1(T) + \int_0^{t} \boldsymbol{g}(\boldsymbol{x}_0, \tau)\, d\tau \tag{6.62}$$

leading to

$$\boldsymbol{x}_1(t + T) - \boldsymbol{x}_1(t) = \boldsymbol{x}_1(T) \tag{6.63}$$

Thus, at the level of the approximation a propagation n orbits into the future yields:

$$\boldsymbol{x}_1(t + nT) = \boldsymbol{x}_1(t) + n\boldsymbol{x}_1(T) \tag{6.64}$$

meaning that, to the leading order, the long-term effects can be captured by understanding the change in $\boldsymbol{x}_1$ over one orbit period of the nominal system. Indeed, it now is possible to take the time average of $\boldsymbol{x}_1$ to understand the predicted long-term behavior:

$$\bar{\boldsymbol{x}}_1 = \lim_{t\to\infty} \frac{1}{t}\int_0^t \boldsymbol{x}_1(\tau)\,d\tau \tag{6.65}$$

$$= \lim_{n\to\infty} \frac{1}{t'+nT}\left[\int_0^{t'} \boldsymbol{x}_1(\tau)\,d\tau + n\int_0^T \boldsymbol{x}_1(\tau)\,d\tau\right] \tag{6.66}$$

$$= \frac{1}{T}\int_0^T \boldsymbol{x}_1(\tau)\,d\tau \tag{6.67}$$

Thus, to initially understand the long-term evolution of the system one only needs to study its evolution over one cycle using the approximate map

$$\boldsymbol{x}(t+nT) = \boldsymbol{x}_0 + n\boldsymbol{x}_1(T) + \int_0^t \boldsymbol{g}(\boldsymbol{x}_0,\tau)\,d\tau \tag{6.68}$$

where $t \leq T$. Such extrapolation of this result is inaccurate, however, since the integral $\boldsymbol{x}_1(T)$ depends strongly on the nominal value $\boldsymbol{x}_0$, and if the full solution is diverging from this initial value the integral will become less and less relevant. This can be fixed by replacing the simple extrapolation of the analytical solution with a more sophisticated mapping approach, updating the nominal solution at every step in time. To represent this dependence a more precise notation is introduced, $\boldsymbol{x}_1(T;\boldsymbol{x}_0)$ which shows that the perturbed solution depends explicitly on the nominal state.

Then the sequence in which the nominal solution is updated using $\boldsymbol{x}_1$ is

$$\boldsymbol{x}(T) = \boldsymbol{x}_0 + \boldsymbol{x}_1(T;\boldsymbol{x}_0) \tag{6.69}$$

$$\boldsymbol{x}(2T) = \boldsymbol{x}(T) + \boldsymbol{x}_1(T;\boldsymbol{x}(T)) \tag{6.70}$$

$$\boldsymbol{x}((n+1)T) = \boldsymbol{x}(nT) + \boldsymbol{x}_1(T;\boldsymbol{x}(nT) \tag{6.71}$$

This directly gives the expected change in the state over one orbit period at an arbitrary step n as

$$\boldsymbol{x}((n+1)T) - \boldsymbol{x}(nT) = \boldsymbol{x}_1(T;\boldsymbol{x}(nT)) \tag{6.72}$$

or

$$\Delta\boldsymbol{x}(nT) = \boldsymbol{x}_1(T;\boldsymbol{x}(nT)) \tag{6.73}$$

This can be interpreted as a net rate of change of the state over one orbit period

$$\frac{\Delta \boldsymbol{x}(nT)}{T} = \frac{1}{T}\boldsymbol{x}_1(T;\boldsymbol{x}(nT)) \tag{6.74}$$

$$= \frac{1}{T}\int_0^T \boldsymbol{g}(\boldsymbol{x}(nT),\tau)\,d\tau \tag{6.75}$$

$$= \bar{\boldsymbol{g}}(\boldsymbol{x}(nT)) \tag{6.76}$$

and this is interpreted as the average rate of change in the state, where now that rate of change can shift from orbit to orbit. Abstracting this to a continuous time system yields:

$$\dot{\boldsymbol{x}} = \bar{\boldsymbol{g}}(\boldsymbol{x}) \tag{6.77}$$

where the function $\bar{\boldsymbol{g}}$ has had its time variation due to orbital motion removed and is computed as

$$\bar{\boldsymbol{g}}(\boldsymbol{x}) = \frac{1}{2\pi}\int_0^{2\pi} \boldsymbol{g}(\boldsymbol{x},M)\,dM \tag{6.78}$$

where the time is replaced with the mean motion, $M = nt$, and the period by $T = 2\pi/n$, essentially assuming that all averaging will be over orbital motion. In this sense this process is equivalent to the first-order Lie Series equations where the “fast” variable has been eliminated. From the analytic continuation side this represents the average rate of change of the system and defines a new nonlinear set of equations for the propagation of the orbit state.

6.6.2 Short-Period Terms

Now consider the solution over one time period of the unperturbed problem. Then the actual solution should be represented with two terms, a secular (or averaged) term represented above, which varies linearly with time over one orbit period, and a periodic term

$$\boldsymbol{x}(t) = \boldsymbol{x}_o + \bar{\boldsymbol{g}}t + \boldsymbol{x}_p(t) \tag{6.79}$$

for $0 \le t \le T$. Note that it is not needed to keep $\bar{\boldsymbol{g}}$ constant, but this eases the derivation. Then the periodic part of the solution $\boldsymbol{x}_p(t)$ repeats itself over one orbit, yielding the proper change in the state over one period. Taking the time derivative of this relationship yields:

$$\boldsymbol{g}(\boldsymbol{x},t) = \bar{\boldsymbol{g}}(\boldsymbol{x}) + \dot{\boldsymbol{x}}_p(t) \tag{6.80}$$

which provides an explicit differential equation for the periodic term:

$$\dot{\boldsymbol{x}}_p = \boldsymbol{g}(\boldsymbol{x}_o,t) - \bar{\boldsymbol{g}}(\boldsymbol{x}_o) \tag{6.81}$$

where only the constant terms inside of the dynamics equations are retained, essentially substituting the nominal initial condition $\boldsymbol{x}_o$ for the full solution. Then a quadrature can be carried out for $\boldsymbol{x}_p$ to find

$$\boldsymbol{x}_p(t) = \int_0^t \boldsymbol{g}(\boldsymbol{x}_o, \tau)\, d\tau - t\bar{\boldsymbol{g}}(\boldsymbol{x}_o) \tag{6.82}$$

This provides an estimate of the periodic solution relative to the secular solution within each time interval, and will be explicitly used in the following.

The solutions to the secular equations given above represent the average evolution of the dynamical system, but comparison between numerical integrations and these secular equations, when starting from the same initial conditions, will in general show a divergence between the two. While this may be due to neglected second-order effects for a strongly perturbed system, often this results because of an inconsistent choice of initial conditions. Carefully considering the original derivation of the mean state $\bar{\boldsymbol{x}}$, it is important to note that the limit $t \to \infty$ was taken. In the later derivation of the secular rates of change of these elements, however, we necessarily limited ourselves to a single cycle of the nominal system. Returning to the decomposition of motion into a mean part and a periodic part, it becomes clear that over a single cycle there may be a non-trivial contribution to the average from the periodic part which does not grow in time, yet which can shift the mean value of the state starting from some initial condition $\boldsymbol{x}_o$. Thus, taking the average of a solution decomposed as Eq. 6.79 over a single cycle yields:

$$\bar{\boldsymbol{x}} = \boldsymbol{x}_o + \frac{1}{2}\bar{\boldsymbol{g}}T + \bar{\boldsymbol{x}}_p \tag{6.83}$$

The average of the periodic part is non-zero and equals:

$$\bar{\boldsymbol{x}}_p = \frac{1}{T}\int_0^T \left[\int_0^{\tau'} \boldsymbol{g}(\boldsymbol{x}_o, \tau)\, d\tau - \tau'\bar{\boldsymbol{g}}(\boldsymbol{x}_o)\right] d\tau' \tag{6.84}$$

$$= \frac{1}{T}\int_0^T \int_0^{\tau'} \boldsymbol{g}(\boldsymbol{x}_o, \tau)\, d\tau\, d\tau' - \frac{T}{2}\bar{\boldsymbol{g}}(\boldsymbol{x}_o) \tag{6.85}$$

So given an initial condition for the dynamical system, $\boldsymbol{x}_o$, which is used in the numerical integrations, the average value of the state over a single period is offset from this and equals:

$$\bar{\boldsymbol{x}} = \boldsymbol{x}_o + \frac{1}{T}\int_0^T \int_0^{\tau'} \boldsymbol{g}(\boldsymbol{x}_o, \tau)\, d\tau\, d\tau' \tag{6.86}$$

The interpretation of this result is that given an initial condition for a state $\boldsymbol{x}_o$ the secular equations should be initialized at the above value of $\bar{\boldsymbol{x}}$ for the mean equations to track the true evolution more closely. This small correction can often reduce deviation between secular and true solutions by a significant amount.

6.6.3 Multiple Averaging Time Scales

For some problems of interest, there may be multiple time scales or frequencies over which relevant dynamical motion occurs. Examples include planetary satellite orbiters, where the orbit period of the spacecraft is often much smaller than the orbit period of the planetary satellite (see Chapter 17). Another example would be orbital dynamics about a slowly rotating asteroid, as again the orbit period may be much shorter than the rotation period [72].

In these situations the two time scales are defined by an orbital rate, n, and the motion of external bodies in the system with their own angular rate N. If $N/n \ll 1$ it is acceptable to hold N constant while an initial average is taken over n. Then the remaining system has a time-varying term associated with N, in general periodic. This term can also be averaged over, usually yielding a greatly simplified set of equations. The usual stipulation is that the time scales between the two periods are sufficiently far apart. The degree of separation that is sufficient should be studied on a case-by-case basis.

6.6.4 Averaging the Lagrange Planetary Equations

A special mention must be made of averaging applied to the Lagrange Planetary Equations. These equations have the general form $\boldsymbol{g}(\boldsymbol{x}, t) = G(\boldsymbol{x}) R_{\boldsymbol{x}}(\boldsymbol{x}, t)$, where $G(\boldsymbol{x})$ is a matrix and does not contain time explicitly. Thus, averaging applied to these equations can be drawn all the way back to the perturbing potential $R(\boldsymbol{x}, t)$, greatly simplifying the derivation of the mean equations. The average potential is defined as

$$\bar{R}(\boldsymbol{x}) = \frac{1}{2\pi} \int_0^{2\pi} R(\boldsymbol{x}, M) \, dM \tag{6.87}$$

where the averaging is generally expressed in terms of mean anomaly. The resulting secular equations are then $\dot{\bar{\boldsymbol{x}}} = G(\boldsymbol{x}) \bar{R}_{\boldsymbol{x}}(\bar{\boldsymbol{x}})$. One main consequence of this approach is that the mean anomaly is removed from the perturbation potential and thus the semi-major axis is constant in the averaged motion and serves as an integral of motion for these equations. If the original system has a Jacobi integral, then that function is still conserved after the averaging and defines an integral of motion distinct from the semi-major axis.

If the singly averaged potential is a function of a second periodic term of period T, or $R(\boldsymbol{x}, t, M) = R(\boldsymbol{x}, t + T, M)$, and sufficient distance between the two time scales exists, another averaging may be performed to yield:

$$\bar{\bar{R}}(\boldsymbol{x}) = \frac{1}{T 2\pi} \int_0^{T} \int_0^{2\pi} R(\boldsymbol{x}, \tau, M) \, dM \, d\tau \tag{6.88}$$

Specific applications of this in will be made in Chapter 17.

6.7 Discrete Orbit Updates

When a dynamical system has a time-varying quantity, an averaging analysis generally holds this time-varying quantity stationary while performing its quadrature over the motions of a satellite. For many systems this is a reasonable approximation which allows the equations to be simplified. Important situations can arise, however, where this is not a good approximation. Specific examples usually include systems where the time variation is of similar magnitude as the orbital variation in time, such as a spacecraft orbiting close to a rotating asteroid. For these systems it is not appropriate to hold the asteroid stationary during the averaging process as significant interactions between the two systems result, yielding large changes in quantities such as the energy or angular momentum.

It is desirable to capture these effects analytically or semi-analytically, as these approaches generally provide more insight into the factors of the system which control the changing orbit. To analyze these effects it is useful to state the dynamical system in terms of the Lagrange Planetary Equations. Assume that the perturbation function has an explicit time variation, $R(\boldsymbol{x}, t)$. If the system has significant interaction over one orbit, then averaging the system can mask these interactions. To capture them consider the Moulton analytic theory to the first order, expressed as a quadrature:

$$\boldsymbol{x}_1(t) = \int_{t_o}^{t} \boldsymbol{f}(\boldsymbol{x}_0(\tau), \tau)\, d\tau \tag{6.89}$$

where both the orbital motion and the physical system are allowed to vary simultaneously. If this is performed over one orbit then one finds explicit formula for the change in the state over this interaction.

$$\Delta\boldsymbol{x} = \int_{t_o}^{t_o+T} \boldsymbol{f}(\boldsymbol{x}_0(\tau), \tau)\, d\tau \tag{6.90}$$

where T is the orbit period. Formally, it is then possible to update the nominal state and re-integrate over a given time-span, leading to a discrete map approximation [130].

These quadratures generally cannot be evaluated in closed form, essentially due to the explicit dependence of the two-body solution on true anomaly and the dependance of the physical system on time. For an analytic expression of the potential, however, it is usually possible to isolate these quadratures as a function of semi-major axis and eccentricity alone, allowing the dependance of the other orbital elements to remain clear. Specific applications of this approach are presented in Chapter 7. Other examples of this approach can be found in [187, 129].

6.8 Phase Space Constraints

The final, and in some sense most precise, constraints that can be placed on a solution arise from the topology of conserved quantities. The most common approach to developing a hard constraint on a trajectory is the use of zero-velocity surfaces. These arise in systems with a Jacobi integral and allow one to specifically map out regions where it is impossible for a particle to enter, as a function of the Jacobi integral value.

For systems where a Jacobi integral exists the generic form that these integrals take is

$$J(\boldsymbol{r}, \boldsymbol{v}) = \frac{1}{2}v^2 - V(\mathbf{r}) \tag{6.91}$$

where V is the potential and $v = |\boldsymbol{v}|$ is the speed relative to the frame in which all time variation in the system is removed (usually a uniformly rotating frame). The concept is quite simple: for a given value of the Jacobi integral, C, the inequality $v^2 \geq 0$ must hold, leading to the more general constraint

$$C + V(\boldsymbol{r}) \geq 0 \tag{6.92}$$

Thus, this provides explicit constraints on the position of the particle, independent of the speed, for a given constant C, which is $V(\boldsymbol{r}) \geq -C$.

A simple example of this is found in the two-body problem and its energy integral, $E = 1/2v^2 - \mu/r$. Now note that the potential is negative definite, with the resulting inequality being expressed as $\mu/r \geq -E$. Next consider the different possibilities, that $E < 0$, $E = 0$ and $E > 0$. First consider $E = 0$, yielding the inequality $\mu/r \geq 0$. This provides no constraint on the system, as the quantity on the left is always positive and only approaches 0 as $r \to \infty$. Similarly for the case $E > 0$, which leads to the inequality $\mu/r \geq 0 \geq -E$. Thus, in these situations there are no constraints placed on the motion, even though additional constraints can be developed for the special case of the two-body problem. Use of zero-velocity curves generally has such null results for specific values of the constant. This is due to the potentials usually being definite in sign, meaning that the inequality will often be trivially true.

More interesting for the two-body problem is the case of negative energy, $E < 0$, as the inequality then becomes $\mu/r \geq -E > 0$, meaning that there are values of r which violate the inequality. Specifically, the system is restricted to $r \leq \mu/|E|$, providing an upper bound on the distance that the solution can get from the origin. While these results are somewhat trivial for the two-body problem, they also apply for more complex systems that do not have specific solutions, such as the restricted three-body problem.

Generally, the presence of additional integrals of motion allow for additional restrictions on the motion, without having to solve for the motion specifically. Although often these restrictions must be combined with some other integral, such as the Jacobi or energy integrals. A good example of this can be found when some component of the angular momentum is conserved. Let's assume that the system conserves angular momentum with a relationship of the form $h = r^2\dot{\theta} = rv\cos\gamma$, and that it has a Jacobi integral of the form $v^2 = 2C + 2V(\boldsymbol{r})$. The presence of such an additional relation allows us to eliminate the speed, v, and transform the angular momentum integral to $h = r\sqrt{2C + 2V(\boldsymbol{r})}\cos\gamma$. Note that the previous zero-velocity constraint can assure us that only regions where $V + C \geq 0$ are considered. If $h \neq 0$, then the sign of the expression cannot change in general and it can be noted that $\cos\gamma \neq 0$ unless $r \to \infty$, and thus that γ lies in the interval $(-\pi/2, \pi/2)$. Now, let us assume that the trajectory has a closest or furthest approach to the origin, i.e., a periapsis or apoapsis passage. Geometrically it is known that $\gamma = 0$ at these points which leads to the relation $h = r_{apse}\sqrt{2C + 2V(\boldsymbol{r}_{apse})}$, implying that there are unique closest and furthest passages, dependent on the number of roots these expressions have. This was seen explicitly in the two-body example given in Chapter 3 where there was a resultant quadratic equation related to this constraint.

Part IV
Appendices

A. Two-Body Orbit Relations

The following is a collection of useful relationships relating position, velocity and time within an orbit for the two-body problem, using both true anomaly and eccentric anomaly. The main application of these relationships is to averaging, and many of these are used in the following appendix. Due to this only results relevant to elliptic orbits are shown, although most of these results will have analogues for parabolic and hyperbolic orbits.

Mean, Eccentric and True Anomaly

Following are the basic relations between mean, M, eccentric, E, and true, f anomaly. Note the usual symbolic definition of the semi-major axis, a, and the eccentricity, e.

$$M = E - e \sin E \tag{A.1}$$

$$\tan \frac{1}{2} E = \sqrt{\frac{1-e}{1+e}} \tan \frac{1}{2} f \tag{A.2}$$

$$\cos E = \frac{e + \cos f}{1 + e \cos f} \tag{A.3}$$

$$\sin E = \frac{\sqrt{1-e^2} \sin f}{1 + e \cos f} \tag{A.4}$$

$$\cos f = \frac{\cos E - e}{1 - e \cos E} \tag{A.5}$$

$$\sin f = \frac{\sqrt{1-e^2} \sin E}{1 - e \cos E} \tag{A.6}$$

Next are the associated differential relationships between these quantities

$$dM = (1 - e\cos E)\,dE \tag{A.7}$$

$$dM = \frac{(1-e^2)^{3/2}}{(1+e\cos f)^2}\,df \tag{A.8}$$

$$dE = \frac{\sqrt{1-e^2}}{1+e\cos f}\,df \tag{A.9}$$

$$df = \frac{\sqrt{1-e^2}}{1-e\cos E}\,dE \tag{A.10}$$

Scalar Kinematic Quantities

Following are the main scalar kinematic quantities of interest, these being the radius r, speed v, and flight path angle γ. Each are expressed both in terms of true and eccentric anomaly.

$$r = \begin{cases} \dfrac{a(1-e^2)}{1+e\cos f} \\ a(1-e\cos E) \end{cases} \tag{A.11}$$

$$v = \begin{cases} \sqrt{\dfrac{\mu}{a(1-e^2)}\,(1+2e\cos f+e^2)} \\ \sqrt{\dfrac{\mu}{a}\left(\dfrac{1+e\cos E}{1-e\cos E}\right)} \end{cases} \tag{A.12}$$

$$\cos\gamma = \begin{cases} \dfrac{1+e\cos f}{\sqrt{1+2e\cos f+e^2}} \\ \sqrt{\dfrac{1-e^2}{1-e^2\cos^2 E}} \end{cases} \tag{A.13}$$

$$\sin\gamma = \begin{cases} \dfrac{e\sin f}{\sqrt{1+2e\cos f+e^2}} \\ \dfrac{e\sin E}{\sqrt{1-e^2\sin^2 E}} \end{cases} \tag{A.14}$$

$$\tan\gamma = \begin{cases} \dfrac{e\sin f}{1+e\cos f} \\ \dfrac{e}{\sqrt{1-e^2}}\sin E \end{cases} \tag{A.15}$$

Vector Kinematic Quantities

Next the main vector kinematic quantities of interest are expressed, these primarily being the position and velocity vectors as well as the defining orientation integrals of the two-body problem – the eccentricity and angular momentum vectors. First the fundamental vectors are presented, specified using the classical orbit elements relative to an inertial frame.

$$\hat{\boldsymbol{H}} = \sin\Omega\sin i\hat{\boldsymbol{x}} - \cos\Omega\sin i\hat{\boldsymbol{y}} + \cos i\hat{\boldsymbol{z}} \tag{A.16}$$

$$\begin{aligned}\hat{\boldsymbol{e}} = &\left(\cos\omega\cos\Omega - \cos i\sin\omega\sin\Omega\right)\hat{\boldsymbol{x}} \\ &+ \left(\cos\omega\sin\Omega + \cos i\sin\omega\cos\Omega\right)\hat{\boldsymbol{y}} \\ &+ \sin\omega\sin i\hat{\boldsymbol{z}}\end{aligned} \tag{A.17}$$

$$\hat{\boldsymbol{e}}_\perp = \hat{\boldsymbol{h}} \times \hat{\boldsymbol{e}} \tag{A.18}$$

$$\begin{aligned}= &-\left(\sin\omega\cos\Omega + \cos i\cos\omega\sin\Omega\right)\hat{\boldsymbol{x}} \\ &- \left(\sin\omega\sin\Omega - \cos i\cos\omega\cos\Omega\right)\hat{\boldsymbol{y}} \\ &+ \cos\omega\sin i\hat{\boldsymbol{z}}\end{aligned} \tag{A.19}$$

These reference directions are all conserved in the two-body problem and can be used to specify the position and velocity vectors.

$$\boldsymbol{r} = \begin{cases} r\left[\cos f\hat{\boldsymbol{e}} + \sin f\hat{\boldsymbol{e}}_\perp\right] \\ \\ a\left[\left(\cos E - e\right)\hat{\boldsymbol{e}} + \sqrt{1-e^2}\sin E\hat{\boldsymbol{e}}_\perp\right] \end{cases} \tag{A.20}$$

$$\boldsymbol{v} = \begin{cases} \sqrt{\frac{\mu}{a(1-e^2)}}\left[-\sin f\hat{\boldsymbol{e}} + \left(e + \cos f\right)\hat{\boldsymbol{e}}_\perp\right] \\ \\ \frac{\sqrt{\mu a}}{r}\left[\sin E\hat{\boldsymbol{e}} + \sqrt{1-e^2}\cos E\hat{\boldsymbol{e}}_\perp\right] \end{cases} \tag{A.21}$$

B. Fourier Series Expansions of Radius Functions

Following are a simple set of results that are of use in carrying out averaging, the expansion of quantities of the form r^n and $1/r^n$ in terms of Fourier series coefficients, taken from [152].

$$(1 + e\cos\tau)^n = \sum_{m=0}^{\infty} a_m^n \cos(m\tau) \tag{B.1}$$

$$\frac{1}{(1 + e\cos\tau)^n} = \sum_{m=0}^{\infty} b_m^n \cos(m\tau) \tag{B.2}$$

where the coefficients are defined as follows:

$$a_0^n = c_0^n \tag{B.3}$$

$$a_k^n = 2\left(\frac{e}{2}\right)^k c_k^n \tag{B.4}$$

$$b_0^n = \frac{\sqrt{1-e^2}}{(1-e^2)^n} f_0^n \tag{B.5}$$

$$b_k^n = (-1)^k 2\left(\frac{e}{2}\right)^k \frac{\sqrt{1-e^2}}{(1-e^2)^n} f_k^n \tag{B.6}$$

The coefficients c_k^n and f_k^n have the general definitions:

$$c_k^n = \begin{cases} \displaystyle\sum_{l=0}^{[(n-k)/2]} \frac{n!}{l!(l+k)!(n-k-2l)!} \left(\frac{e}{2}\right)^{2l} & n \geq k \\ 0 & n < k \end{cases} \tag{B.7}$$

$$f_k^{n+1} = \begin{cases} \displaystyle\frac{(n-k)!(n+k)!}{n!^2} \sum_{l=0}^{[(n-k)/2]} \frac{n!}{l!(l+k)!(n-k-2l)!} \left(\frac{e}{2}\right)^{2l} & n+1 > k \\ \displaystyle\frac{n-k}{n}(1-e^2) f_k^n + 2 f_{k-1}^{n+1} & n+1 \leq k \end{cases} \tag{B.8}$$

$$f_k^1 = \left(\frac{2}{1+\sqrt{1-e^2}}\right)^k \tag{B.9}$$

where $[a]$ denotes the integer part of a. Note that to compute f_k^{n+1} for $n+1 \leq k$ the recursion relation stated above must be solved.

C. Averaging Results

In orbital dynamics averaging is defined as an operator that computes the time average of a quantity over one orbit period while keeping all other orbit elements constant. The periodicity implies that averaging is only taken for elliptic orbits, and for consistency the averaging is performed over mean anomaly. Thus given a quantity $f(\text{æ}, M)$, defined as a function of mean anomaly M in addition to other orbit elements represented as **æ**, the average is defined as

$$\bar{f}(\text{æ}) = \frac{1}{2\pi} \int_0^{2\pi} f(\text{æ}, M)\, dM \tag{C.1}$$

where $\bar{f}(\text{æ})$ is independent of mean anomaly but still a function of the other orbit elements. Although the average is defined with respect to mean anomaly, it is often more convenient to compute averages using the true or eccentric anomaly, using the differential relationships given in Appendix A.

First consider powers of the radius r^n for $n \geq 0$.

$$\overline{r^n} = \frac{1}{2\pi} \int_0^{2\pi} r^n \, dM \tag{C.2}$$

$$= \frac{a^n}{2\pi} \int_0^{2\pi} (1 - e\cos E)^{n+1} \, dE \tag{C.3}$$

Using the Fourier series expansion in Eq. B.1 this equals

$$\overline{r^n} = a^n c_0^{n+1}(e) \tag{C.4}$$

$$= a^n \sum_{l=0}^{\left[\frac{n+1}{2}\right]} \frac{(n+1)!}{l!l!(n+1-2l)!} \left(\frac{e}{2}\right)^{2l} \tag{C.5}$$

where $[n + 1/2]$ denotes the integer part of the expression.

Next, consider $1/r^n$ for $n \geq 2$.

$$\overline{\frac{1}{r^n}} = \frac{(1-e^2)^{3/2}}{2\pi p^n} \int_0^{2\pi} (1+e\cos f)^{n-2}\, df \tag{C.6}$$

Using the Fourier series expansion in Eq. B.1 again yields

$$\overline{\frac{1}{r^n}} = \frac{(1-e^2)^{3/2}}{p^n} c_0^{n-2}(e) \tag{C.7}$$

$$= \frac{(1-e^2)^{3/2}}{p^n} \sum_{l=0}^{\left[\frac{n-2}{2}\right]} \frac{(n-2)!}{l!l!(n-2-2l)!} \left(\frac{e}{2}\right)^{2l} \tag{C.8}$$

Finally, for the special case of $1/r$ the average is

$$\overline{\frac{1}{r}} = \frac{1}{a} \tag{C.9}$$

A few specific examples from the above general formula are

$$\overline{r} = a\left(1 + \frac{1}{2}e^2\right) \tag{C.10}$$

$$\overline{\frac{1}{r^2}} = \frac{1}{a^2\sqrt{1-e^2}} \tag{C.11}$$

$$\overline{\frac{1}{r^3}} = \frac{1}{a^3(1-e^2)^{3/2}} \tag{C.12}$$

More complex averaging needs to occur when gravitational fields are considered. For the most general results it becomes necessary to average inverse powers of radius times the dot product of $\hat{\boldsymbol{r}}$ with an arbitrary unit vector $\hat{\boldsymbol{p}}$. One can always choose the canonical $\hat{\boldsymbol{z}}$ axis to be aligned with the given unit vector, with the result that $\hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{p}} = \sin i \sin(\omega + f)$. If averaging over the true anomaly eliminates the argument of periapsis, resulting in $\sin i$ terms in the final average, these can be re-expressed using the unit vector and the unitized angular moment as $\sin i = \sqrt{1 - (\hat{\boldsymbol{p}} \cdot \hat{\boldsymbol{H}})^2}$. In other cases when the argument of periapsis remains in the averaged expression, it can be related to the dot product of the eccentricity vector with the unit vector. As the average is ultimately expressed in terms of vectors again, the final result is independent of any particular coordinate system. Applying this approach yields the following

$$\overline{\left(\frac{(\hat{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}})^2}{r^3}\right)} = \frac{1}{2a^3(1-e^2)^{3/2}}\left[1-(\hat{\boldsymbol{p}}\cdot\hat{\boldsymbol{H}})^2\right] \tag{C.13}$$

$$\overline{\left(\frac{(\hat{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}})}{r^4}\right)} = \frac{1}{a^4(1-e^2)^{5/2}}\boldsymbol{e}\cdot\hat{\boldsymbol{p}} \tag{C.14}$$

$$\overline{\left(\frac{(\hat{\boldsymbol{p}}\cdot\hat{\boldsymbol{r}})^3}{r^4}\right)} = \frac{3}{4a^4(1-e^2)^{5/2}}\boldsymbol{e}\cdot\hat{\boldsymbol{p}}\left[1-(\hat{\boldsymbol{p}}\cdot\hat{\boldsymbol{H}})^2\right] \tag{C.15}$$

Finally, consider the direct averaging of vectors and their combinations. Here the explicit description of the radius and velocity vectors in terms of orbit elements found in Appendix A are used. Carrying out the averaging operations yields

$$\overline{\left(\frac{\hat{\boldsymbol{r}}\hat{\boldsymbol{r}}}{r^3}\right)} = \frac{1}{2a^3(1-e^2)^{3/2}}\left[\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}+\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_\perp\right] \tag{C.16}$$

$$= \frac{1}{2a^3(1-e^2)^{3/2}}\left[\boldsymbol{U}-\hat{\boldsymbol{H}}\hat{\boldsymbol{H}}\right] \tag{C.17}$$

where the second equation was simplified using the identity $\boldsymbol{U} = \hat{\boldsymbol{e}}\hat{\boldsymbol{e}}+\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_\perp+\hat{\boldsymbol{H}}\hat{\boldsymbol{H}}$. Next, averaging the position and velocity vectors and some of their dyad products yields

$$\overline{\boldsymbol{r}} = -\frac{3}{2}a\boldsymbol{e} \tag{C.18}$$

$$\overline{\boldsymbol{v}} = \boldsymbol{0} \tag{C.19}$$

$$\overline{\boldsymbol{r}\boldsymbol{r}} = \frac{1}{2}a^2\left[(1+4e^2)\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}+(1-e^2)\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_\perp\right] \tag{C.20}$$

$$\overline{\boldsymbol{r}\boldsymbol{v}} = -\frac{1}{2}\tilde{\boldsymbol{H}} \tag{C.21}$$

where $\boldsymbol{H}$ denotes the angular momentum vector. The final averaging result for $\boldsymbol{rv}$ is difficult, and so it described below.

The dyad product of these two vectors can be reduced to the following dyadic

$$\begin{aligned}\boldsymbol{r}\boldsymbol{v} = H\,[&\cos f(\tan\gamma\cos f-\sin f)\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}\\ &+\sin f(\tan\gamma\sin f+\cos f)\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_\perp\\ &+\cos f(\tan\gamma\sin f+\cos f)\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}_\perp\\ &+\sin f(\tan\gamma\cos f-\sin f)\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}]\end{aligned} \tag{C.22}$$

Noting that $\tan\gamma$ and $\sin(mf)$ are both odd functions in true anomaly the factors of $\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}$ and $\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_\perp$ will ultimately be odd in true anomaly, and hence will average to zero. The same is not true of the other factors. Applying trigonometric identities these factors can be rewritten as

$$\frac{1}{2}H\left[(\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}_\perp-\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}})+\frac{\cos(2f)+e\cos f}{1+e\cos f}(\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}_\perp+\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}})\right] \tag{C.23}$$

The first term can be identified with the dyadic $-\frac{1}{2}\tilde{\boldsymbol{H}}$, and thus is a constant. The second term can be averaged over true anomaly and, using the Fourier series expansions, be shown to equal

$$\frac{H}{2}\frac{e^2}{1-e^2}\left[\frac{1}{2}f_2^3 - f_1^3\right](\hat{\boldsymbol{e}}\hat{\boldsymbol{e}}_\perp + \hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}) \tag{C.24}$$

where the f functions are defined in Eq. B.9. Computation of the factors yields $f_2^3 = 3!$ and $f_1^3 = 3!/2$, and hence the term in Eq. C.24 is identically zero, proving the result.

D. Canonical Transformations

The theory of canonical transformations lies at the heart of Hamiltonian dynamics. They serve as an equivalence principle between solutions and coordinate changes in these systems, and in essence make entire classes of Hamiltonian dynamical systems of similar order equivalent. It is important to note that there remains fundamental differences between Hamiltonian systems that are integrable and those that are non-integrable, a topic that goes beyond the current text.

Symplectic Matrix Properties

Define the symplectic unit matrix $J_{2n} \in \mathbf{R}^{2n\times 2n}$ as

$$J_{2n} = \begin{bmatrix} 0_n & I_n \\ -I_n & 0_n \end{bmatrix} \tag{D.1}$$

where I_n and 0_n are $n \times n$ identity and zero matrices, respectively. The $2n$ subscript is suppressed, as the dimension will usually be obvious. This matrix is skew-symmetric, $J^T = -J$. Also note that $JJ = -I$ and that it is orthonormal, or, $JJ^T = J^T J = I$.

A real matrix $M \in \mathbf{R}^{2n\times 2n}$ is defined as a symplectic matrix if it satisfies the equation

$$J = M^T J M \tag{D.2}$$

Based on this definition a series of equivalent statements and identities for symplectic matrices can be recounted. The following list is compiled from [132].

- J is a symplectic matrix.
- I is a symplectic matrix.
- If M is symplectic, then so is M^T, $-M$, and M^{-1}.
- If M_1 and M_2 are symplectic, then $M_1 M_2$ is symplectic.
- The determinant of a symplectic matrix always equals 1, hence an inverse always exists.

- The inverse of a symplectic matrix M is $M^{-1} = -JM^T J$.
- From the previous statements it can be proven that the symplectic matrices of a fixed size form a group.
- Assume a matrix M has the following form:

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \tag{D.3}$$

  where each of the sub-matrices is an $n \times n$ matrix. Then M is a symplectic matrix if and only if the following identities hold: (i) $A^T C$ and $B^T D$ are symmetric, (ii) $D^T A - B^T C = I$, the identity.
- If M is a 2×2 matrix, then it is a symplectic matrix if and only if its determinant equals unity.
- The characteristic equation of a symplectic matrix, $\|\lambda I - M\| = 0$, is a symmetric polynomial in λ.
- The eigenvalues of a symplectic matrix come in inverse pairs, i.e., if λ is an eigenvalue (possibly complex), then so is λ^{-1}. As M is a real matrix, the complex conjugates of the eigenvalues must also be eigenvalues. Thus if λ is an eigenvalue, then so must λ^{-1}, $\bar{\lambda}$, and $\bar{\lambda}^{-1}$. Note that in some cases $\lambda = \bar{\lambda}$ or $\bar{\lambda} = \lambda^{-1}$. In these cases the eigenvalues only need come in pairs.
- If $\boldsymbol{u}$ is a right eigenvector of a symplectic matrix with eigenvalue λ, then $\boldsymbol{v} = -J\boldsymbol{u}$ is a left eigenvector with eigenvalue λ^{-1}.

Tests for Canonical Transformations

There are a number of fundamental tests for whether a transformation is canonical, and thus whether it will transform the Hamiltonian equations of motion back into a similar form. We list a few common tests in the following. Assume an initial Hamiltonian system denoted as $\boldsymbol{x} = [\boldsymbol{q}, \boldsymbol{p}]$, and a candidate transformation $\boldsymbol{X}(\boldsymbol{x}) = [\boldsymbol{Q}(\boldsymbol{x}), \boldsymbol{P}(\boldsymbol{x})]$ that will be tested for whether it is canonical.

- The transformation is canonical if the Jacobian of the transformation, $M = \partial \boldsymbol{X} / \partial \boldsymbol{x}$, is a symplectic matrix.
- The Lagrange brackets are defined for a transformation $\boldsymbol{Q}(u, v) \in \mathbf{R}^n$ and $\boldsymbol{P}(u, v) \in \mathbf{R}^n$ by the operator:

$$[u, v] = \sum_{i=1}^{n} \left(\frac{\partial Q_i}{\partial u} \frac{\partial P_i}{\partial v} - \frac{\partial P_i}{\partial u} \frac{\partial Q_i}{\partial v} \right) \tag{D.4}$$

  A transformation $\boldsymbol{X}(\boldsymbol{x})$ is canonical if and only if the following Lagrange bracket values hold

$$[q_j, q_k] = 0 \tag{D.5}$$

$$[p_j, p_k] = 0 \tag{D.6}$$

$$[q_j, p_k] = \delta_{jk} \tag{D.7}$$

where δ_{jk} is the Kronecker delta function. Note that Lagrange brackets are formally only a function of the original variables. Recall that the Lagrange brackets were also used in the derivation of the Lagrange planetary equations.

- The Poisson brackets can be considered to be the inverse of the Lagrange brackets, and are defined for a transformation $u(\boldsymbol{q}, \boldsymbol{p})$ and $v(\boldsymbol{q}, \boldsymbol{p})$ by the operator:

$$\{u, v\} = \sum_{i=1}^{n} \left(\frac{\partial u}{\partial q_i} \frac{\partial v}{\partial p_i} - \frac{\partial u}{\partial p_i} \frac{\partial v}{\partial q_i} \right) \tag{D.8}$$

A transformation $\boldsymbol{X}(\boldsymbol{x})$ is canonical if and only if the following Poisson bracket values hold

$$\{Q_j, Q_k\} = 0 \tag{D.9}$$
$$\{P_j, P_k\} = 0 \tag{D.10}$$
$$\{Q_j, P_k\} = \delta_{jk} \tag{D.11}$$

Note that the Poisson brackets are formally only a function of the new variables. Recall that the Poisson brackets were also used in the definition of the time derivative of a scalar quantity for a Hamiltonian dynamical system.

- Another important test of whether a proposed transformation is canonical is through the definition of the generating functions. Specifically, given a transformation of the form given above, it is a canonical transformation if and only if the following differential forms are exact after replacement of $\boldsymbol{Q}$ and $\boldsymbol{P}$ as functions of $\boldsymbol{q}$ and $\boldsymbol{p}$.

$$\boldsymbol{P} \cdot d\boldsymbol{Q} - \boldsymbol{p} \cdot d\boldsymbol{q} \tag{D.12}$$
$$\boldsymbol{Q} \cdot d\boldsymbol{P} + \boldsymbol{p} \cdot d\boldsymbol{q} \tag{D.13}$$
$$\boldsymbol{P} \cdot d\boldsymbol{Q} + \boldsymbol{q} \cdot d\boldsymbol{p} \tag{D.14}$$
$$\boldsymbol{Q} \cdot d\boldsymbol{P} - \boldsymbol{q} \cdot d\boldsymbol{p} \tag{D.15}$$

These conditions can all be rigorously related to each other (see [132, 56]). If such a condition is exact, then it equals the differential of a scalar function, called a generating function.

As a simple application of the above tests, consider a Hamiltonian system defined by coordinates $\boldsymbol{q}$ and momenta $\boldsymbol{p}$ and the proposed transformation $\boldsymbol{P} = -\boldsymbol{q}$ and $\boldsymbol{Q} = \boldsymbol{p}$, which swaps the roles of the coordinates and momenta.

The Jacobian of this transformation can be easily shown to be J, the symplectic unit matrix. As this is a symplectic matrix, the transformation is seen to be canonical.

For the Lagrange and Poisson bracket tests, it suffices to only consider one of the elements, say $Q_1 = p_1$ and $P_1 = -q_1$. The first two conditions of each test are trivially satisfied, and the third yields

$$[q_1, p_1] = \frac{\partial p_1}{\partial q_1}\frac{\partial(-q_1)}{\partial p_1} - \frac{\partial(-q_1)}{\partial q_1}\frac{\partial p_1}{\partial p_1} = 1 \tag{D.16}$$

$$\{Q_1, P_1\} = \frac{\partial Q_1}{\partial(-P_1)}\frac{\partial P_1}{\partial Q_1} - \frac{\partial Q_1}{\partial Q_1}\frac{\partial P_1}{\partial(-P_1)} = 1 \tag{D.17}$$

Since the identity transformation is obviously canonical, the above result also shows that just swapping one coordinate-momentum pair is also a canonical transformation.

For the exactness tests, we find the following:

$$\boldsymbol{P} \cdot d\boldsymbol{Q} - \boldsymbol{p} \cdot d\boldsymbol{q} = -d(\boldsymbol{q} \cdot \boldsymbol{p}) \tag{D.18}$$

$$\boldsymbol{Q} \cdot d\boldsymbol{P} + \boldsymbol{p} \cdot d\boldsymbol{q} = 0 \tag{D.19}$$

$$\boldsymbol{P} \cdot d\boldsymbol{Q} + \boldsymbol{q} \cdot d\boldsymbol{p} = 0 \tag{D.20}$$

$$\boldsymbol{Q} \cdot d\boldsymbol{P} - \boldsymbol{q} \cdot d\boldsymbol{p} = -d(\boldsymbol{q} \cdot \boldsymbol{p}) \tag{D.21}$$

all of which are exact differentials and thus canonical.

E. Legendre Polynomials and Associated Functions

The associated Legendre functions are defined by the simple rule

$$P_{lm}(x) = (1 - x^2)^{m/2} \frac{d^m}{dx^m}(P_{l0}(x)) \tag{E.1}$$

The $P_{l0}(x)$ are the Legendre polynomials and are defined as

$$P_{l0} = \frac{1}{2^l l!} \frac{d^l}{dx^l}(x^2 - 1)^l \tag{E.2}$$

These functions and polynomials arise in various expansions of interest, both for the perturbation of a third body and in the spherical harmonic gravity field expansion.

A list of these polynomials and functions up to degree and order 4 are given below:

$$\begin{aligned}
P_{00}(x) &= 1 \\
P_{10}(x) &= x \\
P_{11}(x) &= \sqrt{1 - x^2} \\
P_{20}(x) &= \frac{1}{2}(3x^2 - 1) \\
P_{21}(x) &= 3x\sqrt{1 - x^2} \\
P_{22}(x) &= 3(1 - x^2) \\
P_{30}(x) &= \frac{1}{2}(5x^3 - 3x) \\
P_{31}(x) &= \frac{3}{2}(5x^2 - 1)\sqrt{1 - x^2} \\
P_{32}(x) &= 15x(1 - x^2) \\
P_{33}(x) &= 15(1 - x^2)\sqrt{1 - x^2}
\end{aligned}$$

$$
\begin{aligned}
P_{40}(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3) \\
P_{41}(x) &= \frac{5}{2}(7x^3 - 3x)\sqrt{1 - x^2} \\
P_{42}(x) &= \frac{15}{2}(7x^2 - 1)(1 - x^2) \\
P_{43}(x) &= 105x(1 - x^2)\sqrt{1 - x^2} \\
P_{44}(x) &= 105(1 - x^2)^2
\end{aligned}
$$

F. Elliptic Functions and Integrals

The following sections briefly cover some of the essential qualities and definitions of elliptic functions and integrals. There is a wealth of material pertaining to these quantities not discussed here as it is not needed in the current analysis. See References [34] and [24] for complete discussions of these functions and integrals.

Elliptic Integrals

There are three types of Jacobi elliptic integrals. They are called elliptic integrals of the 1st, 2nd and 3rd kind. The elliptic integral of the 1st kind can be represented as:

$$F(x,k) = \int_0^x \frac{dy}{\sqrt{(1-y^2)(1-k^2y^2)}} \tag{F.1}$$

or its equivalent (assuming $x = \sin\phi$)

$$F(\phi,k) = \int_0^\phi \frac{d\theta}{\sqrt{1-k^2\sin^2\theta}} \tag{F.2}$$

where $k^2 < 1$. This elliptic integral is called complete if $x = 1$ or $\phi = \pi/2$, and is then denoted as:

$$\begin{aligned} K(k) &= F(1,k) && \text{(F.3)} \\ &= F(\pi/2,k) && \text{(F.4)} \\ &= \pi/2 \sum_{n=0}^{\infty} \left(\frac{(2n)!}{2^{2n}(n!)^2}\right)^2 k^{2n} && \text{(F.5)} \end{aligned}$$

The elliptic integral of the 2nd kind can be represented as:

$$E(x,k) = \int_0^x \sqrt{\frac{(1-k^2y^2)}{(1-y^2)}}\, dy \tag{F.6}$$

or its equivalent (assuming $x = \sin\phi$)

$$E(\phi,k) = \int_0^\phi \sqrt{1-k^2\sin^2\theta}\, d\theta \tag{F.7}$$

where $k^2 < 1$. This elliptic integral is called complete if $x = 1$ or $\phi = \pi/2$, and is then denoted as:

$$E(k) = E(1,k) \tag{F.8}$$

$$= E(\pi/2,k) \tag{F.9}$$

$$= \pi/2 \sum_{n=0}^{\infty} \left(\frac{(2n)!}{2^{2n}(n!)^2}\right)^2 \frac{k^{2n}}{2n-1} \tag{F.10}$$

The elliptic integral of the 3rd kind is can be represented as:

$$\Pi(x,n,k) = \int_0^x \frac{dy}{(1+ny^2)\sqrt{(1-y^2)(1-k^2y^2)}} \tag{F.11}$$

or its equivalent (assuming $x = \sin\phi$)

$$\Pi(\phi,n,k) = \int_0^\phi \frac{d\theta}{(1+n\sin^2\theta)\sqrt{1-k^2\sin^2\theta}} \tag{F.12}$$

where $k^2 < 1$, and there are no conditions on n. Again the integral is called complete if $x = 1$ or $\phi = \pi/2$, although there are no simple expansions as there are for the complete elliptic integrals of the 1st and 2nd kind.

The above integrals will be briefly reconsidered in the following section.

Elliptic Functions

The elliptic functions may be defined as the inverse of the elliptic integral of the first kind. Consider the elliptic integral of the 1st kind $F(x_1,k)$. Then the elliptic function denoted by $\mathrm{sn}(\tau,k) = \mathrm{sn}(\tau)$ is defined by the relation:

$$x_1 = \mathrm{sn}[F(x_1,k)] \tag{F.13}$$

or, more suggestively:

$$F(\mathrm{sn}(\tau),k) = \int_0^{\mathrm{sn}(\tau)} \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \tag{F.14}$$

The function $\mathrm{sn}(\tau)$ is periodic with a real period $4K(k)$, so that $\mathrm{sn}\,(\tau + 4K(k)) = \mathrm{sn}(\tau)$. Also note that $\mathrm{sn}(\tau, k = 0) = \sin(\tau)$, $\mathrm{sn}(\tau, k = 1) = \tanh(\tau)$, and $|\mathrm{sn}(\tau)| \leq 1$.

There are three other related elliptic functions, denoted as $\mathrm{cn}(\tau)$, $\mathrm{dn}(\tau)$ and $\mathrm{tn}(\tau)$. These are easily defined given $\mathrm{sn}(\tau)$ as:

$$\mathrm{cn}(\tau) = \pm\sqrt{1 - \mathrm{sn}^2(\tau)} \tag{F.15}$$

$$\mathrm{dn}(\tau) = \sqrt{1 - k^2\mathrm{sn}^2(\tau)} \tag{F.16}$$

$$\mathrm{tn}(\tau) = \frac{\mathrm{sn}(\tau)}{\mathrm{cn}(\tau)} \tag{F.17}$$

Note that the proper sign of cn must be chosen, and that $0 \leq \mathrm{dn} \leq 1$.

It is instructive to rewrite the elliptic integrals in terms of elliptic functions. The elliptic integral of the 1st kind is then simply:

$$F(\tau) = \int_0^\tau d\tau \tag{F.18}$$

which appears to be, and is, trivial.

The elliptic integral of the 2nd kind is then:

$$E(\tau) = \int_0^\tau \mathrm{dn}^2(\tau)\, d\tau \tag{F.19}$$

and the elliptic integral of the 3rd kind is:

$$\Pi(\tau, n) = \int_0^\tau \frac{d\tau}{1 + n\mathrm{sn}^2(\tau)} \tag{F.20}$$

Note that the dependence of the integrals on the k parameter has been suppressed.

In the above definition of the elliptic integrals, there is no a priori bound on the value of τ, and indeed τ may increase arbitrarily. Note that due to the periodicity of the elliptic functions, the integrands will always cycle through the same values, although the total integral will continue to increase with τ. This property of the elliptic integrals is not evident in the usual definitions given in the previous section, where the parameter x can never be greater than 1. Due to this property, the elliptic integrals must be carefully evaluated to ensure that the total integral is being computed, and not just the principal value.

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